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SCHOOL OF CIVIL, ENVIRONMENTAL AND LAND MANAGEMENT  
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**POLITECNICO**  
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## ASSIGNMENT 2: CST FEM CASE STUDY STRUCTURAL ANALYSIS OF PC BRIDGE DECK BEAMS LOAD TESTING

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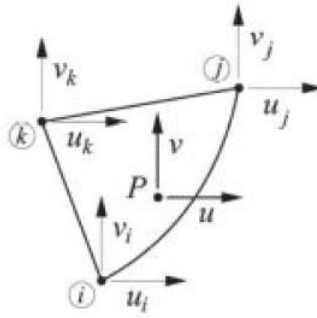


# Chapter 1

## INTRODUCTION

### 1.1 General

A finite element is characterized with a finite number of nodal points having two nodal displacements in the global reference system each. The number of nodal points is associated to the discretization in elements of the continuum. A general representation of a finite element is shown in below figure.



**Figure 1.1:** General Finite Element

The corresponding displacement functions for each element are defined as:

$$\mathbf{u} = \begin{bmatrix} u(x, y) \\ v(x, y) \end{bmatrix}$$

These displacement fields are related to the nodal displacements which, as noted from the above figure, are described as:

$$\mathbf{s}_e = [u_i \quad v_i \quad u_j \quad v_j \quad u_k \quad v_k]$$

The displacement field  $\mathbf{u}$  can further be expressed in terms of a matrix of the shape functions  $\mathbf{N}$  as:

$$\mathbf{u}(x, y) = \mathbf{N}(x, y) \mathbf{s}_e$$

Where the matrix of shape functions can be defined as:

$$\mathbf{N} = \mathbf{N}(x, y) = [N_{ij}(x, y)]$$

The shape function matrix introduces an approximation in the analysis except for if the chosen form of the shape function matrix is capable of capturing the exact form of the actual displacement functions. Nonetheless, depending on the nature of the approximation, the behavior of the model might converge to that of the continuous structure on reducing the size of the elements, or in other words, on adopting a finer discretization of the domain.

## 1.2 Kinematic field

The strain field  $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}(x, y)$  can be associated to the nodal displacements  $\mathbf{s}$  through compatibility and a separate description of the displacement fields as:

$$\boldsymbol{\varepsilon}(x, y) = \begin{bmatrix} \varepsilon_x(x, y) \\ \varepsilon_y(x, y) \\ \gamma(x, y) \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} u(x, y) \\ v(x, y) \end{bmatrix}$$

Now, employing the above discussed relationship between the displacement functions and the shape functions and introducing a linear differential operator matrix  $\mathbf{L}$ , the compatibility equation transforms as:

$$\boldsymbol{\varepsilon}(x, y) = \mathbf{L} \mathbf{u}(x, y) = \mathbf{L} \mathbf{N}(x, y) \mathbf{s}_e = \mathbf{B}(x, y) \mathbf{s}_e$$

Here,  $\mathbf{B}$  is the compatibility matrix described as:

$$\mathbf{B} = \mathbf{B}(x, y) = \mathbf{L} \mathbf{N}(x, y)$$

## 1.3 Static Field

The static field  $\boldsymbol{\sigma} = \boldsymbol{\sigma}(x, y)$  is estimated utilizing equilibrium, constitutive laws, and the element strains as noted above.

$$\boldsymbol{\sigma}(x, y) = \begin{bmatrix} \sigma_x(x, y) \\ \sigma_y(x, y) \\ \tau(x, y) \end{bmatrix} = \mathbf{D} \boldsymbol{\varepsilon}(x, y) = \mathbf{D} \mathbf{B}(x, y) \mathbf{s}_e$$

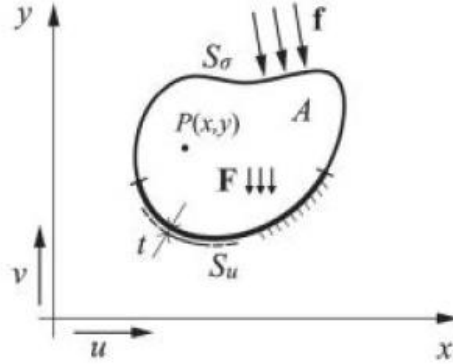
Here,  $\mathbf{D}$  is the constitutive matrix which is constant for a homogeneous material. For an isotropic material, the constitutive law would be in accordance with the type of problem at hand, plane stress or plane strain, and it is stated as:

$$\mathbf{D} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \quad \text{Plane Stress}$$

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \quad \text{Plane Strain}$$

#### 1.4 Principal of Virtual Displacements

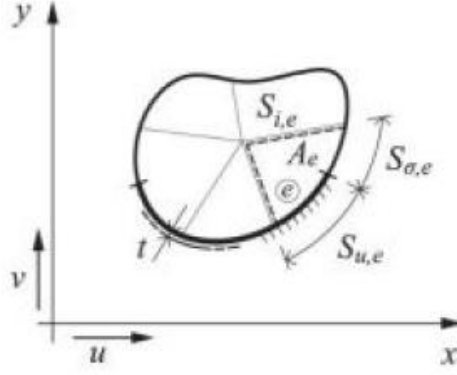
For a continuum, as represented in the following figure, subjected to applied forces on  $S_\sigma$ , imposed displacements on  $S_u$ , and acted on by interfacial forces on  $S_i$ , the principal of virtual displacement is defined as:



**Figure 1.2:** Representation of a general continuum

$$\int_A \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dA = \int_A \delta \mathbf{u}^T \mathbf{F} dA + \int_{S_\sigma} \delta \mathbf{u}^T \mathbf{f} dS + \int_{S_u} \bar{\mathbf{u}}^T \mathbf{r} dS + \int_{S_i} \bar{\mathbf{u}}^T \mathbf{t} dS$$

Now, if the continuum is subdivided into finite elements as illustrated in the following figure, the principle of virtual displacement for each element is given as:



**Figure 1.3:** Subdivision of a continuum into elements

$$\int_{A_e} \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dA = \int_{A_e} \delta \mathbf{u}^T \mathbf{F} dA + \int_{S_{\sigma,e}} \delta \mathbf{u}^T \mathbf{f} dS + \int_{S_{u,e}} \bar{\mathbf{u}}^T \mathbf{r} dS + \int_{S_{i,e}} \bar{\mathbf{u}}^T \mathbf{t} dS$$

The virtual variation on the imposed displacements is null and thus, the corresponding term vanishes. Summing up the contribution of all the elements composing the continuum, the principal of virtual displacements is written as:

$$\sum_e \int_{A_e} \delta \boldsymbol{\varepsilon}^T \boldsymbol{\sigma} dA = \sum_e \int_{A_e} \delta \mathbf{u}^T \mathbf{F} dA + \sum_e \int_{S_{\sigma,e}} \delta \mathbf{u}^T \mathbf{f} dS$$

Here, the term accounting for the interface forces is omitted with reference to the overall interface equilibrium. Now, considering the virtual variation of the displacement and the strain fields as:

$$\delta \mathbf{u}(x, y) = \delta [\mathbf{N}(x, y) \mathbf{s}_e] = \mathbf{N}(x, y) \delta \mathbf{s}_e$$

$$\delta \boldsymbol{\varepsilon}(x, y) = \delta [\mathbf{B}(x, y) \mathbf{s}_e] = \mathbf{B}(x, y) \delta \mathbf{s}_e$$

Consequently, the principle of virtual displacement reduces to an algebraic form as:

$$\sum_e \int_{A_e} \delta \mathbf{s}_e^T \mathbf{B}^T \mathbf{D} \mathbf{B} dA \mathbf{s}_e = \sum_e \int_{A_e} \delta \mathbf{s}_e^T \mathbf{N}^T \mathbf{F} dA + \sum_e \delta \mathbf{s}_e^T \int_{S_{\sigma,e}} \delta \mathbf{s}_e^T \mathbf{N}^T \mathbf{f} dS$$

## 1.5 Stiffness Matrix and Equivalent Nodal Forces

The algebraic form of the principle of virtual displacement can be further modified by taking the terms independent of  $x$  and  $y$  out of the integral as:

$$\sum_e \delta \mathbf{s}_e^T \int_{A_e} \mathbf{B}^T \mathbf{D} \mathbf{B} dA \mathbf{s}_e = \sum_e \delta \mathbf{s}_e^T \int_{A_e} \mathbf{N}^T \mathbf{F} dA \mathbf{s}_e + \sum_e \delta \mathbf{s}_e^T \int_{S_{\sigma,e}} \mathbf{N}^T \mathbf{f} dS \mathbf{s}_e$$

The integrals noticed in the above equation can be physically interpreted in the following way:

$$\mathbf{K}_e = \int_{A_e} \mathbf{B}^T \mathbf{D} \mathbf{B} dA \quad \text{Stiffness matrix}$$

$$\mathbf{f}_{V,e} = \int_{A_e} \mathbf{N}^T \mathbf{F} dA \quad \text{Equivalent Nodal force Vector – Volumetric Forces}$$

$$\mathbf{f}_{S,e} = \int_{S_{\sigma,e}} \mathbf{N}^T \mathbf{f} dS \quad \text{Equivalent Nodal force Vector – Surface Forces}$$

Thus, the equation reduces to:

$$\sum_e \delta \mathbf{s}_e^T \mathbf{K}_e = \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{V,e} + \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{S,e}$$

## 1.6 Assembly and Linear Equilibrium Equations

The vector of nodal displacements of the structure is identified as:

$$\mathbf{s} = [s_1 \quad s_2 \quad s_3 \quad \dots \quad s_n]^T$$

The contribution from each element is assembled into a global form as:

$$\begin{cases} \sum_e \delta \mathbf{s}_e^T \mathbf{K}_e = \delta \mathbf{s}^T \mathbf{K} \mathbf{s} \\ \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{V,e} = \delta \mathbf{s}^T \mathbf{f}_V \\ \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{S,e} = \delta \mathbf{s}^T \mathbf{f}_S \end{cases} \Rightarrow \delta \mathbf{s}^T \mathbf{K} \mathbf{s} = \delta \mathbf{s}^T \mathbf{f}_V + \delta \mathbf{s}^T \mathbf{f}_S$$

On simplifying, a linear equation system is obtained with the solution prescribed as:

$$\mathbf{K} \cdot \mathbf{s} = (\mathbf{f}_V + \mathbf{f}_S) = \mathbf{f} \quad \Rightarrow \quad \mathbf{s} = \mathbf{K}^{-1} \mathbf{f}$$

Where:

- $\mathbf{K}$  is the stiffness matrix of the structure
- $\mathbf{f}$  is the nodal force vector
- $\mathbf{s}$  is the nodal displacement vector

## 1.7 Stress Analysis

The principle of virtual displacements as established till now assumes an equilibrated actual stress field with a compatible virtual kinematic field. The solution obtained through the finite element analysis based on a separate description of the displacement fields is generally an approximation. After the solution of the system in terms of the nodal displacement of the structure  $\mathbf{s}$  is determined, we go back to the definition at the element level for stress analysis. The following path of reasoning is adopted for the same:

$$\text{Displacement Functions} = \mathbf{u}(x, y) = \mathbf{N}(x, y) \mathbf{s}_e$$

$$\text{Compatibility and Strains} = \boldsymbol{\varepsilon}(x, y) = \mathbf{B}(x, y) \mathbf{s}_e$$

$$\text{Equilibrium and Stresses} = \boldsymbol{\sigma}(x, y) = \mathbf{DB}(x, y) \mathbf{s}_e$$

## 1.8 Compatibility and Equilibrium

At the element boundary surface  $S_{u,e}$ , as shown in the last figure, the compatibility is enforced only at the nodes:

$$\int_{S_{u,e}} \delta \bar{\mathbf{u}}^T(x, y) \mathbf{r}(x, y) dS = \delta \mathbf{s}_e^T \int_{S_{u,e}} \delta \mathbf{N}^T(x, y) \mathbf{r}(x, y) dS = \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{r,e} = 0$$

$$\delta \mathbf{s}_e^T = 0 \quad \Rightarrow \text{At node } m, \quad \mathbf{s}_e^m = \bar{\mathbf{s}}_e^m$$

Where:

$\mathbf{f}_{r,e}$  is the equivalent nodal force vector to the reactions

Equilibrium is satisfied in an integral form; however, local violations are possible. At the interface of the element  $S_{i,e}$ , equilibrium is ensured only at the nodes:

$$\sum_e \int_{S_{i,e}} \delta \bar{\mathbf{u}}^T(x, y) \mathbf{r}(x, y) dS = \sum_e \delta \mathbf{s}_e^T \int_{S_{i,e}} \delta \mathbf{N}^T(x, y) \mathbf{r}(x, y) dS = \sum_e \delta \mathbf{s}_e^T \mathbf{f}_{t,e} = 0$$

$$\sum_e \delta \mathbf{s}_e^T \mathbf{f}_{t,e} = 0 \quad \Rightarrow \quad \text{At node } m, \quad \sum_e \mathbf{f}_{t,e} = 0$$

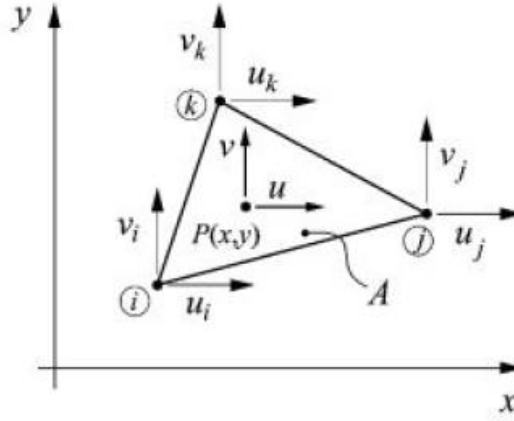
Where:

$\mathbf{f}_{t,e}$  is the equivalent nodal force vector to the reactions

## Chapter 2

### CONSTANT STRAIN TRIANGLE

The subdivision in elements of a 2D body can be done using different elements. One of the simplest and most useful ones is the Constant Strain Triangle (CST). A single 2D CST element is defined below:



$$\mathbf{u} = \begin{bmatrix} u(x, y) \\ v(x, y) \end{bmatrix} \quad \mathbf{s}_e = [u_i \quad v_i \quad u_j \quad v_j \quad u_k \quad v_k]^T$$

As it is observed for a generic element, we can collect the nodal displacements in a vector  $\mathbf{s}$  and then describe the rest of the displacement field using appropriate interpolating functions. In this case we impose the nodal displacements as boundary conditions and assume linear dependence on both the  $x$  and  $y$  directions leading to the following description of the displacement field.

$$\begin{aligned} \mathbf{u}(x_i, y_i) &= u_i & \mathbf{v}(x_i, y_i) &= v_i \\ \mathbf{u}(x_j, y_j) &= u_j & \mathbf{v}(x_j, y_j) &= v_j \\ \mathbf{u}(x_k, y_k) &= u_k & \mathbf{v}(x_k, y_k) &= v_k \end{aligned}$$

$$u(x, y) = a_1 + a_2x + a_3y$$

$$v(x, y) = a_4 + a_5x + a_6y$$



In matrix form we have:

$$\begin{bmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_k & y_k \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} u_i \\ u_j \\ u_k \end{bmatrix}$$

Where:

$$\mathbf{C} = \begin{bmatrix} 1 & x_i & y_i \\ 1 & x_j & y_j \\ 1 & x_k & y_k \end{bmatrix}$$

This is a linear system of equations in which the unknowns are the coefficients  $\alpha_1$ ,  $\alpha_2$  and  $\alpha_3$ .

Using Cramer's rule, we can find the solution:

$$\alpha_i = \frac{\det C_i}{\det C}$$

Where  $C_i$  is the matrix formed by replacing the  $i^{th}$  column of  $\mathbf{C}$  by the vector of constant terms.

Where:

$$\mathbf{C}_1 = \begin{bmatrix} u_i & x_i & y_i \\ u_j & x_j & y_j \\ u_k & x_k & y_k \end{bmatrix} \quad \mathbf{C}_2 = \begin{bmatrix} 1 & u_i & y_i \\ 1 & u_j & y_j \\ 1 & u_k & y_k \end{bmatrix} \quad \mathbf{C}_3 = \begin{bmatrix} 1 & x_i & u_i \\ 1 & x_j & u_j \\ 1 & x_k & u_k \end{bmatrix}$$

$$\det \mathbf{C} = [(x_j y_k - x_k y_j) + (x_k y_i - x_i y_k) + (x_i y_j - x_j y_i)] = 2A$$

And:

$$\alpha_1 = \frac{1}{2A} [(x_j y_k - x_k y_j)u_i + (x_k y_i - x_i y_k)u_j + (x_i y_j - x_j y_i)u_k]$$

$$\alpha_2 = \frac{1}{2A} [(y_j - y_k)u_i + (y_k - y_i)u_j + (y_i - y_j)u_k]$$

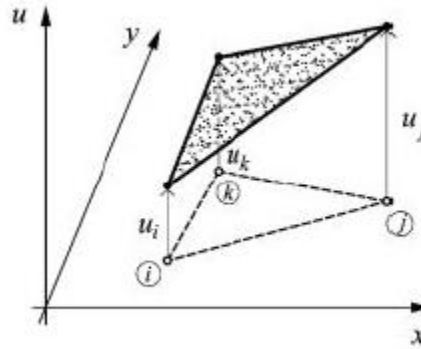
$$\alpha_3 = \frac{1}{2A} [(x_k - x_j)u_i + (x_i - x_k)u_j + (x_j - x_i)u_k]$$

We can simplify the expression making the following observations:

$$u(x, y) = \frac{1}{2A} [(a_i + b_i x + c_i y)u_i + (a_j + b_j x + c_j y)u_j + (a_k + b_k x + c_k y)u_k]$$

## 2.1 Shape Functions

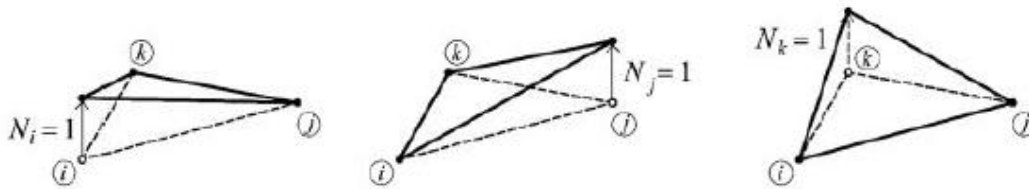
Up until now, we haven't defined the set of shape functions  $N_i(x, y)$  that allow to describe the displacement field as a function of the nodal displacements. However, we can see this dependence in the description of the displacement field. We now give an alternative but equivalent way to describe the displacement field by defining the shape functions.



$$\mathbf{u}(x, y) = \frac{1}{2A} [N_i(x, y)u_i + N_j(x, y)u_j + N_k(x, y)u_k]$$

$$N_i(x, y) = \frac{1}{2A} (a_i + b_i x + c_i y)$$

For how we have defined them, the shape function assume this form:



The displacement field become:

$$\mathbf{u}(x, y) = \begin{bmatrix} N_i & 0 & N_j & 0 & N_k & 0 \\ 0 & N_i & 0 & N_j & 0 & N_k \end{bmatrix} [u_i \quad v_i \quad u_j \quad v_j \quad u_k \quad v_k]^T = \mathbf{N}(x, y) \mathbf{s}$$

## 2.2 Strain Field

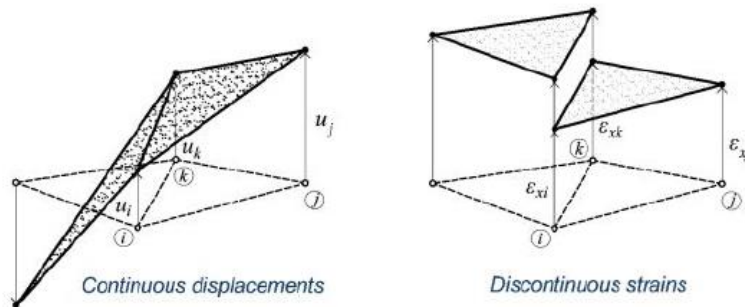
Having defined the displacement field inside the element we use the compatibility equations to get the strains:

$$\boldsymbol{\varepsilon}(x, y) = \begin{bmatrix} \varepsilon_x(x, y) \\ \varepsilon_y(x, y) \\ \gamma(x, y) \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} u(x, y) \\ v(x, y) \end{bmatrix}$$

$$= \mathbf{L} \mathbf{u}(x, y) = \mathbf{L} \mathbf{N}(x, y) \mathbf{s} = \mathbf{B}(x, y) \mathbf{s}$$

$$\mathbf{B}(x, y) = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 & \frac{\partial N_j}{\partial x} & 0 & \frac{\partial N_k}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} & 0 & \frac{\partial N_j}{\partial y} & 0 & \frac{\partial N_k}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} & \frac{\partial N_j}{\partial y} & \frac{\partial N_j}{\partial x} & \frac{\partial N_k}{\partial y} & \frac{\partial N_k}{\partial x} \end{bmatrix} = \frac{1}{2A} \begin{bmatrix} b_i & 0 & b_j & 0 & b_k & 0 \\ 0 & c_i & 0 & c_j & 0 & c_k \\ c_i & b_i & c_j & b_j & c_k & b_k \end{bmatrix}$$

As it can be seen, the matrix B is made of constant terms, so the strains inside an element are constant (doesn't depend on the coordinate of a point inside the element), hence the name **CONSTANT STRAIN TRIANGLE**. Another important thing to note is that the strain field will present discontinuities in the boundaries between elements if the slope of the displacement function changes.



**Figure 2.1:** Discontinuity of strain field

## 2.3 Stress Field

The CST formulation is valid for any constitutive law we decide to adopt, note that the fact that the strains are constant does not necessarily mean that the stresses will be so too. This will depend on the material and on the imposed strains, if the material is homogeneous and the imposed strains constant then we can conclude that the stress field will be constant over the element.

$$\sigma(x, y) = \mathbf{D}[\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0(x, y)] = \mathbf{D}\mathbf{B}\mathbf{s} - \mathbf{D}\boldsymbol{\varepsilon}_0(x, y)$$

## 2.4 Stiffness Matrix

Now that the compatibility and constitutive matrices have been defined for the CST, we can go back to the solution of the PVD system, in particular to the definition of the stiffness matrix of the single element. Assuming a plane stress or plane strain with linear elastic homogeneous isotropic material and small displacement hypothesis, it is possible to consider B and D matrices constant over the element and so:

$$\mathbf{K} = t \int_A \mathbf{B}^T \mathbf{D} \mathbf{B} dA = tA \mathbf{B}^T \mathbf{D} \mathbf{B}$$

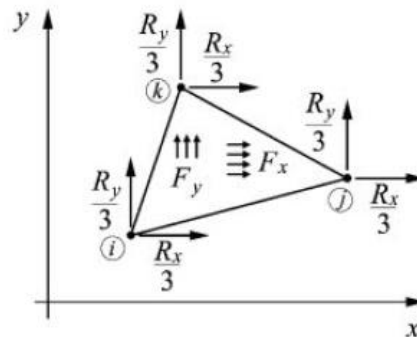
## 2.5 Equivalent nodal forces vector

The load type that will be treated are:

- Volume forces
- Surface forces
- Temperature

### 2.5.1 Volume forces

$$\mathbf{F} = \begin{bmatrix} F_x(x, y) \\ F_y(x, y) \end{bmatrix}$$



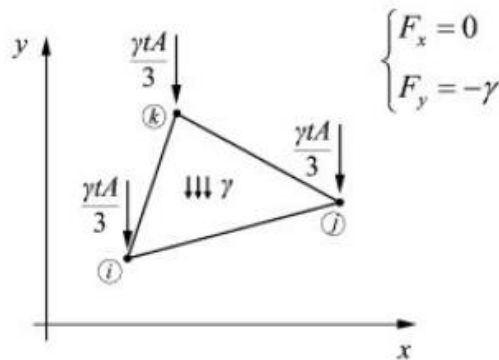
$$\mathbf{f}_V = t \int_A \mathbf{N}^T \mathbf{F} dA = t \int_A \begin{bmatrix} N_i & 0 \\ 0 & N_i \\ N_j & 0 \\ 0 & N_j \\ N_k & 0 \\ 0 & N_k \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix} dA$$

Let's consider constant volume forces. We can take F out of the integral.

$$\mathbf{f}_V = -t \int_A \mathbf{N}^T dA \mathbf{F}$$

$$\int_A N_i(x, y) dA = \frac{A}{3} \Rightarrow \mathbf{f}_V = \frac{tA}{3} \begin{bmatrix} F_x \\ F_y \\ F_x \\ F_y \\ F_x \\ F_y \end{bmatrix} = \frac{1}{3} \begin{bmatrix} R_x \\ R_y \\ R_x \\ R_y \\ R_x \\ R_y \end{bmatrix}$$

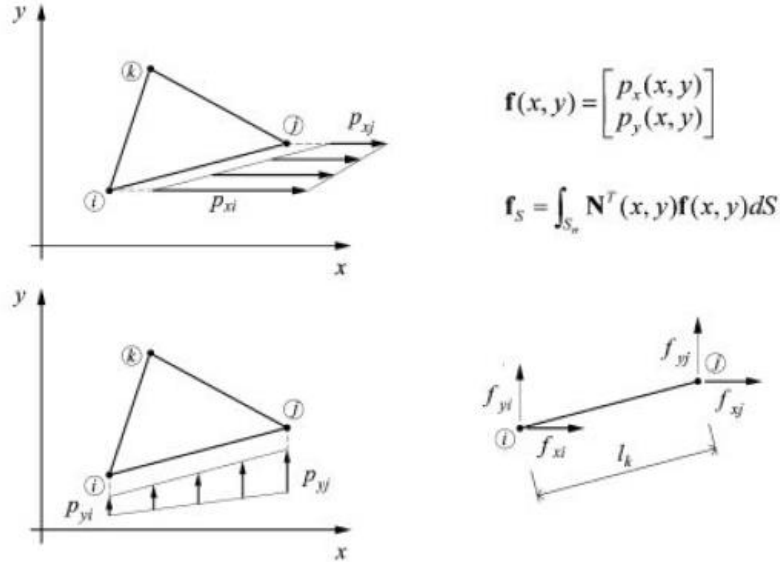
**Self-Weight:**



In this case the only volume force present is given by the density of the material. The force is the density times the volume of the element:  $F_y = \gamma V = \gamma A t$

$$\mathbf{f}_V = \frac{tA}{3} \begin{bmatrix} F_x \\ F_y \\ F_x \\ F_y \\ F_x \\ F_y \end{bmatrix} = -\frac{\gamma A t}{3} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

## 2.5.2 Surface forces



Let's consider an internal coordinate  $\xi = \frac{s}{l_k}$  where  $s$  is abscissa along the surface starting from node  $i$ .

First just consider the load along  $x, p_x$ :

Displacement function

$$u(\xi) = (1 - \xi)u_i + \xi u_j$$

Virtual Displacements

$$\delta u(\xi) = (1 - \xi)\delta u_i + \xi \delta u_j$$

Load function Principle of Virtual Displacements

$$p_x(\xi) = (1 - \xi)p_{x_i} + \xi p_{x_j} \quad f_{x_i}\delta u(\xi) + f_{x_j}\delta u_j = \int_0^1 p_x(\xi) \delta u(\xi) l_k d\xi$$

Let's fix node  $j$  ( $\delta u_j = 0$ ) and impose a virtual displacement at node  $i$ .

$$\delta u(\xi) = (1 - \xi)\delta u_i$$

The Principle of Virtual Displacements became:

$$f_{x_i} \delta u(\xi) = \int_0^1 p_x(\xi) \delta u(\xi) l_k d\xi = \delta u(\xi) l_k \int_0^1 [(1-\xi)p_{x_i} + \xi p_{x_j}](1-\xi) d\xi$$

$$f_{x_i} = l_k \int_0^1 [(1-\xi)p_{x_i} + \xi p_{x_j}](1-\xi) d\xi \Rightarrow f_{x_i} = \frac{l_k}{6} (2p_{x_i} + p_{x_j})$$

The same can be done for  $\delta u_i = 0$  and for the load along y. The final result is:

$$f_{x_i} = \frac{l_k}{6} (2p_{x_i} + p_{x_j})$$

$$f_{x_j} = \frac{l_k}{6} (2p_{x_j} + p_{x_i})$$

$$f_{y_i} = \frac{l_k}{6} (2p_{y_i} + p_{y_j})$$

$$f_{y_j} = \frac{l_k}{6} (2p_{y_j} + p_{y_i})$$

### 2.5.3 Thermal load

Let's consider imposed strains:

$$f_\epsilon = t \int_A \mathbf{B}^T \mathbf{D} \boldsymbol{\epsilon}_0(x, y) dA$$

Let's consider a uniform temperature change in plane stress hypothesis:

$$\boldsymbol{\epsilon}_0 = \alpha \Delta T [1 \quad 1 \quad 0]^T$$

$$\mathbf{D} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

The result is:

$$\mathbf{f}_\varepsilon = \frac{E(1+\nu)t}{2(1-\nu^2)} \alpha \Delta T \begin{bmatrix} b_i \\ c_i \\ b_j \\ c_j \\ b_k \\ c_k \end{bmatrix}$$

Where the coefficient  $b_i, c_i, b_j, c_j, b_k, c_k$  are the element of the compatibility matrix **B**.



## Chapter 3

### TYPE OF ANALYSIS

The current and subsequent chapters are devoted to elaborating the implementation of the CST-FEM Code developed using the FORTRAN language.

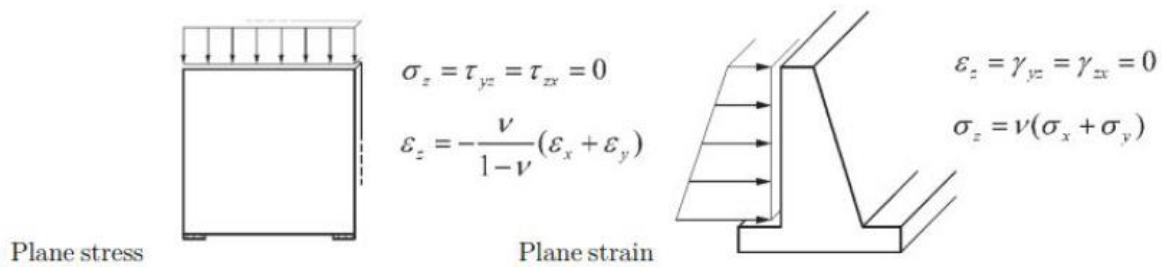
The first step to consider is the type of analysis required to be performed. In this program, we implemented *Plane Stress* and *Plane Strain* analysis. In order to keep track of which one we are dealing with we use the variable **ICODE**.

$$\text{ICODE} = \begin{cases} 0 & \Rightarrow \text{Plane Stress} \\ 1 & \Rightarrow \text{Plane Strain} \end{cases}$$

The following to be kept in consideration:

- Plane stress: is used when the out-of-plane dimension is much smaller than the in-plane dimensions ( $z \ll 1$ ).
- Plane strain: is used when the out-of-plane dimension is much larger than the in-plane dimensions ( $z \rightarrow \infty$ ).

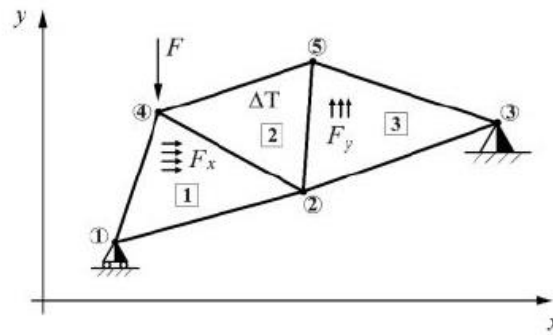
The choice influences the constitutive matrix **D** and, by consequence, the stiffness matrix **K** and the vector of nodal forces.



## Chapter 4

### GEOMETRY

The first step in analyzing a structure is defining its shape and mechanical properties. To do so, it needs to be divided into elements with uniform properties that can be idealized as straight lines and a specific material and cross-section. Restraints or joints must be specified. We collect and organize this information in the **GEOMET** subroutine of the code.



**Figure 4.1:** Example of a 2D structure

#### 4.1 Nodes and Coordinates

To identify the position of each beam element 2 nodes are needed. We begin by defining the nodes and their position in the plane by listing their global  $x$  and  $y$  coordinates. It's also important to know the total number of nodes which is denoted as **NNODE**. To organize this information in the input phase, the **COORD** matrix is used. This is a matrix formed by a number of rows equal to the number of nodes and 2 columns.

| COORD (NNODE,2) |             |             |
|-----------------|-------------|-------------|
| Node            | $x$         | $y$         |
| 1               | $x_1$       | $y_1$       |
| 2               | $x_2$       | $y_2$       |
| 3               | $x_3$       | $y_3$       |
| ...             | ...         | ...         |
| NNODE           | $x_{NNODE}$ | $y_{NNODE}$ |

This matrix is also used to give an ID number to each node, this will then be used to assign the correct nodes to each element.

## 4.2 Elements

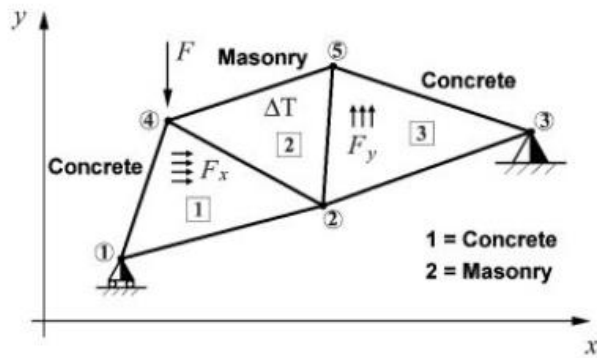
The following step is to describe the elements that compose the system. Each element must have 3 nodes assigned to it and to do so we use the incidence matrix or **INC** which is defined in the input by numbering the elements and assigning the nodes that generate it. The total number of elements is denoted as **NELE** and so the **INC** matrix has  $(\text{NELE}) \times 3$ .

| IN (NELE,3) |              |              |              |
|-------------|--------------|--------------|--------------|
| Element     | Node 1       | Node 2       | Node 3       |
| 1           | $N_{11}$     | $N_{12}$     | $N_{13}$     |
| 2           | $N_{21}$     | $N_{22}$     | $N_{23}$     |
| 3           | $N_{31}$     | $N_{32}$     | $N_{33}$     |
| ...         | ...          | ...          | ...          |
| NELE        | $N_{NELE,1}$ | $N_{NELE,2}$ | $N_{NELE,3}$ |

Note that even though each element requires 3 nodes, **NNODE** is not necessarily three times **NELE** since the elements share nodes when the frame is continuous, but they require separate nodes if there's an internal link. This aspect will be discussed in detail in the following chapter.

## 4.3 Material Properties

To specify the material of each element we use the same method seen for the cross sections; in this case, we use the matrix **CMAT** to define each material. This matrix will have 5 columns and **NMAT** rows, which is the total number of different materials present in the frame. Each material is here defined using 5 properties: Young's modulus, Poisson's ratio, thermal expansion coefficient, specific weight and out-of-plane thickness.



**Figure 4.2:** Example of different materials inside a system

| CMAT (NMAT,4) |   |       |          |          |   |
|---------------|---|-------|----------|----------|---|
| Material      | E | $\nu$ | $\alpha$ | $\gamma$ | t |
| 1             |   |       |          |          |   |
| 2             |   |       |          |          |   |
| 3             |   |       |          |          |   |
| ...           |   |       |          |          |   |
| NMAT          |   |       |          |          |   |

The vector “**IMAT**” is then used to assign to each element its corresponding material.

| IMAT (NMAT) |          |
|-------------|----------|
| Element     | Material |
| 1           |          |
| 2           |          |
| 3           |          |
| ...         | ...      |
| NELE        | ...      |

## Chapter 5

# DEGREE OF FREEDOM AND UNKNOWN NODAL DISPLACEMENTS

Once the geometry and mechanical properties of the frame are defined, we must establish the structural behavior, this will be done on the **SCODE** subroutine. We begin by identifying the global degrees of freedom of our structure. These nodal displacements will be the unknowns of the problem. The numbering of the degrees of freedom relies on the **IDOF** matrix that describes the two nodal displacements of each node in the system.

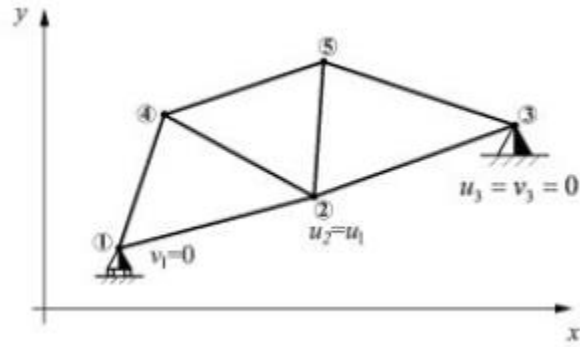
| IDOF (NNODE,2) |   |   |
|----------------|---|---|
| Node           | u | v |
| 1              |   |   |
| 2              |   |   |
| 3              |   |   |
| ...            |   |   |
| NNODE          |   |   |

To build this matrix, three main steps are required to account for external restraints and internal links. In order to better illustrate these steps, we refer to the following structure:

This system is characterized by 5 nodes and a total of 3 constrains and 1 link.

### 5.1 Initialization

We begin by setting all the terms equal to zero, the value of each term will be changed according to the links and restraints specified in the input file.



**Figure 5.1:** Example of constrains and links

| IDOF (NNODE,2) |   |   |
|----------------|---|---|
| Node           | u | v |
| 1              | 0 | 0 |
| 2              | 0 | 0 |
| 3              | 0 | 0 |
| 4              | 0 | 0 |
| 5              | 0 | 0 |

## 5.2 Restraints and Links

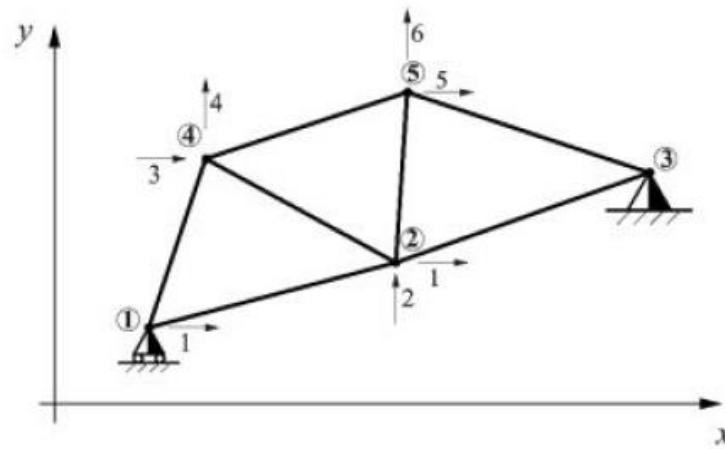
The following step is to consider the external constraints on each node. For each of these, we will change the value in the matrix from 0 to -1. In the example structure, the external restraints block the vertical displacement of node 1 and all the degrees of freedom of node 3. We must also consider the internal links by using the Master-Slave approach. In each internal link, we define two separate nodes and call the one with a lower ID number as Master and the other as Slave. Then we impose that the linked degrees of freedom of the Slave are equal to those of the Master.

In this case, the internal hinge is linking horizontal displacements of nodes 1 and 2. Node 1 is the Master and so we must impose the horizontal displacements of node 2 to be equal to those of node 1. To do so, we change the value from 0 to the ID number of the Master which in this case is 1. After this step the IDOF matrix should be the following:

| IDOF (NNODE,2) |    |    |
|----------------|----|----|
| Node           | u  | v  |
| 1              | 0  | -1 |
| 2              | 1  | 0  |
| 3              | -1 | -1 |
| 4              | 0  | 0  |
| 5              | 0  | 0  |

### 5.3 Numbering the degrees of freedom

The number of DOFs are now counted sequentially from the zeros in the **IDOF** matrix. In order to impose the link between the master and the slave node the DOF label of the slave node is to be same as that of the master node. For the system considered the consequent **IDOF** matrix is illustrated in the following table. The numerical implementation of the above-stated procedure is condensed as:



**Figure 5.2:** Example of numbering DOF inside a frame

(a) if the component  $j$  at the node  $i$  is not restrained

$$d_{ij} = d_{ij,max} + 1$$

(b) if the component  $j$  at the node  $i$  is restrained

$$d_{ij} = -1$$

(c) if the component j at the node i is linked to component h at node k < i

$$d_{ij} = d_{hk}$$

Where:

- $i = 1, \dots, \text{NNODE}$
- $j = 1, 2, 3$
- $d_{ij, \max} = \max_{k \leq i, h \leq j} d_{hk}$
- $\text{NDOF} = d_{\max} = \max_{i,j} d_{ij}$

This value coincides with the largest value inside the **IDOF** matrix. With regards to the example, the degrees of freedom in the global reference system are:

| IDOF (NNODE,2) |    |    |
|----------------|----|----|
| Node           | u  | v  |
| 1              | 1  | -1 |
| 2              | 1  | 2  |
| 3              | -1 | -1 |
| 4              | 3  | 4  |
| 5              | 5  | 6  |

## 5.4 Unknown Nodal Displacements

Once we know the degrees of freedom of the structure, we can define the vector of the unknown nodal displacements **VDISP** which naturally has a dimension equal to **NDOF**. This vector will give me the value of the displacement along each degree of freedom once the system is solved. It's important to note that **NDOF** gives the size of the problem.



| VDISP(NDOF,1) |              |
|---------------|--------------|
| DOF           | Displacement |
| 1             | $s_1$        |
| 2             | $s_2$        |
| ...           | ...          |
| NDOF          | $s_{NDOF}$   |

It's important also to define the direction of the displacements in the global reference system as this will influence the sign of the applied loads as well.

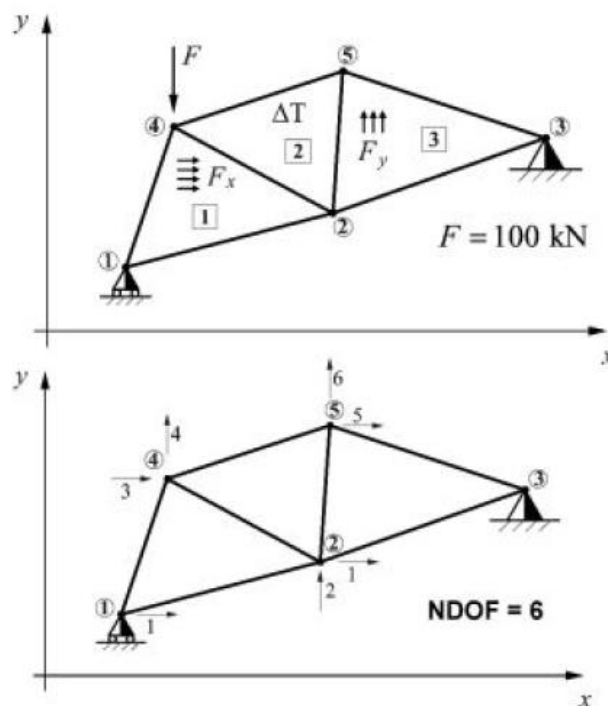
## Chapter 6

### LOADS

Once the structural behavior has been determined, the loads acting on our structure have to be defined. In order to get the nodal load vector as the input of the system we will consider nodal loads, surface loads, self-weight and thermal actions. The **LOADS** subroutine is devoted to this section. Let's consider the following forces on the structure to better illustrate the procedure.

#### 6.1 Concentrated nodal loads

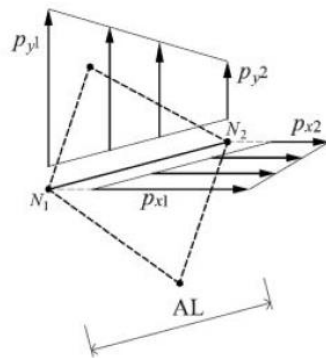
The first set is to build the vector of nodal loads **VLOADS**, this has dimension **NDOF** and assigns the applied load to the corresponding degree of freedom along which it acts. It's important to use the correct sign convention, the load is positive if it acts in the same direction as the nodal displacement in the global reference frame and negative if the direction is opposite.



| VLOADS (NDOF,1) |      |
|-----------------|------|
| DOF             | LOAD |
| 1               | 0    |
| 2               | 0    |
| 3               | 0    |
| 4               | -100 |
| ...             | 0    |
| NDOF            | ...  |

## 6.2 Surface loads

The implementation of surface loads was discussed in detail in the Chapter 2. The same is recalled here.



$$jd1 = IDOF(N1, 1)$$

$$jd2 = IDOF(N1, 2)$$

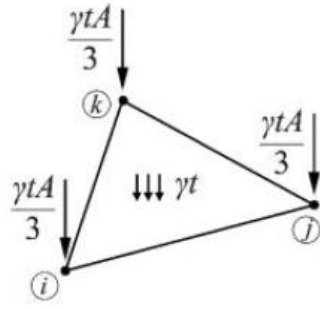
$$jd3 = IDOF(N2, 1)$$

$$jd4 = IDOF(N2, 2)$$

$$\begin{cases} VLOADS(jd1) = \frac{AL}{6} (2p_{x1} + p_{x2}) \\ VLOADS(jd2) = \frac{AL}{6} (2p_{y1} + p_{y2}) \\ VLOADS(jd3) = \frac{AL}{6} (2p_{x2} + p_{x1}) \\ VLOADS(jd4) = \frac{AL}{6} (2p_{y2} + p_{y1}) \end{cases}$$

## 6.3 Self Weight

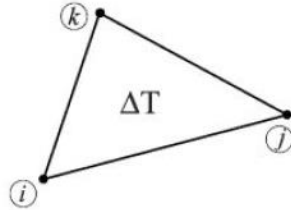
The implementation of loads due to self-weight was discussed in detail in the Chapter 2. The same is recalled here.



$$\begin{aligned} & \mathbf{f} \\ & \begin{aligned} jd1 &= IDOF(N1, 2) \\ jd2 &= IDOF(N2, 2) \\ jd3 &= IDOF(N3, 2) \end{aligned} \\ & \begin{cases} VLOADS(jd1) = -\frac{\gamma At}{3} \\ VLOADS(jd2) = -\frac{\gamma At}{3} \\ VLOADS(jd3) = -\frac{\gamma At}{3} \end{cases} \end{aligned}$$

## 6.4 Thermal Variation

The implementation of loads due to thermal effects was discussed in detail in the Chapter 2. The same is recalled here.



$$\begin{aligned} \mathbf{f}_\varepsilon &= \frac{E(1+\nu)t}{2k} \alpha \Delta T \begin{bmatrix} b_i & c_i & b_j & c_j & b_k & c_k \end{bmatrix}^T \\ k &= \begin{cases} 1 - \nu & ICODE = 0 \\ (1 + \nu)(1 - 2\nu) & ICODE = 1 \end{cases} \\ jd1 &= IDOF(Nh, 1) \\ jd2 &= IDOF(Nh, 2) \\ \begin{cases} VLOADS(jd1) = \frac{E(1+\nu)t}{2k} \alpha \Delta T \cdot b_h \\ VLOADS(jd2) = \frac{E(1+\nu)t}{2k} \alpha \Delta T \cdot c_h \end{cases} \\ h &= i, j, k \end{aligned}$$

## Chapter 7

### ASSEMBLY

First we need to compute the compatibility matrix **B**, the constitutive matrix **D** and the element stiffness matrix **ST**. This is done in the subroutine **MKK**. The element stiffness matrix is obtained by:

$$\mathbf{ST}_{(6,6)} = t \mathbf{A} \mathbf{B}^T \mathbf{D} \mathbf{B}$$

Once the stiffness matrix and the vector of equivalent nodal forces for each element have been computed, the next step is combining all these quantities into a global system for the whole structure to obtain the global governing equilibrium equation of the system. To do so, the global stiffness matrix and the global load vector have to be assembled. This part is devoted to the subroutine **ASSEMB**.

We introduce the assembly matrix  $\mathbf{Z}_m(6, \text{NDOF})$ , one for each element, to make this passage.

If the  $j$  degree of freedom of the element coincides with the  $i$  degree of freedom of the system, value of that component of the matrix is equal to 1, otherwise is 0.

$$\mathbf{Z}_m = z_{ij}^m$$

- $z_{ij}^m = 1$  if  $j_{\text{element}} \rightarrow i_{\text{system}}$
- $z_{ij}^m = 0$  otherwise

Once the assembly matrix has been obtained:

$$\text{Equilibrium Equation (Element)} \quad \Rightarrow \quad \mathbf{ST} \mathbf{s}_m = \mathbf{f}_m$$

$$\text{Replace} \quad \mathbf{s}_m = \mathbf{Z}_m^T \mathbf{s} \quad \Rightarrow \quad \mathbf{ST} \mathbf{Z}_m^T \mathbf{s} = \mathbf{f}_m$$

$$\text{Pre - multiply each side by } \mathbf{Z}_m \quad \Rightarrow \quad \mathbf{Z}_m \mathbf{ST} \mathbf{Z}_m^T \mathbf{s} = \mathbf{Z}_m \mathbf{f}_m$$

$$\text{Sum over the structural system} \quad \Rightarrow \quad \sum_m \mathbf{Z}_m \mathbf{ST} \mathbf{Z}_m^T \mathbf{s} = \sum_m \mathbf{Z}_m \mathbf{f}_m$$

$$\text{Equilibrium equation (System)} \quad \Rightarrow \quad \mathbf{K} \mathbf{s} = \mathbf{f}$$

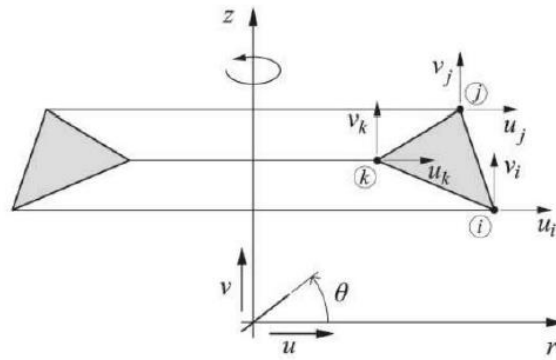
$$\mathbf{K} = \sum_m \mathbf{z}_m \mathbf{S}^T \mathbf{z}_m^T$$

$$\mathbf{f} = \sum_m \mathbf{z}_m f_m$$

## Chapter 8

### FORMULATION OF AXISYMMETRIC PROBLEM

Axisymmetric problems refer to geometrically axially symmetric bodies; these are solids of revolution about a central axis, defined in the cylindrical coordinate plane, with variables  $r$ ,  $\theta$ ,  $z$ . These bodies however are mathematically two-dimensional, in that they are independent of the  $\theta$  variable. We can thus formulate the problem in 2 dimensions, with the displacement field a function of  $r$  and  $z$ . Considering the CST problem, our nodal displacement vector  $s$ , consists of 6 components, 2 displacement components for each node.



#### DISPLACEMENT FIELD

Displacement functions:

$$\mathbf{u}(r, z) = \begin{bmatrix} u(r, z) \\ v(r, z) \end{bmatrix}$$

Nodal displacements:

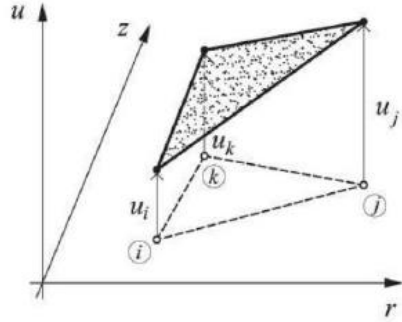
$$\mathbf{s} = [u_i \quad v_i \mid u_j \quad v_j \mid u_k \quad v_k]^T$$

We can also make some simplifying conclusions about our displacements, strains and stresses:

$$u_\theta = w = 0 \quad \gamma_{r\theta} = \gamma_{\theta r} = 0 \quad \tau_{r\theta} = \tau_{\theta r} = 0$$

#### 8.1 Shape Functions

The shape functions remain the same as those for the 2D case, except that we substitute our  $x$  and  $y$  values for our analogous coordinates in our new reference system  $r$  and  $z$ .

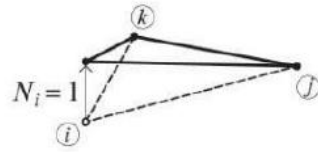


$$u(r, z) = N_i(r, z)u_i + N_j(r, z)u_j + N_k(r, z)u_k$$

$$N_i(r, z) = \frac{1}{2A}(a_i + b_i r + c_i z)$$

Like before we can write the displacements field as a linear system of equations, and thus, can express them in matrix form:

$$\begin{cases} u(r, z) = N_i(r, z)u_i + N_j(r, z)u_j + N_k(r, z)u_k \\ v(r, z) = N_i(r, z)v_i + N_j(r, z)v_j + N_k(r, z)v_k \end{cases}$$



$$\mathbf{u}(r, z) = \begin{bmatrix} u(r, z) \\ v(r, z) \end{bmatrix} = \begin{bmatrix} N_i & 0 & N_j & 0 & N_k & 0 \\ 0 & N_i & 0 & N_j & 0 & N_k \end{bmatrix} \cdot \begin{bmatrix} u_i \\ v_i \\ u_j \\ v_j \\ u_k \\ v_k \end{bmatrix} = \mathbf{N}(r, z) \mathbf{s}$$

$$\mathbf{N}(r, z) = [\mathbf{N}_i(r, z) \mid \mathbf{N}_j(r, z) \mid \mathbf{N}_k(r, z)] \quad \mathbf{N}_i(r, z) = \begin{bmatrix} N_i(r, z) & 0 \\ 0 & N_i(r, z) \end{bmatrix}$$

## 8.2 Strains and Compatibility Matrix

The strains for the axisymmetric problem however differ from the plane stress and strain case in that they include a 4th term, the circumferential normal strain  $\epsilon_\theta$ , which is uniquely independent from the other terms and depends on the radial coordinate.



$$\varepsilon_{\theta}(r, z) = \frac{2\pi(r + \Delta r) - 2\pi r}{2\pi r} = \frac{\Delta r}{r} = \frac{u}{r}$$

We thus write a strains vector  $\varepsilon(x, y)$  with 4 terms. Like the previous 2D formulations it includes partial derivative terms, and thus utilizes the differential operator in cylindrical coordinates  $\mathbf{L}$ .

$$\boldsymbol{\varepsilon}(r, z) = \begin{bmatrix} \varepsilon_r(r, z) \\ \varepsilon_z(r, z) \\ \varepsilon_{\theta}(r, z) \\ \gamma_{rz}(r, z) \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial r} \\ \frac{\partial v}{\partial z} \\ u \\ \frac{\partial u}{\partial z} + \frac{\partial v}{\partial r} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial r} & 0 \\ 0 & \frac{\partial}{\partial z} \\ 1 & 0 \\ \frac{r}{\partial z} & \frac{\partial}{\partial r} \end{bmatrix} \cdot \begin{bmatrix} u(r, z) \\ v(r, z) \end{bmatrix} = \mathbf{L}\mathbf{u}(r, z)$$

We can thus arrive at our compatibility matrix by writing the displacement vector as a product of the shapes function matrix  $\mathbf{N}(r, z)$  and our nodal displacement vector  $\mathbf{s}$ , and grouping terms.

$$\boldsymbol{\varepsilon}(r, z) = \mathbf{L}\mathbf{u}(r, z) = \mathbf{L}\mathbf{N}(r, z)\mathbf{s} = \mathbf{B}(r, z)\mathbf{s}$$

$$\mathbf{B}(r, z) = [\mathbf{B}_i(r, z) \mid \mathbf{B}_j(r, z) \mid \mathbf{B}_k(r, z)]$$

$$\mathbf{L} = \begin{bmatrix} \frac{\partial}{\partial r} & 0 \\ 0 & \frac{\partial}{\partial z} \\ 1 & 0 \\ \frac{r}{\partial z} & \frac{\partial}{\partial r} \end{bmatrix} \quad \mathbf{B}_i(r, z) = \begin{bmatrix} \frac{\partial N_i}{\partial r} & 0 \\ 0 & \frac{\partial N_i}{\partial z} \\ N_i & 0 \\ \frac{r}{\partial z} & \frac{\partial N_i}{\partial r} \end{bmatrix} = \frac{1}{2A} \begin{bmatrix} b_i & 0 \\ 0 & c_i \\ N_i^* & 0 \\ c_i & b_i \end{bmatrix}$$

The compatibility matrix however is not constant since the strain  $\varepsilon_{\theta}$  is not constant. The formulation includes the term  $N_i$ , which is the term  $N_i$  scaled by  $2A$ .

$$N_i^*(r, z) = 2A N_i(r, z) = a_i + b_i r + c_i z \quad \frac{N_i^*(r, z)}{r} = \frac{a_i}{r} + b_i + c_i \frac{z}{r}$$

### 8.3 Stresses

Our stresses utilize the constitutive law, and take into account the effects from imposed displacements such as thermal effects. The material stiffness matrix  $\mathbf{D}$  for an isotropic material is given by the following matrix:

$$\mathbf{D} = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1 & \frac{\nu}{1-\nu} & \frac{\nu}{1-\nu} & 0 \\ \frac{\nu}{1-\nu} & 1 & \frac{\nu}{1-\nu} & 0 \\ \frac{\nu}{1-\nu} & \frac{\nu}{1-\nu} & 1 & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2(1-\nu)} \end{bmatrix}$$

Whilst the thermal imposed strains vector is given by:

$$\boldsymbol{\varepsilon}_0 = \alpha \Delta T \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

We can thus write our stress vector as:

$$\boldsymbol{\sigma}(r, z) = \begin{bmatrix} \sigma_r(r, z) \\ \sigma_z(r, z) \\ \sigma_\theta(r, z) \\ \tau_{rz}(r, z) \end{bmatrix} \quad \boldsymbol{\sigma}(r, z) = \mathbf{D}[\boldsymbol{\varepsilon}(r, z) - \boldsymbol{\varepsilon}_0(r, z)] = \mathbf{D}\mathbf{B}(r, z)\mathbf{s} - \mathbf{D}\boldsymbol{\varepsilon}_0(r, z)$$

Due to the presence of non-constant circumferential normal strain, the stress field is not constant over the element even for homogeneous material and constant imposed strains. This is a difference between the axisymmetric formulation and the plane stress or strain problem.

## 8.4 Stiffness Matrix

Once we have the values for stress and strain, we can calculate the potential energy of the system. The double derivative of this equation concerning the displacements is our global stiffness **K**. This is expressed as an integral over the volume consisting of both our compatibility matrix **B** and our material stiffness matrix **D**.

$$\mathbf{K} = \int_V \mathbf{B}^T(r, z) \mathbf{D} \mathbf{B}(r, z) dV$$

The issue with solving this integral analytically is due to the term involving the radius term on the denominator. We are left with logarithmic solutions involving the radius ratio of the nodal circles such as  $\ln(r_i/r_j)$ . As the ratio approaches unity, the logarithmic expression tends to zero and can become inaccurate. The solution is to compute the integral numerically using Guldino's theorem, which computes the exact volume of a toroidal element with only one integration point.

$$V = \int_V dV = \int_0^{2\pi} \int_A r dA d\theta = 2\pi A r_G \quad r_G = \bar{r} = \frac{r_i + r_j + r_k}{3}$$

Utilizing this technique, the stiffness matrix integral resolves to:

$$\mathbf{K} = 2\pi \int_A \mathbf{B}^T(r, z) \mathbf{D} \mathbf{B}(r, z) r dA \cong 2\pi \underbrace{\bar{\mathbf{B}}^T \mathbf{D} \bar{\mathbf{B}} \bar{r} A}_{\mathbf{K}'}$$

Where we have also defined a new compatibility matrix given below as a function of our new variables  $\bar{r}$ ,  $\bar{z}$  the mean element radius and height.

## 8.5 Equivalent Nodal Forces

We can use this integration technique to solve for the equivalent nodal forces which involve an integral over the volume.

### 8.5.1 Volume Forces (i.e. self-weight)

We integrate the production of the shape functions and the body forces over the volume. The shape functions are changed to be functions of  $\bar{r}$ ,  $\bar{z}$

$$\mathbf{f}_V = \int_V \mathbf{N}^T(r, z) \mathbf{F} dV = 2\pi \int_A \mathbf{N}^T(r, z) \mathbf{F} r dA \cong 2\pi \underbrace{\bar{\mathbf{N}}^T \bar{\mathbf{F}} \bar{r} A}_{\mathbf{f}'_V}$$
$$\bar{\mathbf{N}} = \mathbf{N}(\bar{r}, \bar{z}) \quad \bar{\mathbf{F}} = \mathbf{F}(\bar{r}, \bar{z})$$

### 8.5.2 Imposed Strains (i.e. thermal strains)

The integral for the nodal forces stemming from the imposed strains is as follows:

$$\mathbf{f}_\varepsilon = \int_V \mathbf{B}^T(r, z) \mathbf{D} \boldsymbol{\varepsilon}_0(r, z) dV = 2\pi \int_A \mathbf{B}^T(r, z) \mathbf{D} \boldsymbol{\varepsilon}_0(r, z) r dA \cong 2\pi \underbrace{\bar{\mathbf{B}}^T \mathbf{D} \bar{\boldsymbol{\varepsilon}}_0 \bar{r} A}_{\mathbf{f}'_\varepsilon}$$
$$\bar{\mathbf{B}} = \mathbf{B}(\bar{r}, \bar{z}) \quad \bar{\boldsymbol{\varepsilon}}_0 = \boldsymbol{\varepsilon}_0(\bar{r}, \bar{z})$$

### 8.5.3 Surface Forces

We observe that the equivalent nodal forces  $\mathbf{f}_S$  can be computed more efficiently by integration over the sides of the element.

$$\mathbf{f}_S = \int_{S_\sigma} \mathbf{N}^T(r, z) \mathbf{f} dS = 2\pi \int_{\ell_\sigma} \mathbf{N}^T(r, z) \mathbf{f} r d\ell = 2\pi \mathbf{f}'_S \quad \mathbf{f}'_S = \int_{\ell_\sigma} \mathbf{N}^T(r, z) \mathbf{f} r d\ell$$

### 8.5.4 Nodal Forces

A uniform line load  $q_i$  on a nodal circle of radius  $r_i$  produces the nodal load  $2\pi r_i q_i$ . The net sum of this force in the axial direction is zero however if the force  $q_i$  acts radially.

## 8.6 Equilibrium Equations

The general equilibrium equation for the body is given by the following equation:

$$\mathbf{K}\mathbf{s} = \mathbf{f}_V + \mathbf{f}_S + \mathbf{f}_\varepsilon + \mathbf{f}_N$$

Substituting the formulations derived in the previous section yields a common multiplier term  $2\pi$ , which can be factored out, or the equation simply viewed as an integral with theta limits from zero to one radian.

$$2\pi \mathbf{K}' \mathbf{s} = 2\pi (\mathbf{f}'_V + \mathbf{f}'_S + \mathbf{f}'_\varepsilon + \mathbf{f}'_N)$$

$$\mathbf{K}' \mathbf{s} = \mathbf{f}'_V + \mathbf{f}'_S + \mathbf{f}'_\varepsilon + \mathbf{f}'_N$$

## 8.7 Additional Comments

A few additional comments regarding the numerical analysis of the axisymmetric problem:

- The only rigid body motion is the z-direction translation due to axial symmetry. This can be avoided by placing a z-direction restraint
- A imposed radial displacement of  $u = 0$  should be prescribed for all nodes that lie on the z axis ( $r = 0$ )
- The indeterminate form  $0/0$  may arise when the circumferential strain  $\varepsilon_\theta$  is computed at nodal circles lying on the z-axis ( $r = 0$ ) for core elements. To avoid this the most practical approach is to apply L'Hopitals rule which equates the circumferential strain to the radial strain.

$$\varepsilon_\theta(0, z) = \lim_{r \rightarrow 0} \frac{\partial u / \partial r}{\partial r / \partial r} = \lim_{r \rightarrow 0} \frac{\partial u}{\partial r} = \lim_{r \rightarrow 0} \varepsilon_r = \varepsilon_r(0, z) \quad \Rightarrow \quad \varepsilon_\theta(0, z) \equiv \varepsilon_r(0, z)$$

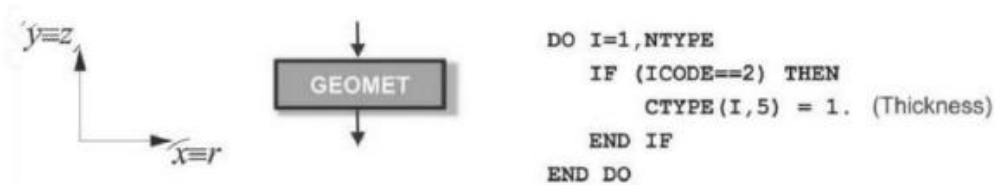
# Chapter 9

## IMPLEMENTATION OF AXISYMMETRIC PROBLEM

We now extend the CST code developed earlier to 3D axisymmetric problems. We do this initially by calling the identifier ICODE=2 which lets the program known to initiate the axisymmetric formulation.

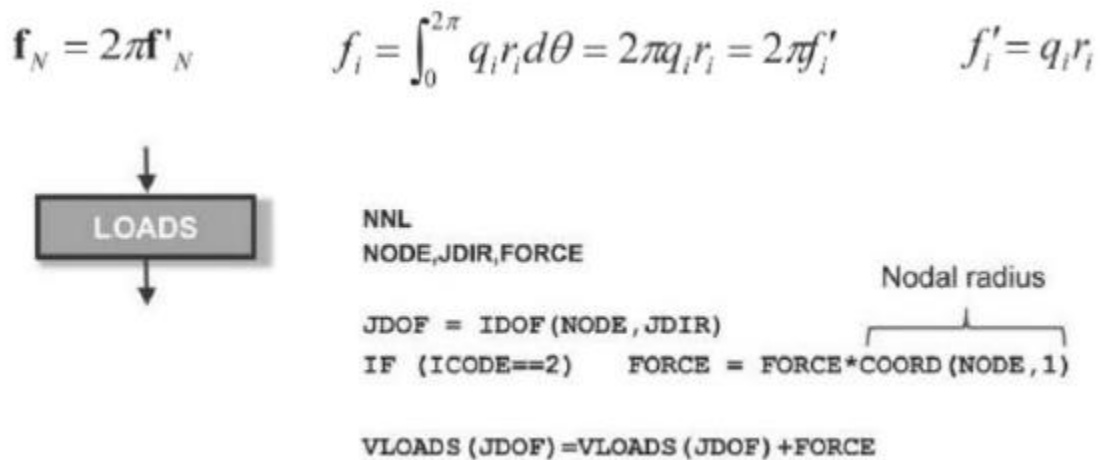
### 9.1 Geometry

In our GEOMET subroutine the thickness of the elements is taken as 1.



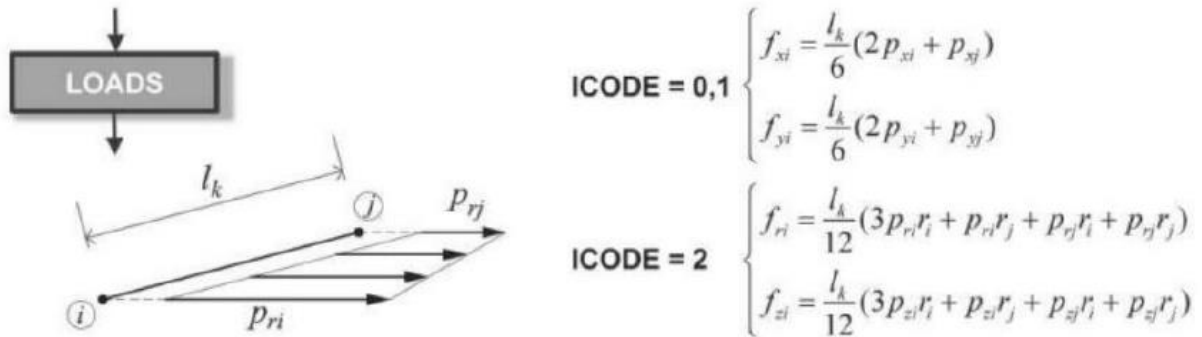
### 9.2 Nodal Loads

For the nodal forces in our LOADS subroutine:



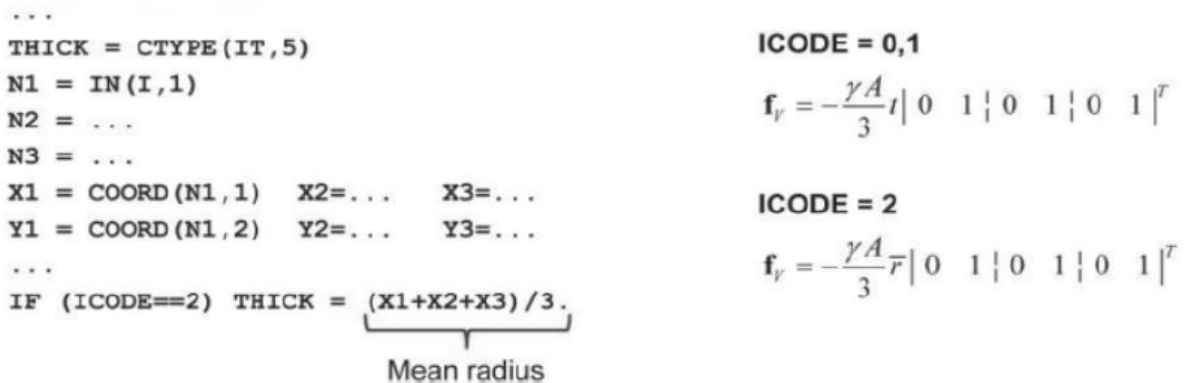
### 9.3 Surface Loads

For our surface loads, in our LOADS subroutine we must implement a different formulation for the equivalent nodal loads as shown below



### 9.4 Volume Loads

The volume loads, in the LOADS subroutine, is modified by utilizing the mean radius value



### 9.5 Thermal Loads

The thermal loads, in the LOADS subroutine, we must modify our variables and make use of the mean radius and height variables.

$$\mathbf{f}_\epsilon = \mathbf{B}^T \mathbf{D} \epsilon_0 \bar{r} A \quad \Rightarrow \quad \mathbf{f}_\epsilon = \frac{E \alpha \Delta T \cdot \bar{r}}{2k(\nu)} \begin{bmatrix} b_i^* & c_i & b_j^* & c_j & b_k^* & c_k \end{bmatrix}^T$$

$$k(\nu) = 1 - 2\nu$$

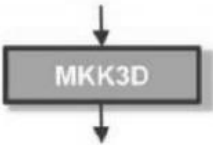
$$b_i^* = b_i + \bar{N}_i^* / \bar{r}$$

$$\bar{N}_i^* / \bar{r} = (a_i / \bar{r} + b_i + c_i \bar{z} / \bar{r})$$

**ICODE = 2**

## 9.6 Stiffness Matrix

To calculate the element stiffness matrix we have the **MKK2D** subroutine for the plane stress and strain case, and the **MKK3D** subroutine for the 3D axisymmetric case. The **MKK3D** now utilizes a compatibility matrix **BB** of size [4,6], and the stiffness matrix **DD** of size [4,4], the same outlined in the previous chapter.



$$\mathbf{K}_{6 \times 6} = \int_A \mathbf{B}^T \mathbf{D} \mathbf{B} r dA = A \cdot \bar{r} \cdot \bar{\mathbf{B}}^T \mathbf{D} \bar{\mathbf{B}}$$

$$\begin{matrix} 6 \times 6 & & 6 \times 4 & 4 \times 4 & 4 \times 6 \end{matrix}$$

**ICODE = 2**

$$\begin{matrix} \mathbf{BB}(4, 6) \\ \mathbf{DD}(4, 4) \\ \mathbf{ST}(6, 6) \end{matrix}$$

$\Rightarrow$

$$\begin{matrix} \text{WRITE}(9, *) \\ \text{WRITE}(9, *) \\ \text{WRITE}(9, *) \end{matrix}$$



## Chapter 10

# SOLUTION OF THE LINEAR SYSTEM OF EQUATION

The algorithm implemented for solving the structural system influences the overall computational efficiency of the system and the accuracy and reliability of the results. Thus, the selection of the solution method becomes critical for our analysis. Direct inversion is a straightforward method for solving a system of equation expressed as  $Ax=B$ . However, the direct inversion of the coefficient matrix  $A$  is computationally inefficient and thus this method is rarely adopted. Thus, two alternative solution methods that can be employed are extensively discussed hereafter.

### 10.1 Solution Methods

#### *10.1.1 Direct Elimination Methods*

Direct elimination methods are based on factoring the coefficient matrix  $A$  into a product of a lower and an upper triangular matrix. Besides the round-off error this method generates the exact solution in a finite number of elementary arithmetic operations. However, it may be inadequate for solving a system comprised of large number of equations because of the high number of arithmetic operations involved and the accumulation of the round-off error. The procedure for the direct elimination method is as shown:

$$Ax = b$$

$$A = LU$$

$$\Rightarrow LUx = b$$

Where:

- $L$  is the lower triangular matrix
- $U$  is the upper triangular matrix

Following this decomposition theorem, the system is rewritten as:

$$L y = b$$

$$U x = y$$

The solution method involves two major steps:

1. Decomposition by forward or backward elimination usually through the Gauss or Cholesky method
2. Solution by forward and/or backward substitution

The fundamental procedure behind the forward and backward substitution is described below.

#### Forward Substitution:

For the decomposed system defined as  $Ly = b$ , the forward substitution is carried out as:

$$y_i = \frac{b_i - \sum_{k=1}^{i-1} l_{ik} y_k}{l_{ii}} \quad i = 1, \dots, n$$

Here:

$$l_{ii} \neq 0$$

$$\det L \neq 0$$

#### Backward Substitution:

For the decomposed system defined as  $Ux = y$ , the backward substitution is carried out as:

$$x_i = \frac{b_i - \sum_{k=i+1}^n u_{ik} x_k}{u_{ii}} \quad i = 1, \dots, n$$

$$u_{ii} \neq 0$$

$$\det U \neq 0$$

The detailed algorithm to be implemented for these steps are laid out in the subsequent discussion.

#### Gauss Elimination:

In the Gauss elimination method, the unknown  $x_j$  can be eliminated from any equation  $i = k$  as:

$$a'_{ir} = a_{ir} - \frac{a_{ij}}{a_{kj}} a_{kr}$$

$$a_{kj} \neq 0$$

$$i \neq k, \text{ and}$$

$$r = 1, \dots, n$$

For the case of  $r = j$ , we get  $a'_{ij} = 0$ . On sequential elimination from  $x_1$  to  $x_{n-1}$  after  $n-1$  elimination steps the original linear equation system is reduced to an upper triangular form:

$$\tilde{A} \equiv U$$

$$\tilde{b} \equiv y = L^{-1}b$$

$$\tilde{A}x = \tilde{b} \Rightarrow Ux = y$$

### Compact Procedure:

The computational efficiency of the elimination procedure described above can be improved on dividing the equation  $i$  by the pivotal term  $a_{ii}$  before eliminating the unknown  $x_i$ . The corresponding algorithm is illustrated below:

$$Ax = b \quad W = [A|b]$$

### Forward Elimination

$$\tilde{A}x = \tilde{b} \quad \tilde{W} = [\tilde{A}|\tilde{b}]$$

$$\begin{cases} \tilde{w}_{ik} = \frac{w_{ik}}{w_{ii}} & i = 1, \dots, n-1 \\ \tilde{w}_{jk} = w_{jk} - \frac{w_{ik}}{w_{ii}} w_{ji} & k = i, \dots, n+1 \end{cases}$$

### Backward Substitution

$$x_i = \frac{\tilde{b}_i - \sum_{k=i+1}^n \tilde{a}_{ik} x_k}{\tilde{u}_{ii}} = \tilde{b}_i - \sum_{k=i+1}^n \tilde{a}_{ik} x_k \quad i = 1, \dots, n$$

### Cholesky Method

If the matrix of coefficients  $A$  is symmetric and positive definite, its factorization can be carried out as:

$$\mathbf{A} = \mathbf{LU} = \mathbf{LL}^T$$

Where the lower triangular matrix  $\mathbf{L}$  is computed as:

$$l_{ii} = \sqrt{a_{ii} - \sum_{k=1}^{i-1} l_{ik}^2}$$

$$l_{ji} = \frac{a_{ji} - \sum_{k=1}^{i-1} l_{jk} l_{ik}}{l_{ii}} \quad j = i + 1, \dots, n$$

The solution by Forward and Backward substitution is now obtained as:

$$\begin{cases} \mathbf{Ly} = \mathbf{b} \\ \mathbf{Ux} = \mathbf{L}^T \mathbf{x} = \mathbf{y} \end{cases}$$

These elimination methods intrinsically differ in the way that the matrix of coefficient  $\mathbf{A}$  is factored into  $\mathbf{L}$  and  $\mathbf{U}$ . The computational efficiencies of these elimination methods could be compared for a rationalized selection. The Cholesky method is numerically stable and accurate and it requires fewer floating-point operations and less workspace relative to the Gauss elimination method. However, the computational efficiencies of these two methods are comparable when the symmetry of the matrix  $\mathbf{A}$  is exploited. The compact Gauss elimination method is adopted for obtaining the solution of our structural system defined with the equation.

$$\mathbf{Ks} = \mathbf{f}$$

### 10.1.2 Iterative Methods

Iterative methods entails the selection of an initial approximation that is progressively refined to a better approximation with the aid of a suitable algorithm, until we reach a solution within a predetermined error tolerance. This method is advantageous with respect to the direct elimination methods owing to the fact that it is relatively insensitive to the accumulation of the round-off errors and moreover, the operations involved are uniform and simple. The iterative process may however converge, or diverge slowly.

## 10.2 Conditioning

Conditioning reflects the sensitivity of any function  $f(x)$  to changes in its argument  $x$  and it only depends on the function to be evaluated. For a system of linear equations

such as  $\mathbf{K}\mathbf{s} = \mathbf{f}$  of our structural system, if small errors in the stiffness matrix  $\mathbf{K}$  or in the load vector  $\mathbf{f}$  have little effects on the displacement vector  $\mathbf{s}$ , the system is well-conditioned. However, if the effects of these small errors are large then the system is ill-conditioned.

The condition number is a criteria that enables this classification of the system. It allows an a priori estimate of the upper bound of the relative error of the solution. The smaller the condition number, the more well-conditioned the system is. The condition number is defined as:

$$\text{Cond}(\mathbf{K}) = \|\mathbf{K}\| \|\mathbf{K}^{-1}\|$$

Where

$\|\cdot\|$  is the matrix norm

However, artificial ill-conditioning or skewness is possible, where high condition numbers could be seen even for well-conditioned systems. This misleading error estimate is resulting from isolated stiff regions raising the condition number of  $\mathbf{K}$ . With the help of appropriate scaling and preconditioning of the stiffness matrix, this issue could be averted.

### Scaling and Preconditioning:

Taking into consideration the system of equation of our structural problem,  $\mathbf{K}\mathbf{s} = \mathbf{f}$ , the following operations are made for scaling:

$$\mathbf{D}_1\mathbf{K}\mathbf{s} = \mathbf{D}_1\mathbf{f}$$

Now,  $\mathbf{s}$  and  $\mathbf{D}_1\mathbf{f}$  are substituted in this equation as:

$$\mathbf{s} = \mathbf{D}_2\bar{\mathbf{s}}$$

$$\mathbf{D}_1\mathbf{f} = \bar{\mathbf{f}}$$

$$\Rightarrow \mathbf{D}_1\mathbf{K}\mathbf{D}_2\bar{\mathbf{s}} = \bar{\mathbf{f}}$$

The scaled system can now be rewritten as:

$$\bar{\mathbf{K}} \bar{\mathbf{s}} = \bar{\mathbf{f}}$$

Here  $\mathbf{D}_1$  and  $\mathbf{D}_2$  are the preconditioning matrices defined as:

$$\mathbf{D}_1 = \mathbf{D}_2 = \mathbf{D}$$

$$d_{ii} = \frac{1}{\sqrt{k_{ii}}}$$

The estimate of the maximum number of significant digits that could be lost in the solution process due to ill-conditioning can be made as:

$$\bar{m} = \log [\text{Cond}(\bar{\mathbf{K}})]$$

This manipulation makes the diagonal coefficients of the scaled stiffness matrix as unity. Moreover, the nature of the problem remains unaltered even after scaling and if a system is truly ill-conditioned, it remains so even after preconditioning. Scaling helps in preventing artificial ill-conditioning and thus the unscaled stiffness matrix can be used to solve the system.

### Diagonal Decay Error Test

On factorizing a matrix into triangular form it is noted that the ratio of a diagonal term in the factorized matrix to the corresponding term in the original matrix decreases in size as one moves down the diagonal. If this ratio goes to zero the matrix is singular and if it is negative, the matrix is not positive definite. The diagonal decay of the stiffness matrix  $\mathbf{K}$  can be utilized as an approximate estimate of its condition number. The procedure is as follows:

$$\delta_i = \frac{k_{ii}}{\bar{k}_{ii}}$$

Where

- $k_{ii}$  is the diagonal  $i$ th term of the original matrix, and
- $\bar{k}_{ii}$  is the diagonal term just before the elimination of the  $i$ th DOF

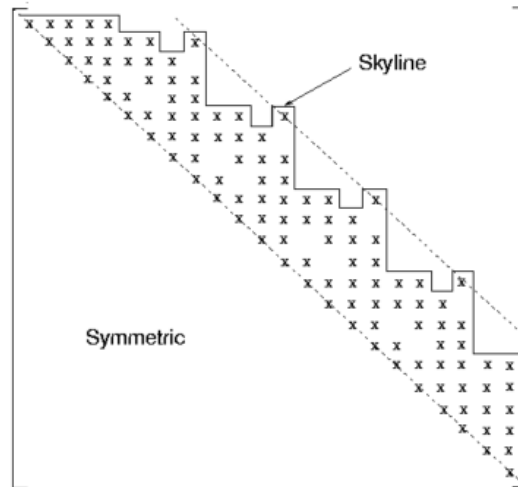
The estimate of the maximum number of significant digits that could be lost in the solution process due to ill-conditioning can be made as:

$$\max \delta_i = 10^m$$

$$\begin{cases} 0 \leq m \leq 2 & \text{well – conditioned system} \\ 3 \leq m \leq 9 & \text{possible ill – conditioning or modelling problem} \\ m \geq 10 & \text{ill – conditioned system} \end{cases}$$

### 10.3 Skyline Method

The system of equations noted in finite element analysis often results in symmetric matrices with several zero-valued elements. The skyline method is advantageous for such sparse matrices that can be made more compact so as to reduce the cost of computation. The stiffness matrix obtained after the analysis can be ordered so that it fits into a banded matrix. An example of such a matrix highlighting the skyline is shown in Figure 4.1.



**Figure 10.1:** Skyline structure

The half-bandwidth, HBW, of a sparse symmetric matrix is expressed as:

$$HBW = \max_{i=1, \dots, n} |j - i|$$

where  $j$  is the column number of the last non-zero element in the  $i^{th}$  row, counting below the diagonal. A vector **ISKY** is formed that stores the half-bandwidth of each column counting from

the diagonal. Then, to account for the non-zero diagonal entries, the **ISKY** vector is updated by adding 1 to each element. Lastly, the cumulative sum of each row of the **ISKY** vector is stored to obtain the total number of non-zero elements in the stiffness matrix, **NSKY**. As a result, the stiffness matrix in vectorial form that has to be defined for the skyline method has a dimension equal to **NSKY**.

The stiffness matrix in vectorial form is now extracted from the stiffness matrix making use of the non-cumulative vector **ISKY**. The non-zero elements of the stiffness matrix for each column are thus stored one after another in this vectorial form. The system of equations has been reduced to a simpler form and it is solved with the Gaussian elimination method described above. This approach significantly reduces the computational cost and time. The skyline method is an extremely powerful tool for carrying out procedures such as the evolutionary structural optimization, as a consequence of the several iterations that have to be performed.



# Chapter 11

## STRESS ANALYSIS

The formulation of the stresses and strains generated in the CST element as a result of the applied loading is discussed in this chapter. Following this, the corresponding stress and strain contribution coming from these elements on the nodes is calculated. Lastly, the strain energy evaluation is also highlighted.

### 11.1 Element Stresses and Strains

#### *Element Strains:*

The compatibility associating the strain to the displacement fields of the CST element is as outlined in the Section 1.2 and 1.3. Following this, the compatibility matrix can be further decomposed into the matrices for each node as:

$$\mathbf{B} = [\mathbf{B}_i \mid \mathbf{B}_j \mid \mathbf{B}_k] \quad \text{where} \quad \mathbf{B}_i = \begin{bmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{bmatrix} = \frac{1}{2A} \begin{bmatrix} b_i & 0 \\ 0 & c_i \\ c_i & b_i \end{bmatrix}$$

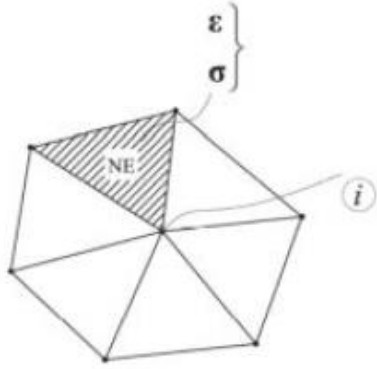
#### *Element Stresses:*

The element stresses are estimated utilizing the constitutive laws and the element strains as noted above. The constitutive law exploited here is as suggested in the Section 2.3.

$$\begin{aligned} \boldsymbol{\sigma}(x, y) &= \begin{bmatrix} \sigma_x(x, y) \\ \sigma_y(x, y) \\ \tau_{xy}(x, y) \end{bmatrix} = \begin{bmatrix} d_{11} & d_{12} & 0 \\ d_{21} & d_{22} & 0 \\ 0 & 0 & d_{33} \end{bmatrix} \cdot \left\{ \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} - \begin{bmatrix} \varepsilon_{0x}(x, y) \\ \varepsilon_{0y}(x, y) \\ \gamma_{0xy}(x, y) \end{bmatrix} \right\} \\ &= \mathbf{D}[\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_0(x, y)] = \mathbf{D}\mathbf{B}\mathbf{s} - \mathbf{D} \boldsymbol{\varepsilon}_0(x, y) \\ \boldsymbol{\sigma}(x, y) &= \frac{1}{2A} \begin{bmatrix} d_{11}b_i & d_{12}c_i & d_{11}b_j & d_{12}c_j & d_{11}b_k & d_{12}c_k \\ d_{21}b_i & d_{22}c_i & d_{21}b_j & d_{22}c_j & d_{21}b_k & d_{22}c_k \\ d_{33}b_i & d_{33}c_i & d_{33}b_j & d_{33}c_j & d_{33}b_k & d_{33}c_k \end{bmatrix} \cdot \mathbf{s} - \begin{bmatrix} d_{11}\varepsilon_{0x}(x, y) + d_{12}\varepsilon_{0y}(x, y) \\ d_{21}\varepsilon_{0x}(x, y) + d_{22}\varepsilon_{0y}(x, y) \\ d_{33}\gamma_{0xy}(x, y) \end{bmatrix} \end{aligned}$$

## 11.2 Nodal Stresses and Strains

The contribution of the stresses and strains obtained from each CST element into the node that it shares with the other elements has to be identified as shown in the following figure. The average nodal values of stress and strain over the set of elements sharing the node  $i$  is calculated as:



$$\varepsilon_i = \frac{\sum_{e \rightarrow i} w_e \varepsilon_{i,e}}{\sum_{e \rightarrow i} w_e}$$

$$\sigma_i = \frac{\sum_{e \rightarrow i} w_e \sigma_{i,e}}{\sum_{e \rightarrow i} w_e}$$

where the weights that determine the method adopted for the average computation are defined as:

$$w_e = \begin{cases} 1 & \text{Airthemetic} \\ V_e & \text{Volumetric} \\ E_e & \text{Energy based} \end{cases}$$

## 11.3 Strain Energy

The strain energy for the CST element is evaluated as:

$$E_e = \frac{1}{2} \int_A t \varepsilon^T \sigma dA = \frac{1}{2} A_e t_e \varepsilon_e^T \sigma_e = \frac{1}{2} \mathbf{s}_e^T \mathbf{K}_e \mathbf{s}_e$$

Lastly, the total strain energy,  $E_2$ , estimated as the summation of the strain energy for all elements should be checked with the work done by external forces,  $E_1$ .

$$E_1 = \frac{1}{2} \mathbf{s}^T \mathbf{f} \quad E_2 = \sum_e E_e$$

## Chapter 12

### SPECIALIZED PROBLEMS

Not all structures can be modeled properly with the base code only. Many specialized problems have to be taken into account when designing, for this reason some changes must be implemented in the code so that the results can better reflect the behavior of the problem at hand.

#### 12.1 Elastic Constraints

In some cases structures might not be restrained to solid, immovable objects but to softer ones that behave like springs; these might be other flexible and extensible structures or soil. To account for their contribution we must introduce their stiffness into the system.

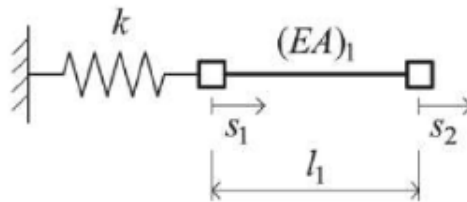
The following is the stiffness matrix for a 1D spring, the 2 degrees of freedom are related to the 2 nodes connected by the spring.

$$\mathbf{K}_e = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

Beams work much the same way:

$$\mathbf{K}_1 = \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \quad \text{where } k_1 = \frac{(EA)_1}{l}$$

Therefore the stiffness coefficients for beams and springs can easily be assembled.

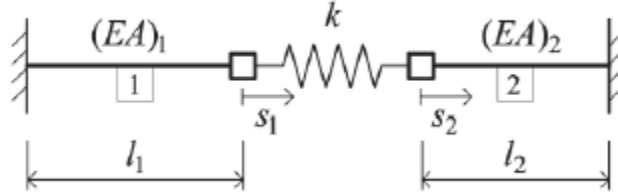


$$\mathbf{K} = \begin{bmatrix} k_1 + k & -k_1 \\ -k_1 & k_1 \end{bmatrix}$$

If the spring is an external constraint then one of its degrees of freedom will be restrained and the stiffness coefficient will only factor in in the global stiffness matrix of the structure in the diagonal.

$$k_{ii}^* = k_{ii} + k$$

If the spring is internal however both of the degrees of freedom take effect and the



Assembly procedure must include both diagonal terms and coupling.

$$\mathbf{K}_1 = \begin{bmatrix} k_1 & -k_1 \\ -k_1 & k_1 \end{bmatrix} \quad \mathbf{K}_2 = \begin{bmatrix} k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$$

Where

$$k_1 = \frac{(EA)_1}{l_1} \quad \text{and} \quad k_2 = \frac{(EA)_2}{l_2}$$

- Case (I):  $k = 0$

$$\mathbf{K} = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$$

- Case (II):  $k \neq 0$

$$\mathbf{K} = \begin{bmatrix} k_1 + k & -k \\ -k & k_2 + k \end{bmatrix}$$

The degrees of freedom of the springs are always assigned at the nodes therefore in the assembly procedure it's possible to account for all the elastic constraints and restraints with the following equations to be applied in the global reference frame.

$$k_{ii}^* = k_{ii} + k$$

$$k_{jj}^* = k_{jj} + k$$

$$k_{ij}^* = k_{ij} - k$$

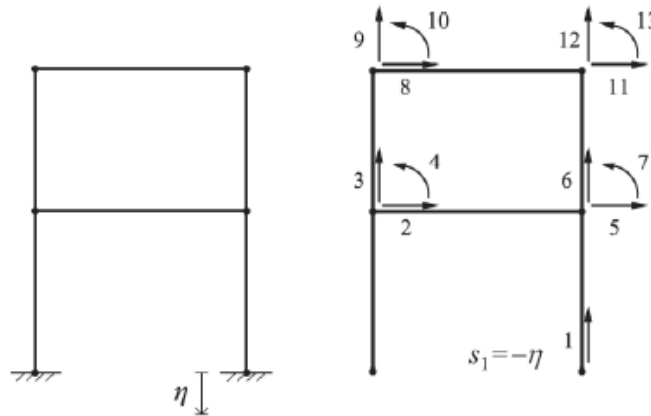
$$k_{ji}^* = k_{ji} - k$$

## 12.2 Imposed Displacements

The structural system under analysis might be subjected to imposed displacements that can be either absolute or relative. The imposed displacements have to be accounted for in the compatibility equations rather than the linear equilibrium equations. The imposed displacements generate reaction forces in the system and are to be accounted for in the analysis. The formulations pertaining to the two classes of imposed displacements as stated earlier are noted in the discussion that follows.

### 12.2.1 Imposed Absolute Displacements

The structural model, as shown below, could be subjected to an imposed absolute displacement in a particular DOF of the system. A typical example of this case would be foundation settlement. The degree of freedom on which the displacement is imposed is freed in the analysis. The structural system is characterized with  $n$  equilibrium equations corresponding to the  $n$  degrees of freedom, and  $m$  compatibility equations relating to the  $m$  number of imposed absolute displacements. Thus, effectively, the system has  $(n+m)$  linear equations, equilibrium and compatibility, with  $(n+m)$  unknowns, displacements and reactions. For the case illustrated below,  $n = 13$  and  $m = 1$ .



The corresponding displacement vector, reaction vector, force vector, and the stiffness matrix are partitioned as:

$$\mathbf{s} = \begin{bmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \end{bmatrix} \quad \mathbf{r} = \begin{bmatrix} \mathbf{0} \\ \mathbf{r}_2 \end{bmatrix} \quad \mathbf{f} = \begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} \\ \mathbf{K}_{21} & \mathbf{K}_{22} \end{bmatrix}$$

The compatibility and equilibrium equations, respectively, are written as:

$$\mathbf{s}_2 = \bar{\mathbf{s}}$$

$$\mathbf{f} + \mathbf{r} = \mathbf{K} \mathbf{s}$$

Now, substituting the compatibility condition in the equilibrium equations we obtain the following set of equations:

$$\mathbf{f}_1 = \mathbf{K}_{11}\mathbf{s}_1 + \mathbf{K}_{12}\bar{\mathbf{s}}$$

$$\mathbf{f}_2 + \mathbf{r}_2 = \mathbf{K}_{12}^T\mathbf{s}_1 + \mathbf{K}_{22}\bar{\mathbf{s}}$$

The unknowns of the system  $\mathbf{s}$  and  $\mathbf{r}_2$  are evaluated as:

$$\mathbf{s}_1 = \mathbf{K}_{11}^{-1}(\mathbf{f}_1 + \hat{\mathbf{f}}_1)$$

$$\mathbf{r}_1 = \mathbf{K}_{12}^T\mathbf{s}_1 + \mathbf{K}_{22}\bar{\mathbf{s}} - \mathbf{f}_2$$

Where  $\hat{\mathbf{f}}_1$  represents the nodal forces that would equivalently result in similar response of the structure as the imposed displacements, given by:

$$\hat{\mathbf{f}}_1 = -\mathbf{K}_{12}\bar{\mathbf{s}}$$

The algorithm implemented is noted below, highlighting the modifications carried out in the vector of nodal loads and on the stiffness matrix.

$$f_j^* = f_j + k_{ji}\eta_i \quad j \neq i$$

$$f_i^* = \bar{k} \eta_i$$

$$k_{ij}^* = k_{ji}^* = 0 \quad j \neq i$$

$$k_{ii}^* = \bar{k}$$

Where:

$i$  is associated to the DOF on which the absolute displacement is imposed

$\eta$  is the imposed absolute displacement

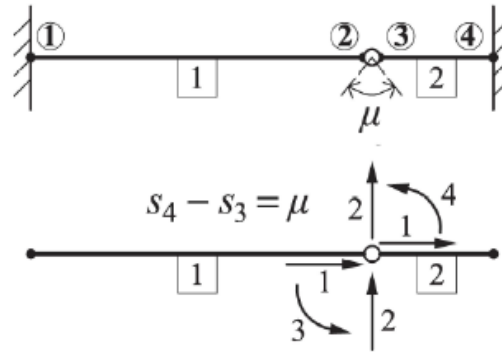
### 10.2.2 Imposed Relative Displacements

The structural model, as shown below, could be subjected to an imposed relative displacement between two degrees of freedom of the system. An example of this case could be flexural distortion as in this case. The degrees of freedom on which the relative displacement is imposed is freed in the analysis. The structural system is characterized with  $n$  equilibrium equations corresponding to

the  $n$  degrees of freedom, and  $m$  compatibility equations relating to the  $m$  number of imposed relative displacements. Thus, effectively the system has  $(n+m)$  linear equations with  $(n+m)$

$$\mathbf{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \quad \mathbf{r} = \begin{bmatrix} 0 \\ r_2 \\ r_3 \end{bmatrix} \quad \mathbf{f} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} & \mathbf{K}_{13} \\ \mathbf{K}_{21} & \mathbf{K}_{22} & \mathbf{K}_{23} \\ \mathbf{K}_{31} & \mathbf{K}_{32} & \mathbf{K}_{33} \end{bmatrix}$$

unknowns. For the case illustrated below,  $n = 4$  and  $m = 1$ .



The corresponding displacement vector, reaction vector, force vector, and the stiffness matrix are partitioned as:

The compatibility and equilibrium equations, respectively, are written as:

$$\mathbf{s}_3 - \mathbf{s}_2 = \Delta \bar{\mathbf{s}}$$

$$\mathbf{f} + \mathbf{r} = \mathbf{K} \mathbf{s}$$

Now, replacing the compatibility equations in the equilibrium ones we identify the relations as:

$$\mathbf{f}_1 = \mathbf{K}_{11}\mathbf{s}_1 + \mathbf{K}_{12}\mathbf{s}_2 + \mathbf{K}_{13}(\mathbf{s}_1 + \Delta \bar{\mathbf{s}})$$

$$\mathbf{f}_2 + \mathbf{r}_2 = \mathbf{K}_{12}^T \mathbf{s}_1 + \mathbf{K}_{22}\mathbf{s}_2 + \mathbf{K}_{23}(\mathbf{s}_2 + \Delta \bar{\mathbf{s}})$$

$$\mathbf{f}_3 + \mathbf{r}_3 = \mathbf{K}_{13}^T \mathbf{s}_1 + \mathbf{K}_{23}^T \mathbf{s}_2 + \mathbf{K}_{33}(\mathbf{s}_2 + \Delta \bar{\mathbf{s}})$$

The system is solved to determine the unknowns of the problem,  $\mathbf{s}$  and  $\mathbf{r}_3$  as:

$$\begin{bmatrix} \mathbf{s}_1 \\ \mathbf{s}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{11} & \mathbf{K}_{12} + \mathbf{K}_{13} \\ \mathbf{K}_{11}^T + \mathbf{K}_{13}^T & \mathbf{K}_{22} + \mathbf{K}_{23}^T + \mathbf{K}_{23} + \mathbf{K}_{33} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{f}_1 - \mathbf{K}_{13} \Delta \bar{\mathbf{s}} \\ \mathbf{f}_2 + \mathbf{f}_3 - \mathbf{K}_{23} \Delta \bar{\mathbf{s}} \end{bmatrix}$$

$$\mathbf{r}_3 = \mathbf{K}_{13}^T \mathbf{s}_1 + (\mathbf{K}_{23}^T + \mathbf{K}_{33})\mathbf{s}_2 + \mathbf{K}_{33}\Delta \bar{\mathbf{s}} - \mathbf{f}_3$$

The algorithm implemented is noted below, highlighting the modifications carried out in the vector of nodal loads and on the stiffness matrix.

$$f_h^* = f_h + k_{hj}\mu_{ij} \quad h \neq i, \quad h \neq j$$

$$f_i^* = f_i + f_j - (k_{ij} + k_{jj} + \bar{k})\mu_{ij}$$

$$f_j^* = \bar{k} \mu_{ij}$$

$$k_{ih}^* = k_{ih} + k_{jh} = k_{hi}^* \quad h \neq i, \quad h \neq j$$

$$k_{hj}^* = k_{jh}^* = 0 \quad h \neq i, \quad h \neq j$$

$$k_{ii}^* = k_{ii} + 2k_{ij} + k_{jj} + \bar{k}$$

$$k_{jj}^* = \bar{k}$$

$$k_{ij}^* = k_{ji}^* = -\bar{k}$$

Here  $i$  and  $j$  are associated to the DOFs on which the relative displacement is imposed.

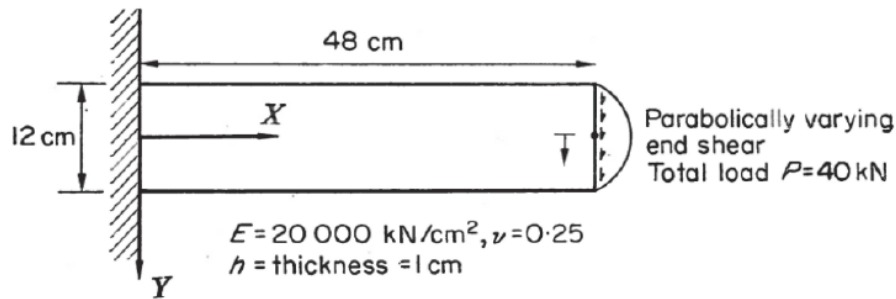


## Chapter 13

### BENCHMARKS

#### 13.1 Cantilever beam

In the following benchmark, the effect of refining a mesh is studied against given reference. The beam is shown below:

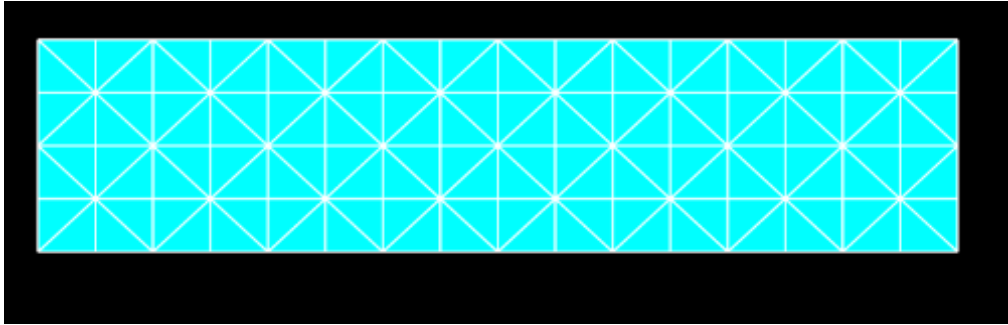


**Figure 13.1:** Fixed cantilever loaded at the end

The beam was modelled using 2 mesh sizes. The first with 128 elements (1) and the second with 512 elements (2). Considering the theory, the finer mesh should result into better approximation of the reference. This benchmark has a larger number of elements and therefore, to solve it, the automatic meshing functionality is used. The plane stress case is employed. The cantilever is loaded with a parabolic shear at the end that is approximated with linearly varying surface loads depending on the size of the mesh. This surface load will vary linearly from the value of the parabolic shear in one node to the next one, this means that a lower number of elements in the mesh leads to a greater underestimation of the load. The objective is to compare the results of the code with the ones from the theory of finite elements and as an upper bound with the beam theory.

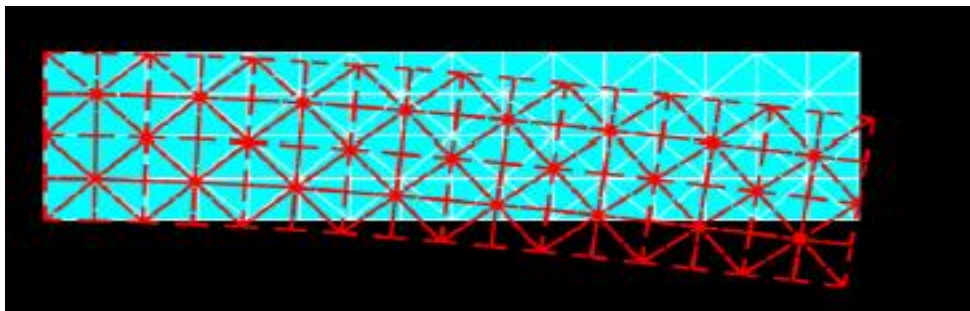
##### 13.1.1 Coarse Mesh

In this case the element count is lower, with just 128 CST elements and the mesh is in a periodic rhomboidal shape. The linear surface loads applied to the end go from 0 kN, to 375 kN, to 500 kN in the mid-span and back.

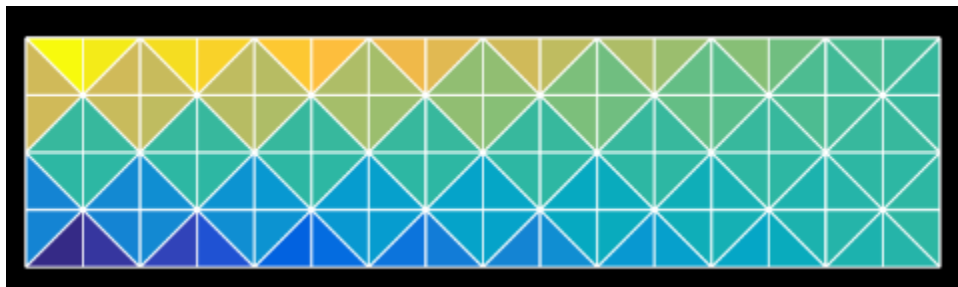


**Figure 13.2** Fixed cantilever coarser mesh

The deformed configuration and stress distribution is:



(a) Deformed Shape



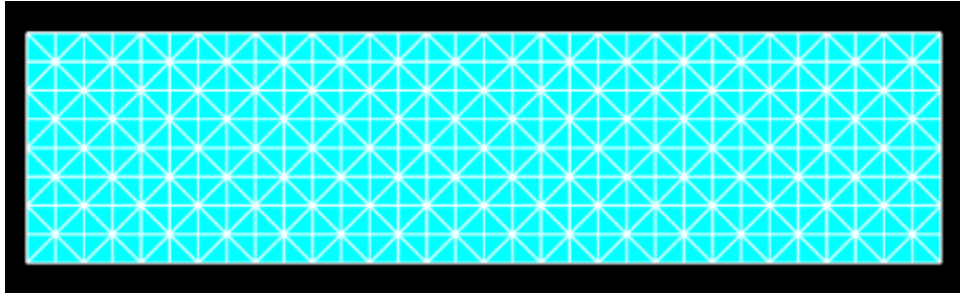
(b) Stress distribution  $\sigma_x$

**Figure 13.3:** Fixed cantilever coarse mesh deformed configuration and stress distribution

The results tabulated at the end of this section.

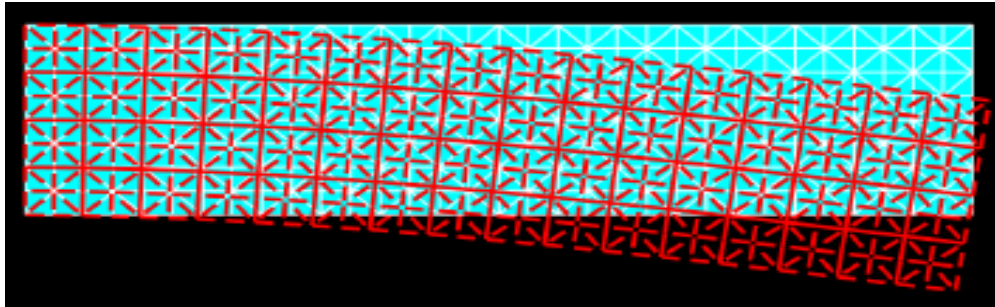
### 13.1.2 Finer Mesh

In this case the element count is higher, with 512 CST and the mesh is in a periodic X shape.

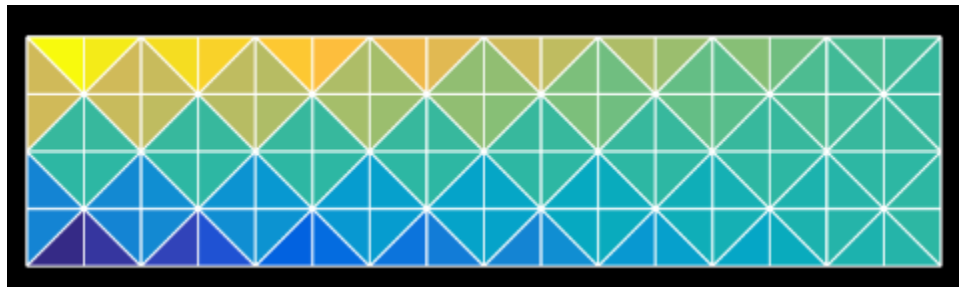


**Figure 13.4:** Fixed cantilever finer mesh

The deformed configuration and stress distribution is:



(a) Deformed Shape



(b) Stress distribution  $\sigma_x$

**Figure 13.5:** Fixed cantilever fine mesh deformed configuration and stress distribution

The results tabulated at the end of this section.

In this case the linear surface loads applied to the end go from 0 kN, to 218.75 kN, to 375 kN, to 468.75 kN, to 500 kN in the mid-span and back. The deformed configuration is given in the following figure.

The results are gathered for both the cases from the outputs of the various tests and are gathered in the following table. The deflection refers to the average of the values on the top and the bottom of the beam. The stress refers to the value at the coordinates  $X = 12$  cm and  $Y = 6$  cm in the reference system.

| Case             | Tip Deflection [mm] | $\sigma_x [MN/m^2]$ |
|------------------|---------------------|---------------------|
| Coarser mesh     | 4.2964              | 565.05              |
| Coarse benchmark | 4.5834              | 512.25              |

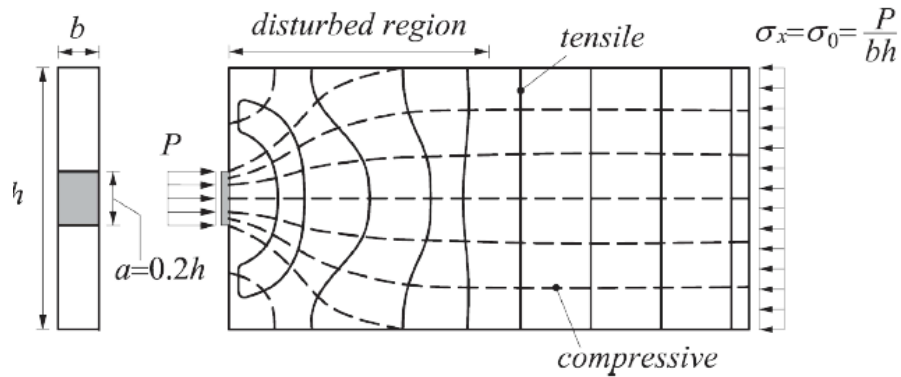
**Table 13.1:** Results- Cantilever Benchmark for coarse mesh

| Case           | Tip Deflection [mm] | $\sigma_x [MN/m^2]$ |
|----------------|---------------------|---------------------|
| Fine mesh      | 5.0561              | 578.8               |
| Fine benchmark | 5.1282              | 573.42              |
| Beam theory    |                     |                     |

**Table 13.2:** Results - Cantilever Benchmark for finer mesh

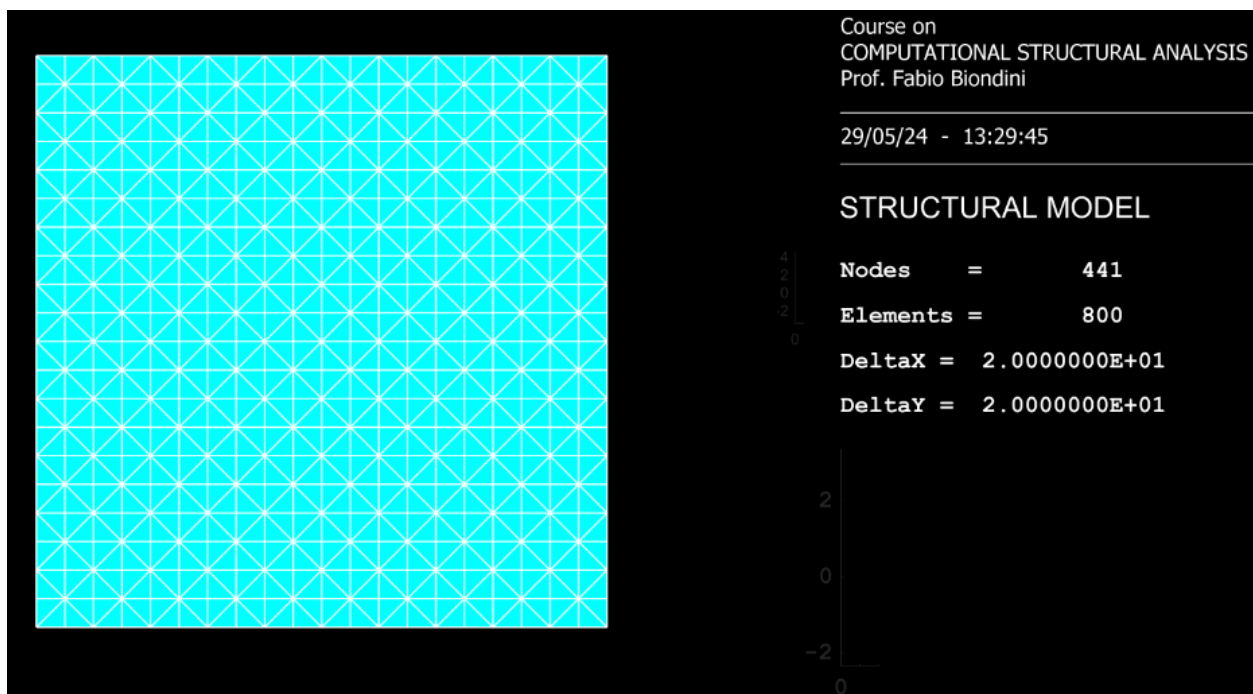
### 13.2 Load effect on anchorage zone

Saint Venant's principal grants clarity on the values of internal stresses in beams far from the application of loads. Localized effects of concentrated loads have an area of effect around the point of application. In this case study the dissipation of these effects will be analyzed inside a beam with the same dimensions. In the following figure, the current problem is illustrated as a example of the way the region close to the loaded end is disturbed. For this application the left end of the beam is loaded with a distributed horizontal surface force and on the right end is entirely supported to resist the force uniformly.

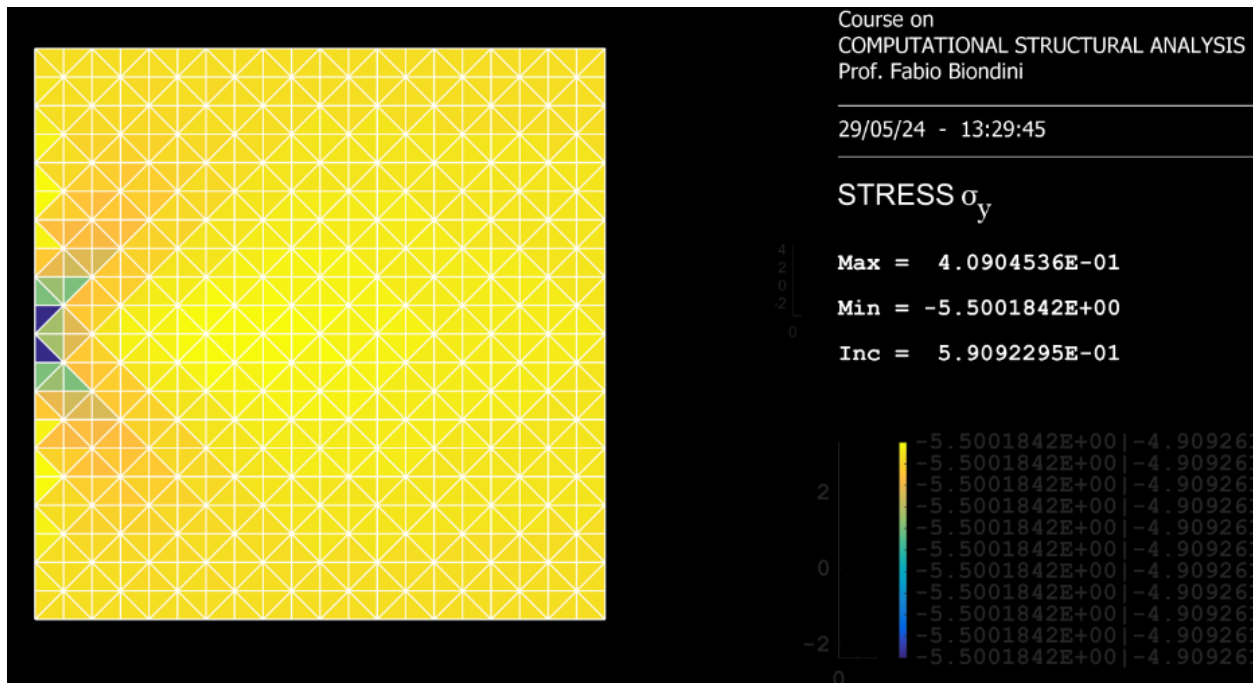


**Figure 13.6:** Disturbed region

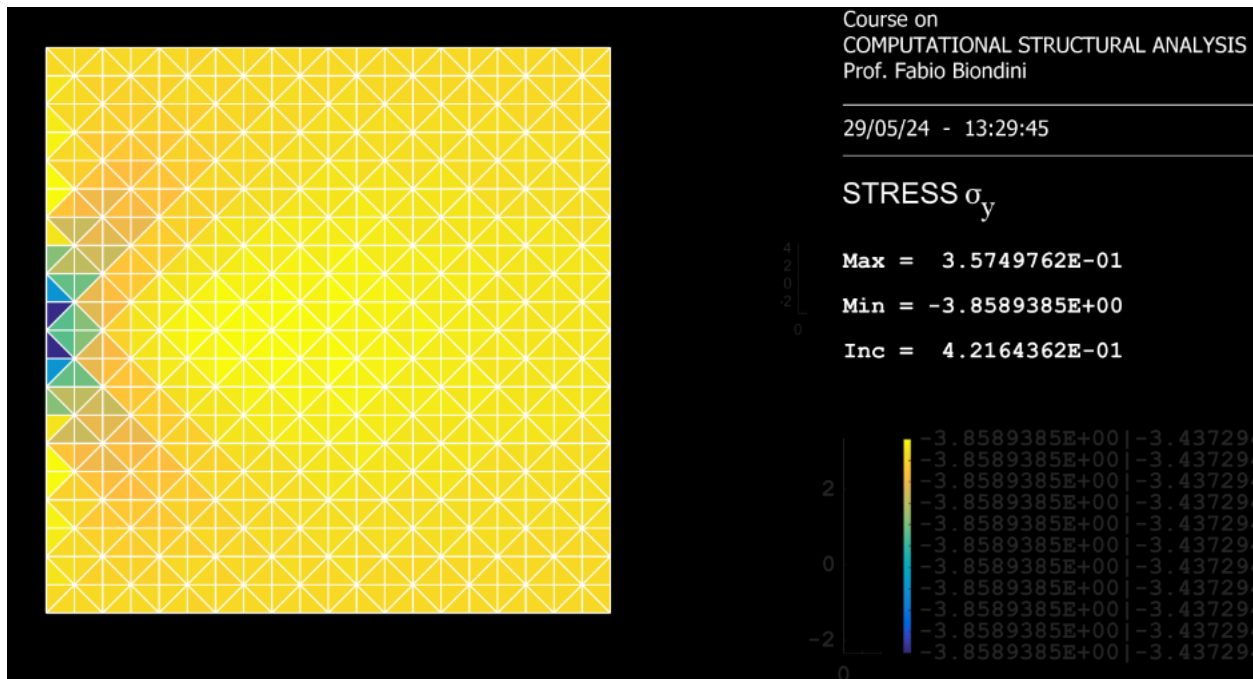
In the following figure, the disturbed region is visualized in the problem at hand ( $a/h = 0.1$ ). The color indicates the vertical stress  $\sigma_y$  and the color grading goes from lighter to darker to indicate lower (white is negative) to higher values. In the subsequent figure, the deformed configuration of the same problem is displayed.



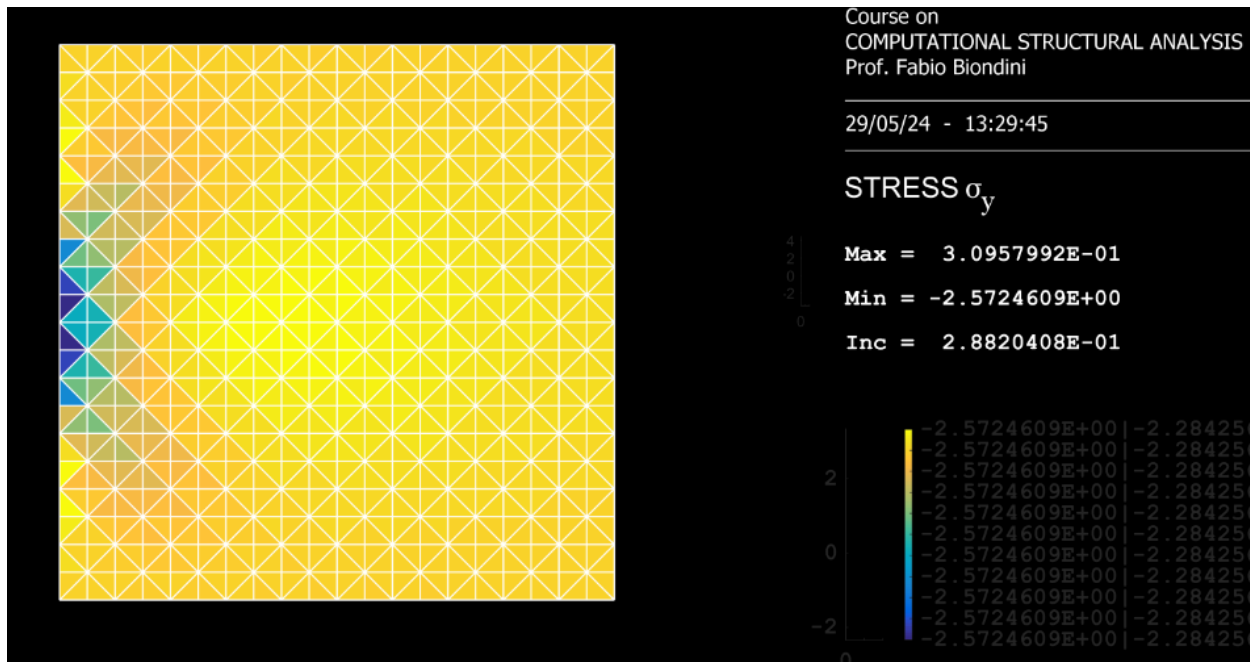
**Figure 13.7:** Model of compressed beam



(a)  $a/h = 0.1$



(b)  $a/h = 0.2$

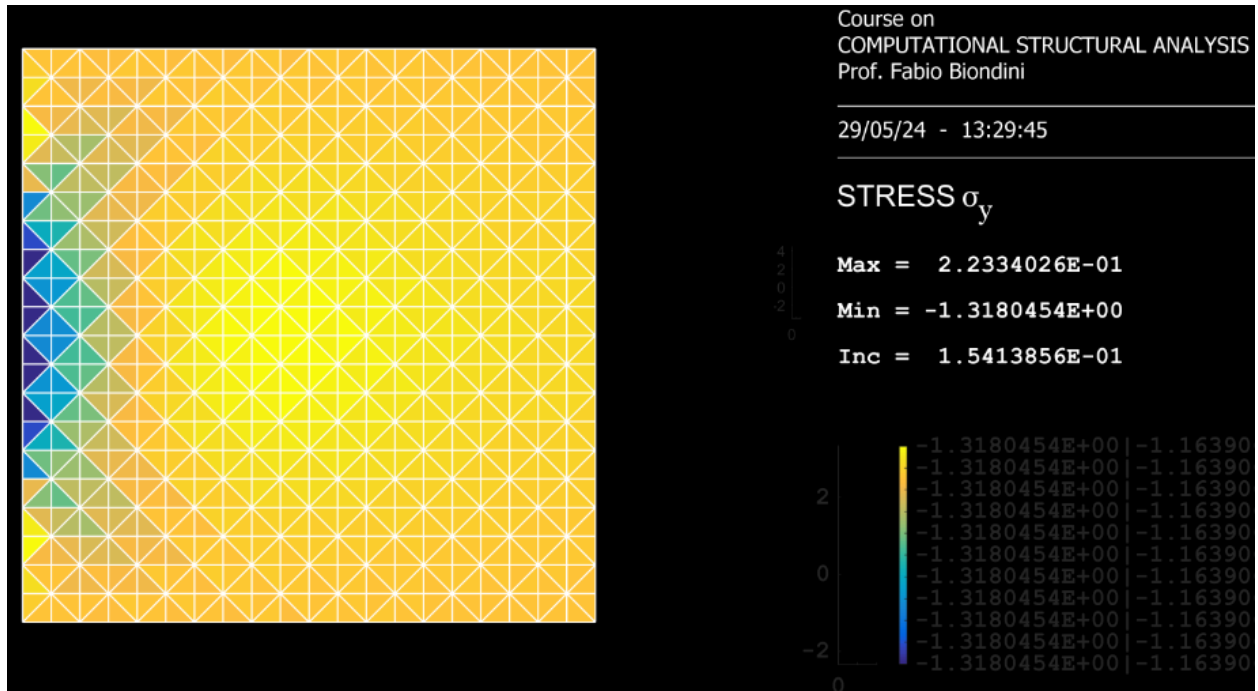


(c)  $a/h = 0.3$

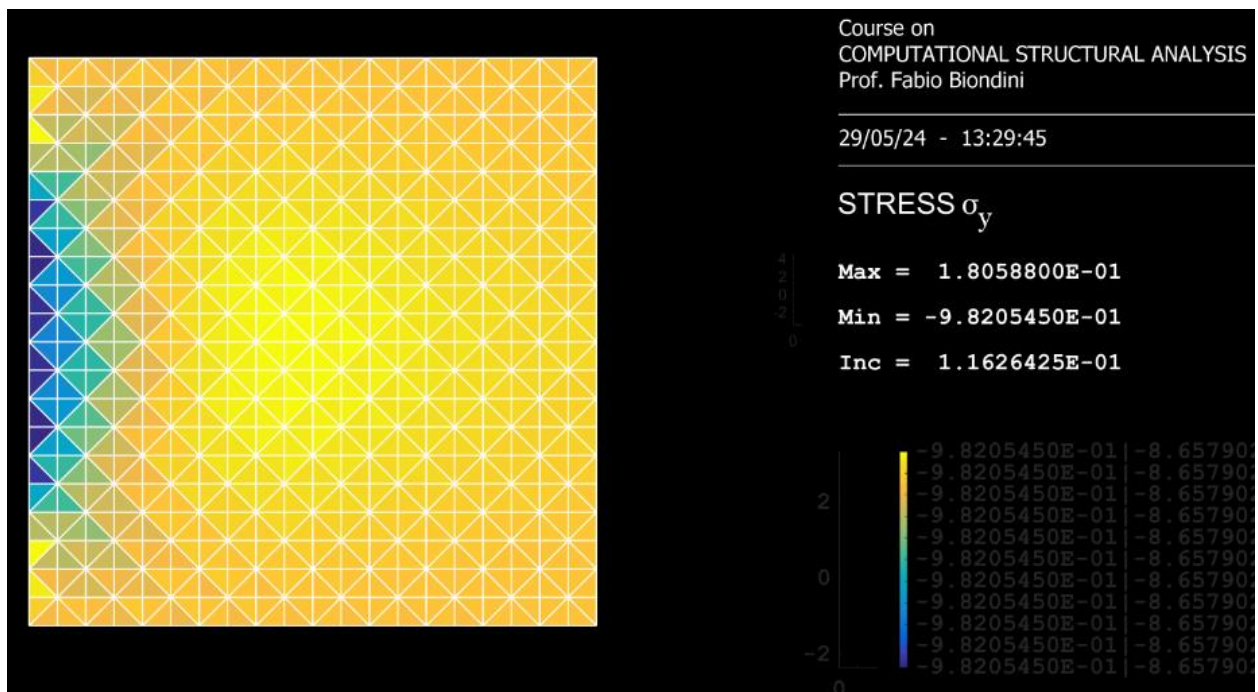


(d)  $a/h = 0.4$



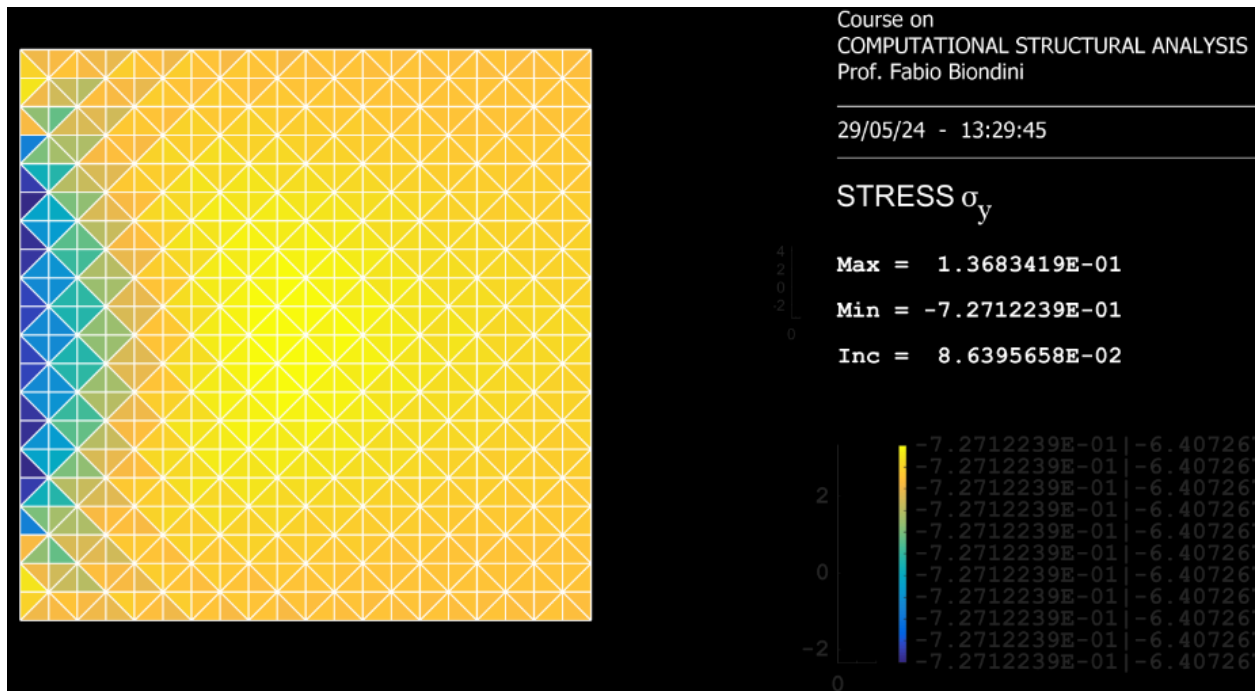


(e)  $a/h = 0.5$

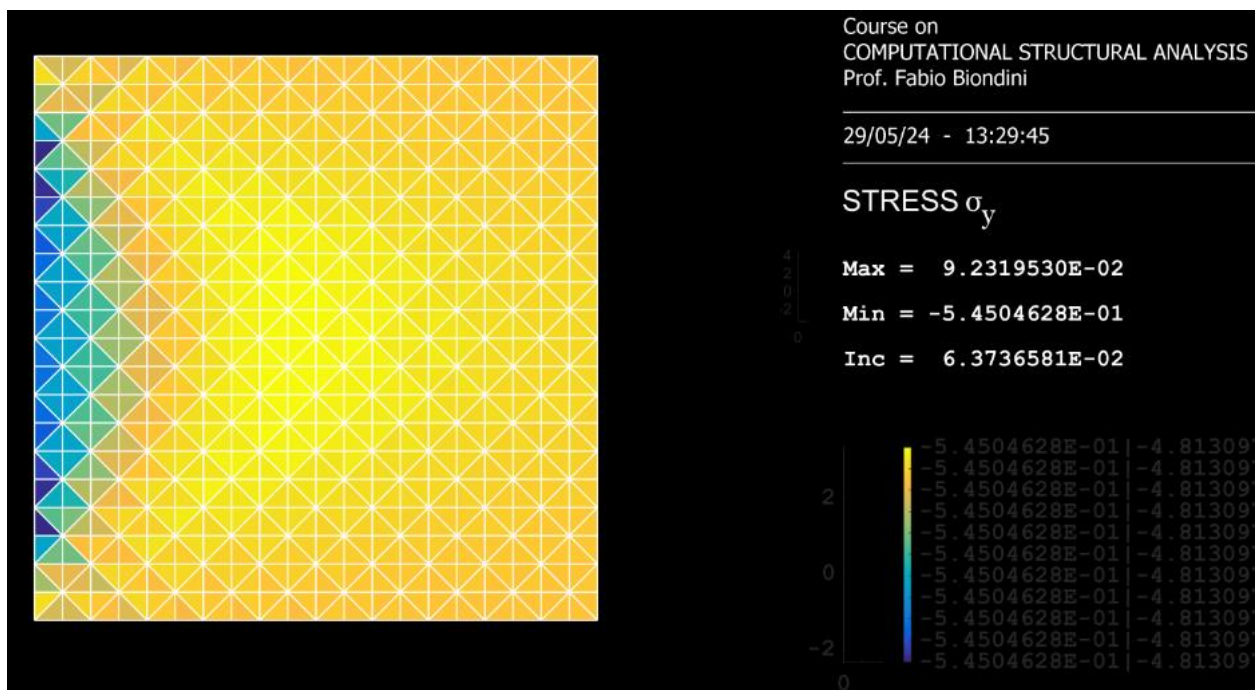


(f)  $a/h = 0.6$

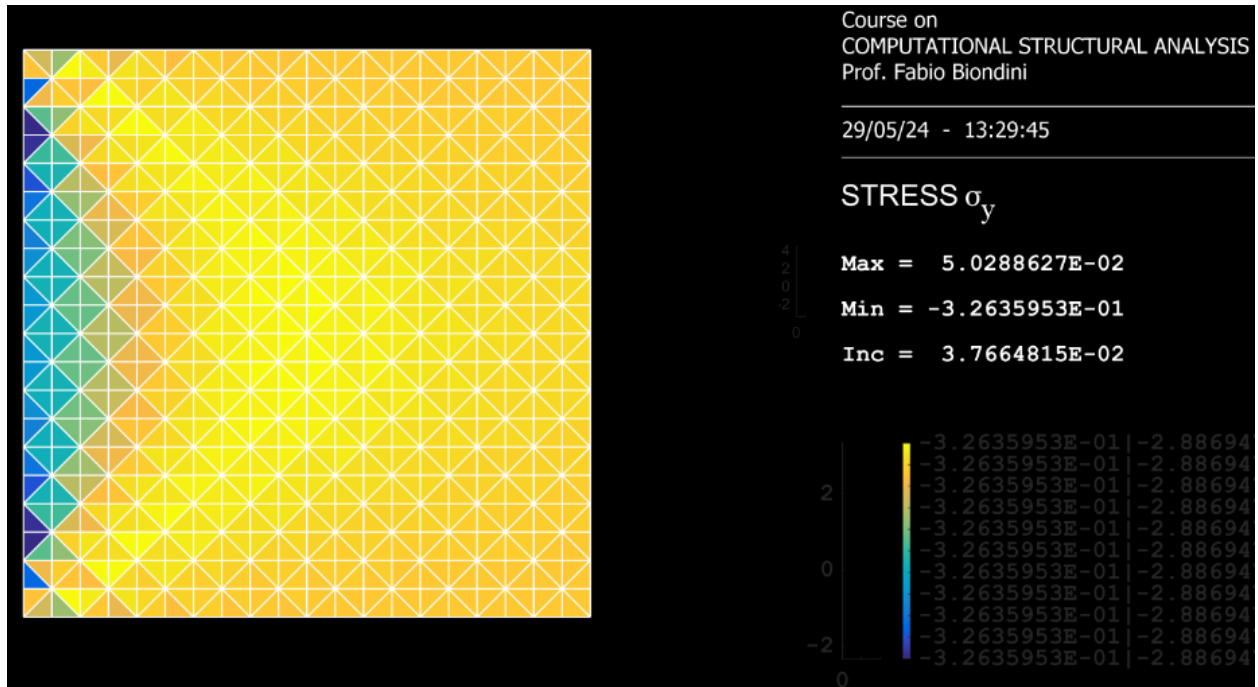




(g)  $a/h = 0.7$



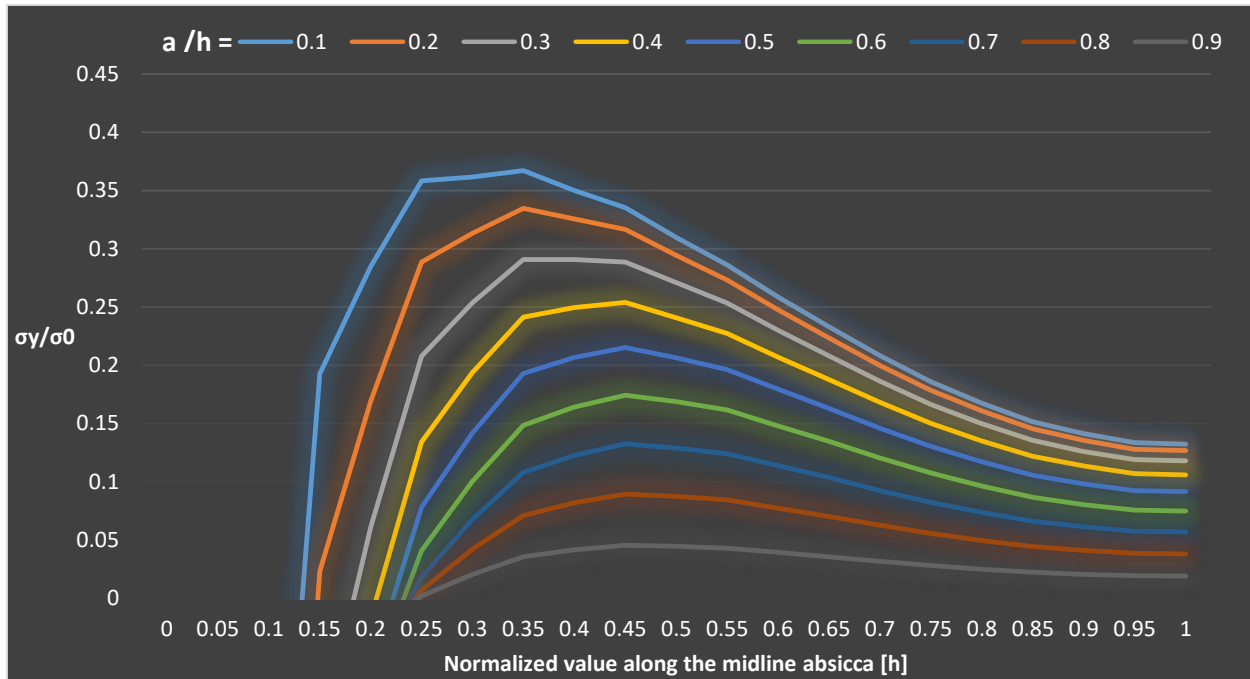
(h)  $a/h = 0.8$



(i)  $a/h = 0.9$

**Figure 13.8:**  $\sigma_y$  in compressed beam

The last evaluation is done to compare different loading conditions. As the area of application of the loads becomes wider the disturbance becomes less intense and widespread and the vertical stresses dissipate. In the following figure the values of the vertical stresses in the mean line of the beam are compared for different loading conditions. The values of the internal stresses are normalized and studied only in the area of most disturbance, within a length equal to the thickness of the beam.



**Figure 13.9:** Disturbance dissipation for different load combinations

### 13.3 Deep Beam

In this second application, the study is on deep beams and the internal distribution of stresses depending on the height of the element. Four different cases are studied for this problem and all focus on horizontal stresses  $\sigma_x$  on the mean line. The study is on deep beams that all share the properties in the following table. The element is always simply supported by 2 small blocks of size  $c = 0.11 l$  and subjected to a distributed load  $w$  that acts as a vertical surface force on the top of the beam for a length equal to  $l$ .

|                                  |       |
|----------------------------------|-------|
| <b>L[m]</b>                      | 1     |
| <b>w [KN/m<sup>2</sup>]</b>      | 10    |
| <b>E [MPa]</b>                   | 10000 |
| <b><math>\nu</math> [-]</b>      | 0.3   |
| <b><math>\alpha</math> [-]</b>   | 0     |
| <b>Weight [KN/m<sup>3</sup>]</b> | 0     |
| <b>b [m]</b>                     | 1     |

**Table 13.2:** Element properties shared in all 4 cases

The following sections contain the information relative to the theoretical and numerical solution to the problem as well as additional information on each case.

13.3.1 Case A with  $h = 0.25l$

In this case to reduce the disturbance on the edges and obtain the desired values in the mid-span the boundary conditions must be defined differently than other cases. While the beam is still simply supported in this case only 1 node for each support acts as a restraint, in this application it is the one in the center of the supports.

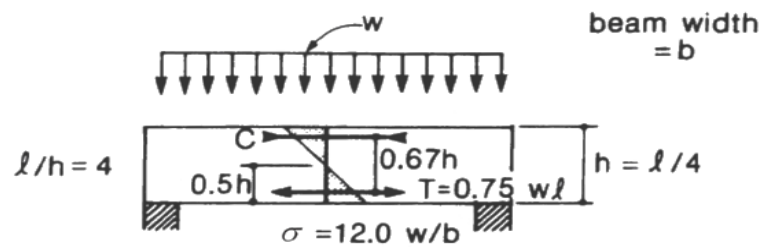
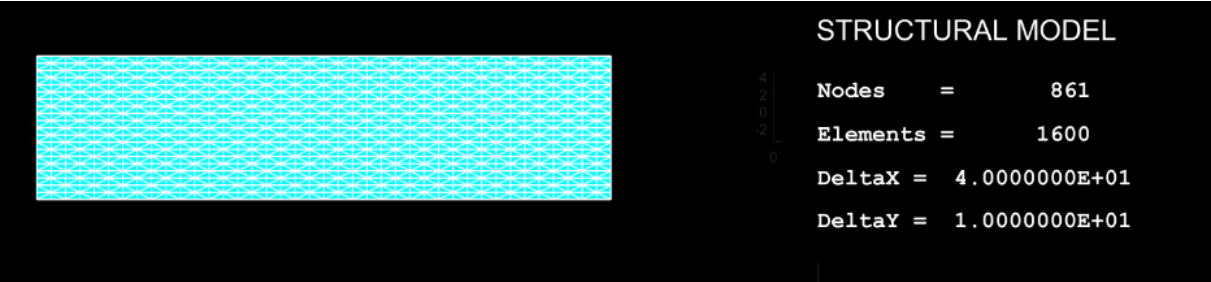
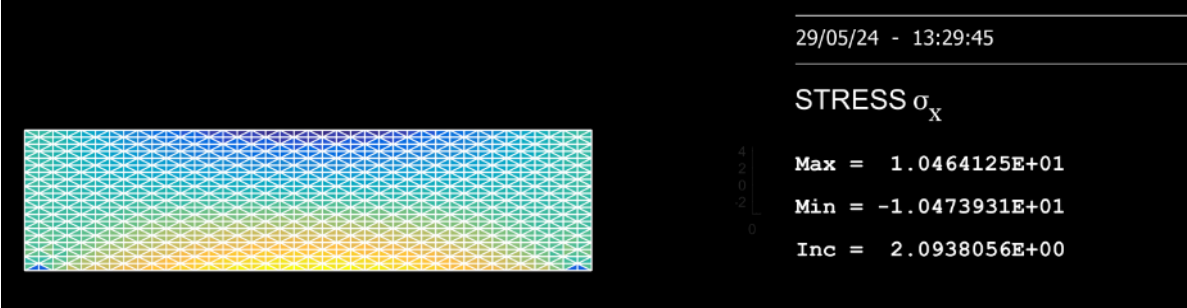


Figure 13.10: Problem definition Case A

The results of the test are shown in the following figure and the numerical values are contained in the following table.

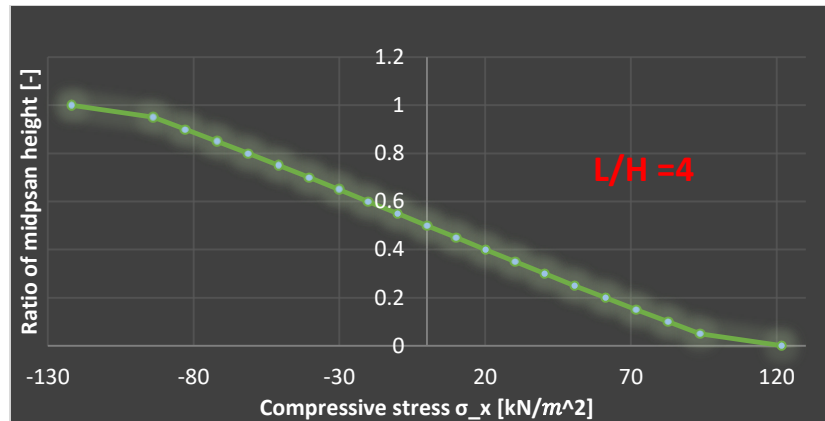


(a) Mesh model for  $l/h = 0.4$



(b)  $\sigma_x$  for  $l/h = 4$

Figure 13.11: Deformed configuration and stress distribution scaled Case A



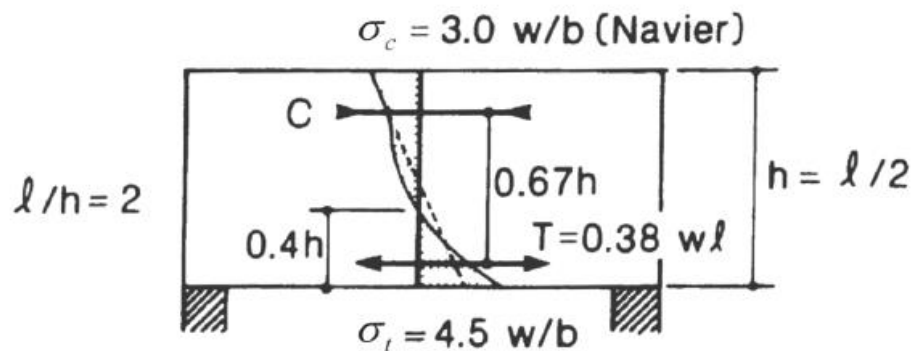
**Figure 13.12:** Stress distribution in beam with  $h = 0.25 l$

| $\sigma_x [MN/m^2]$ | Top     | Bottom   | Neutral axis depth [m] |
|---------------------|---------|----------|------------------------|
| Theoretical         | 120     | -120     | 0.5                    |
| Numerical           | 121.749 | -121.786 | 0.5                    |

**Table 13.3:** Numerical results for Case A

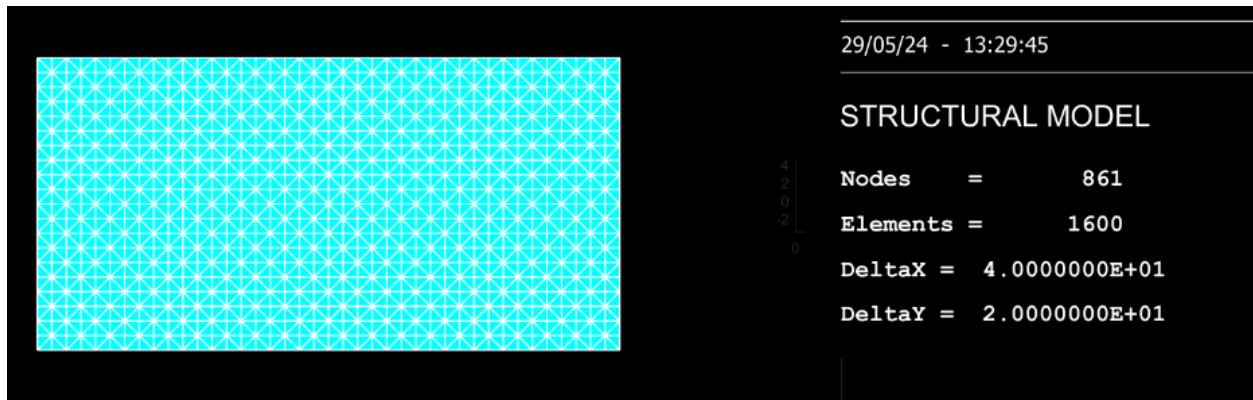
### 13.3.2 Case B with $h = 0.5l$

In this second case the beam is simply supported by the whole size of the supports. The results for this case don't match the theoretical ones exactly, probably because of the boundary conditions introducing disturbances in the deep beam.<sup>m2</sup>

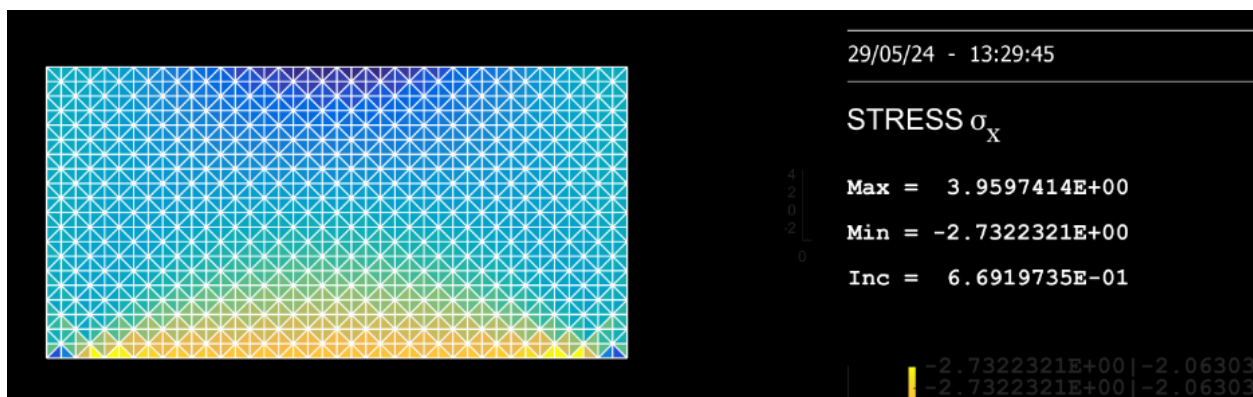


**Figure 13.13:** Problem definition Case B

The results of the test are shown in the following figure and the numerical values are contained in the following table. It is noted that the values obtained here are in good agreement with the linear stress distribution proposed by Navier.

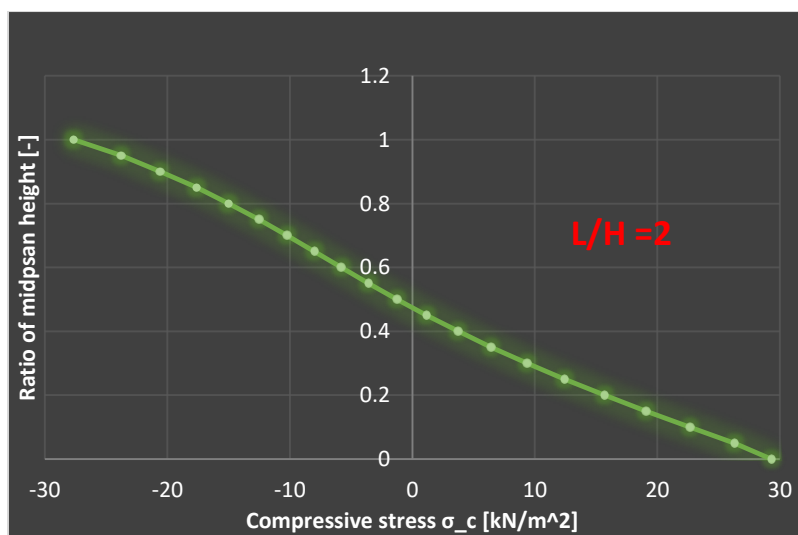


(a) Mesh model for  $l/h = 2$



(b)  $\sigma_x$  for  $l/h = 2$

**Figure 13.14:** Deformed configuration and stress distribution scaled Case B



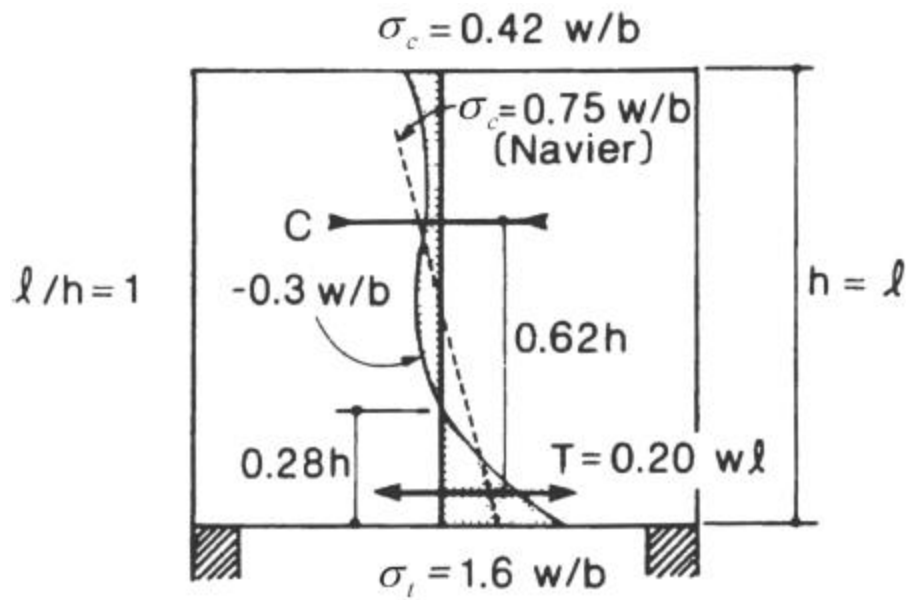
**Figure 13.15:** Stress distribution in beam with  $h = 0.5l$

| $\sigma_x [MN/m^2]$ | Top    | Bottom  | Neutral axis depth [m] |
|---------------------|--------|---------|------------------------|
| Theoretical         | 45     | -30     | 0.4                    |
| Numerical           | 29.328 | -27.669 | 0.486                  |

**Table 13.4:** Numerical results for Case B

### 13.3.3 Case C with $h = l$

In the third case the depth of the beam is equal to the distance between the midspan of the supports.



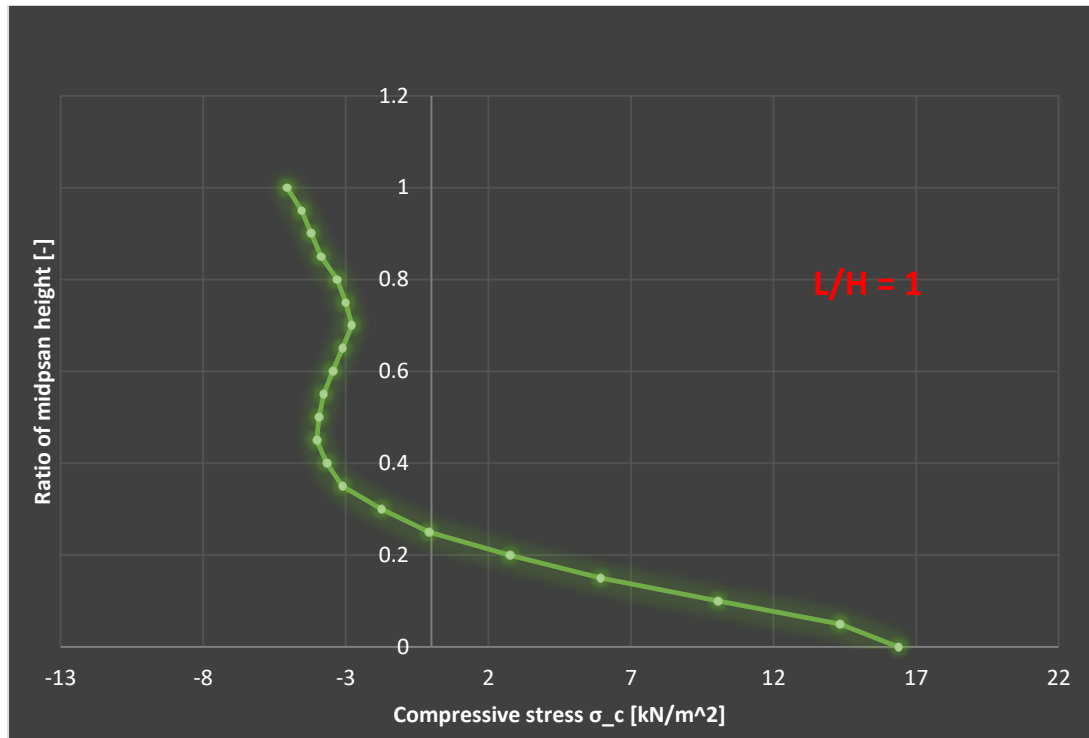
**Figure 13.16:** Problem definition Case C

The results of the test are shown in the following figure and the numerical values are contained in the following table.









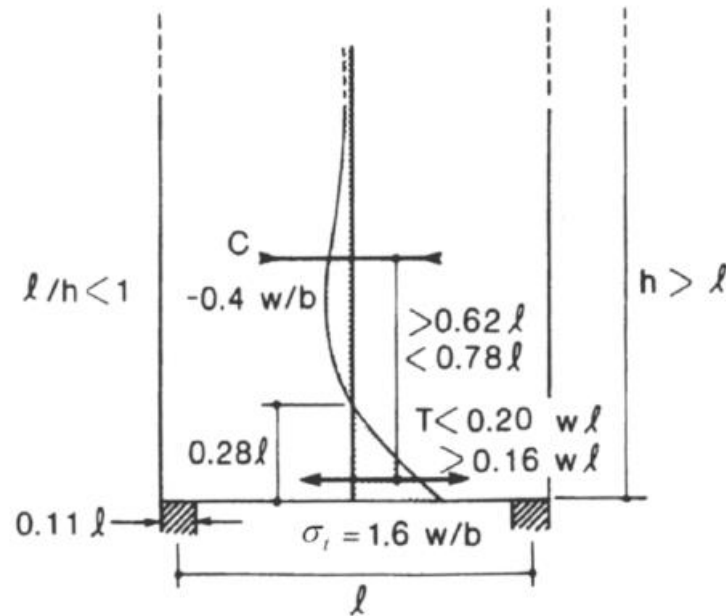
**Figure 13.18:** Stress distribution in beam with  $h = l$

| $\sigma_x [MN/m^2]$ | Top    | Bottom | Neutral axis depth [m] |
|---------------------|--------|--------|------------------------|
| Theoretical         | 16     | -42    | 0.28                   |
| Numerical           | 14.803 | -5.059 | 0.286                  |

**Table 13.5:** Numerical results for Case C

### 13.3.4 Case D with $h > l$

In the fourth case the depth of the beam is double the distance between the midspan of the supports.

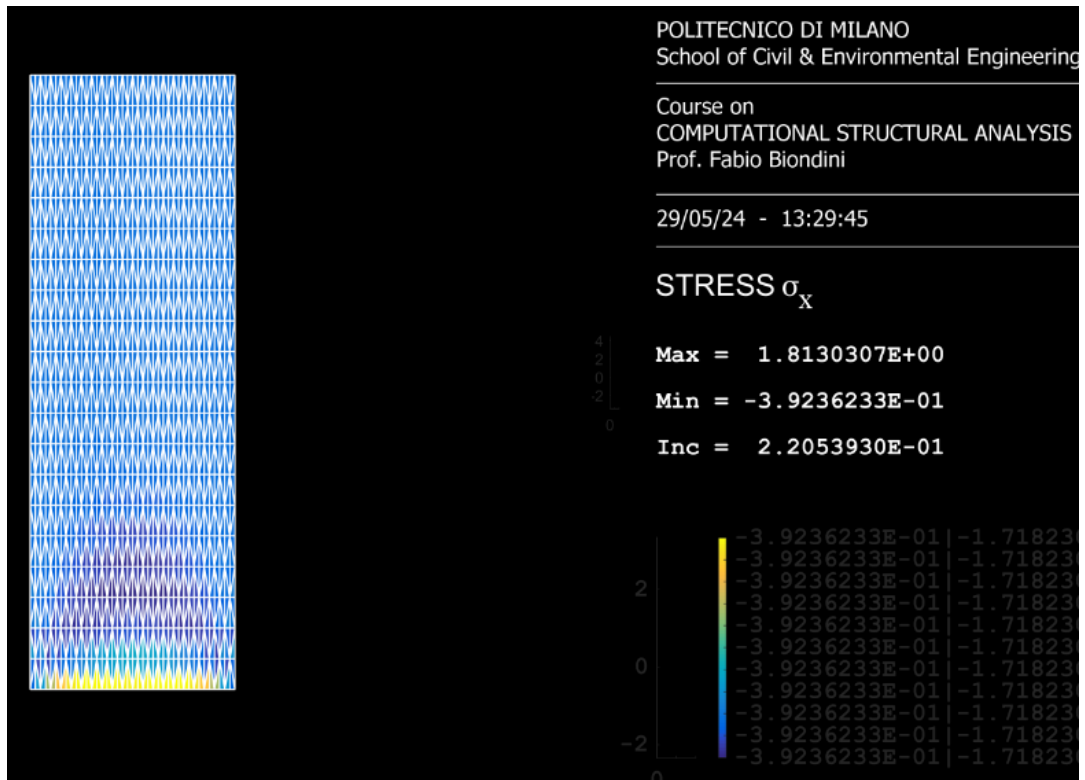


**Figure 13.19:** Problem definition Case D

The results of the test are shown in the following figure and the numerical values are contained in the following table.

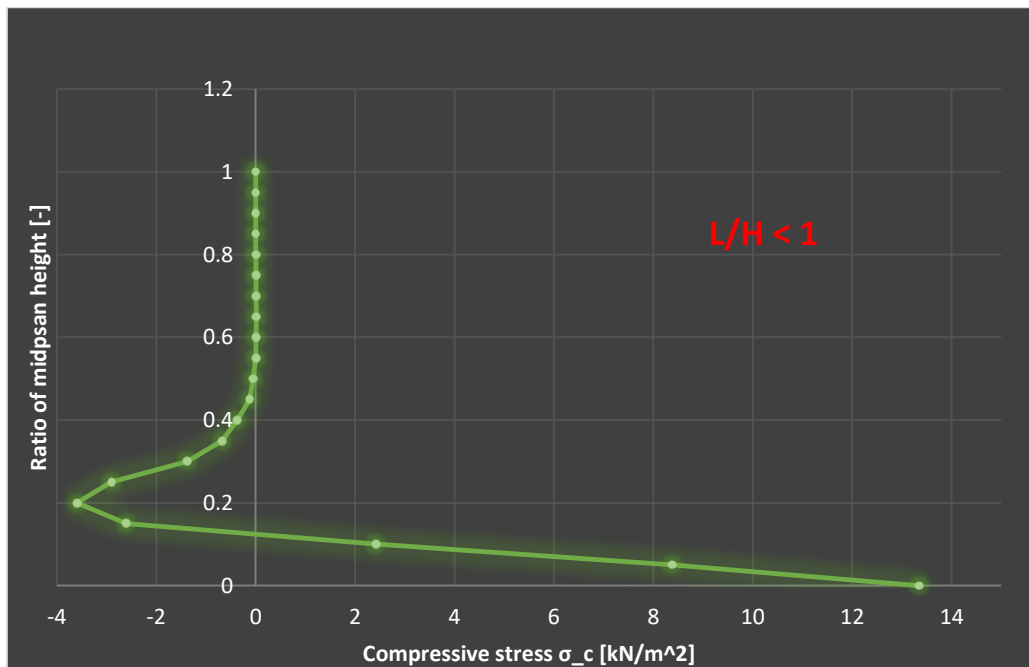


(a) Mesh model for  $l/h < 1$



(b)  $\sigma_x$  for  $l/h < 1$

**Figure 13.20:** Deformed configuration and stress distribution scaled Case D



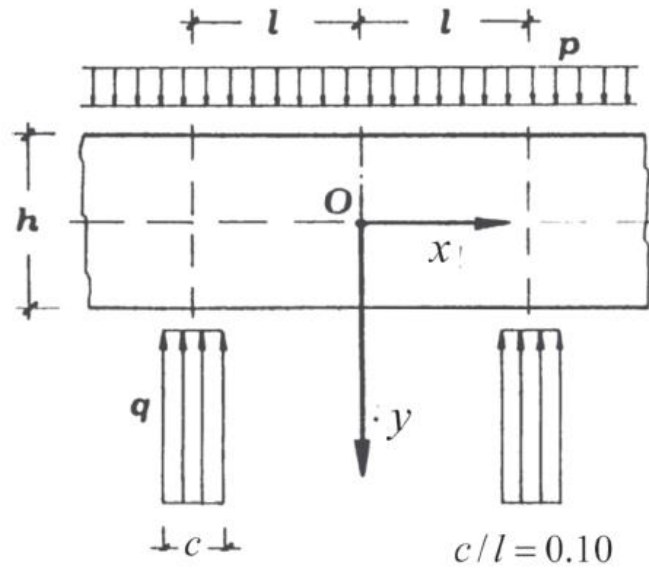
**Figure 13.21:** Stress distribution in beam with  $h > l$

| $\sigma_x[MN/m^2]$ | Top    | Bottom | Neutral axis depth [m] |
|--------------------|--------|--------|------------------------|
| Theoretical        | 16     | 0      | 0.28                   |
| Numerical          | 13.342 | 0      | 0.166                  |

**Table 13.6:** Numerical results for Case D

### 13.4 Continuous Deep Beam

In the last application, internal stresses are analyzed inside long deep beams subjected to distributed loads and supported by multiple vertical restraints. To reproduce the conditions illustrated in the following, the element studied is completely restrained on its sides preventing horizontal movement and the total width is equal to  $4l$ .



**Figure 13.22:** Problem definition

For this application multiple cases are studied for different beam depths, the value observed is the internal horizontal stress on both axes of the mid-span and the supports. In the following table are properties shared by all cases.

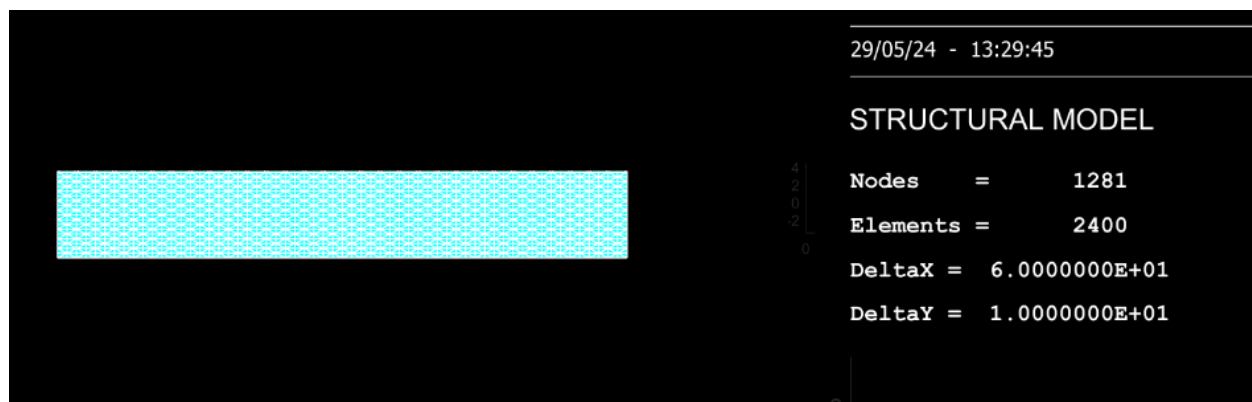
|                                  |       |
|----------------------------------|-------|
| <b>L[m]</b>                      | 0.12  |
| <b>w [KN/m<sup>2</sup>]</b>      | 10    |
| <b>E [MPa]</b>                   | 20000 |
| <b><math>\nu</math> [-]</b>      | 0     |
| <b><math>\alpha</math> [-]</b>   | 0     |
| <b>Weight [KN/m<sup>3</sup>]</b> | 0     |
| <b>b [m]</b>                     | 0.01  |

**Table 13.7:** Element properties shared in all cases

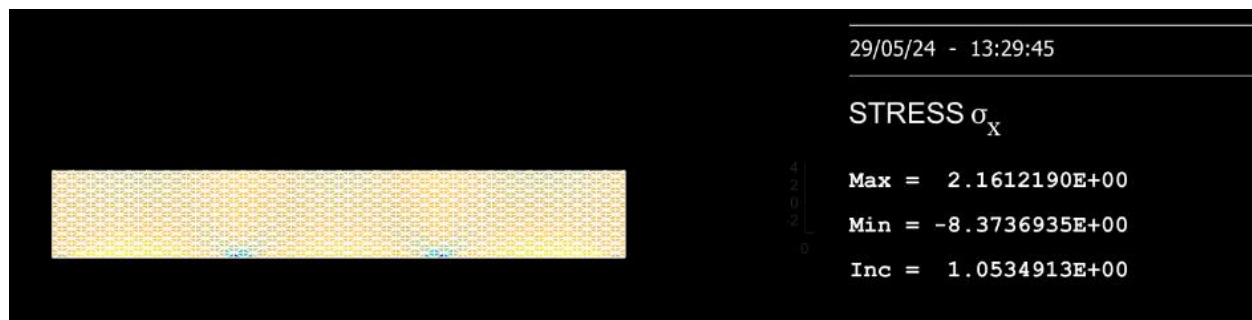
The following sections contain the information relative to the theoretical and numerical solution to the problem as well as additional information on each case.

### 13.4.1 Case A with $h = l$

In the first case the depth of the beam is equal to half the mid-span between the supports. The values of the horizontal stresses  $\sigma_x$  is normalized by the value of the surface force applied on the top.

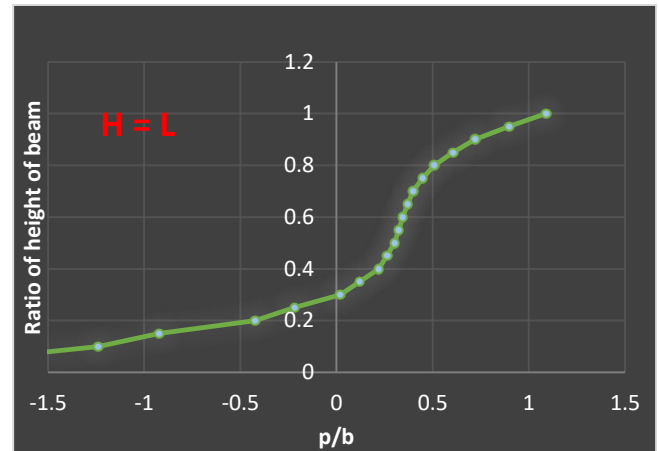
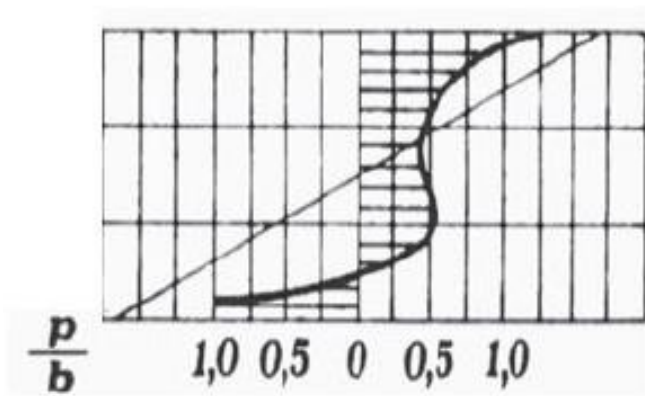


(a) Mesh model for  $h = l$

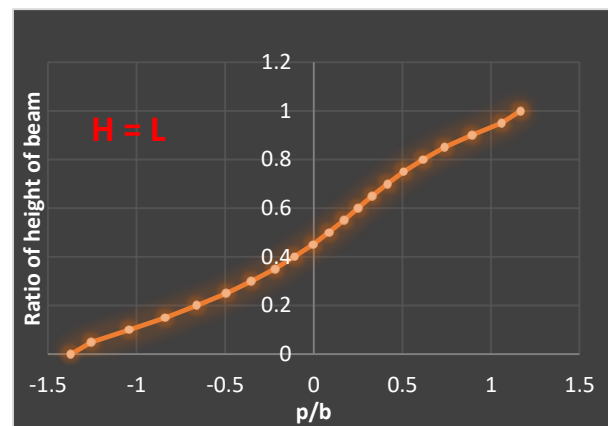
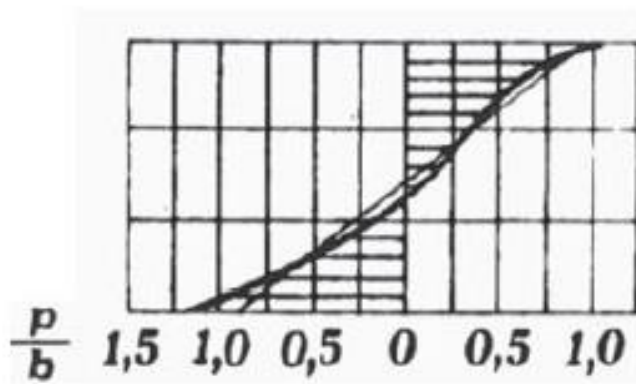


(b)  $\sigma_x$  for  $h = l$

**Figure 13.23:** Deformed configuration and stress distribution scaled Case A



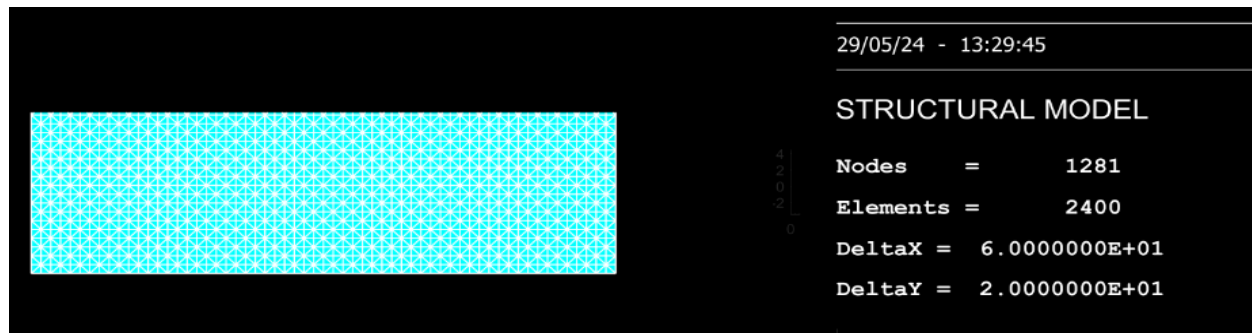
**Figure 13.24:** Horizontal stress distribution - Midspan - Case A



**Figure 13.25:** Horizontal stress distribution - Support - Case A

### 13.4.2 Case B with $h = 2l$

In the second case the depth of the beam is equal to the mid-span between the supports. The values of the horizontal stresses  $\sigma_x$  is normalized by the value of the surface force applied on the top.

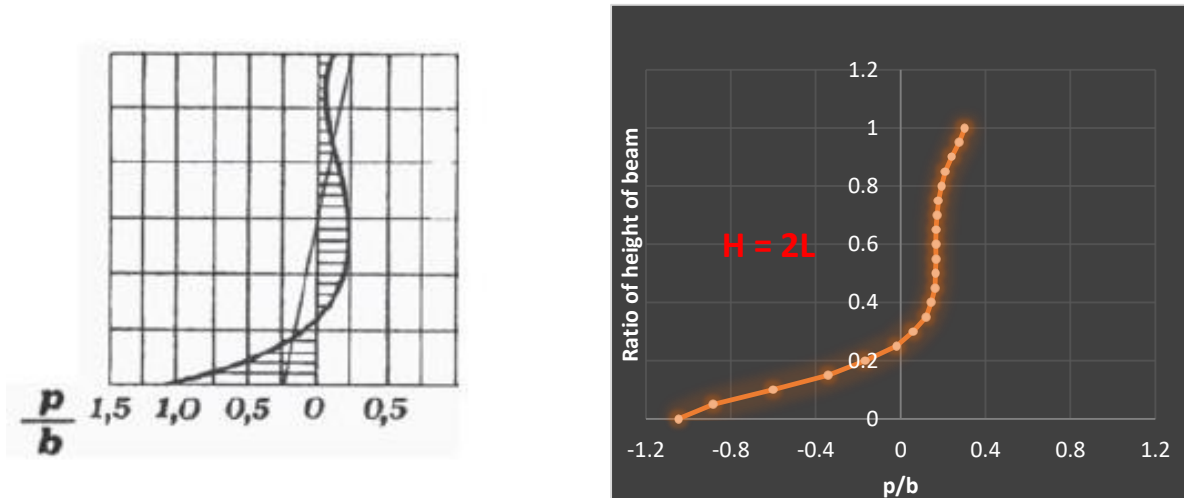


(a) Mesh model for  $h = 2l$



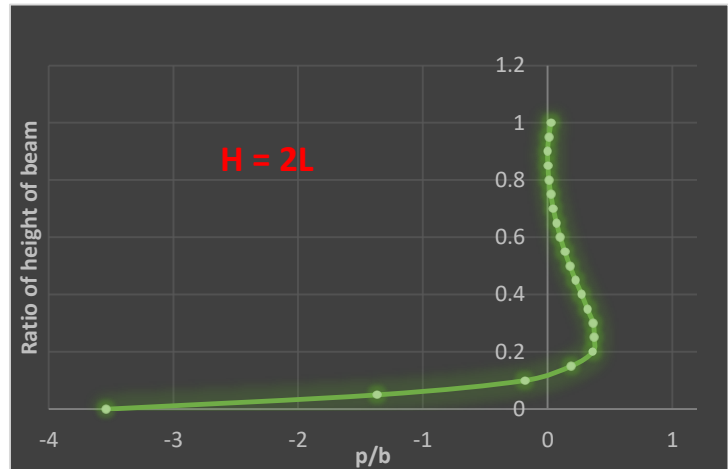
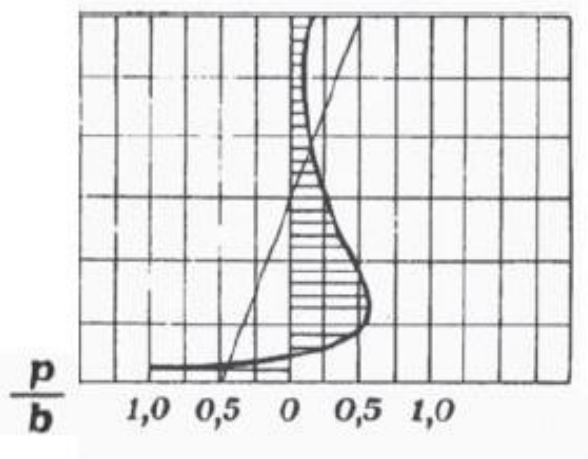
(b)  $\sigma_x$  for  $h = 2l$

**Figure 13.26:** Deformed configuration and stress distribution scaled Case B



**Figure 13.27:** Horizontal stress distribution - Midspan - Case B

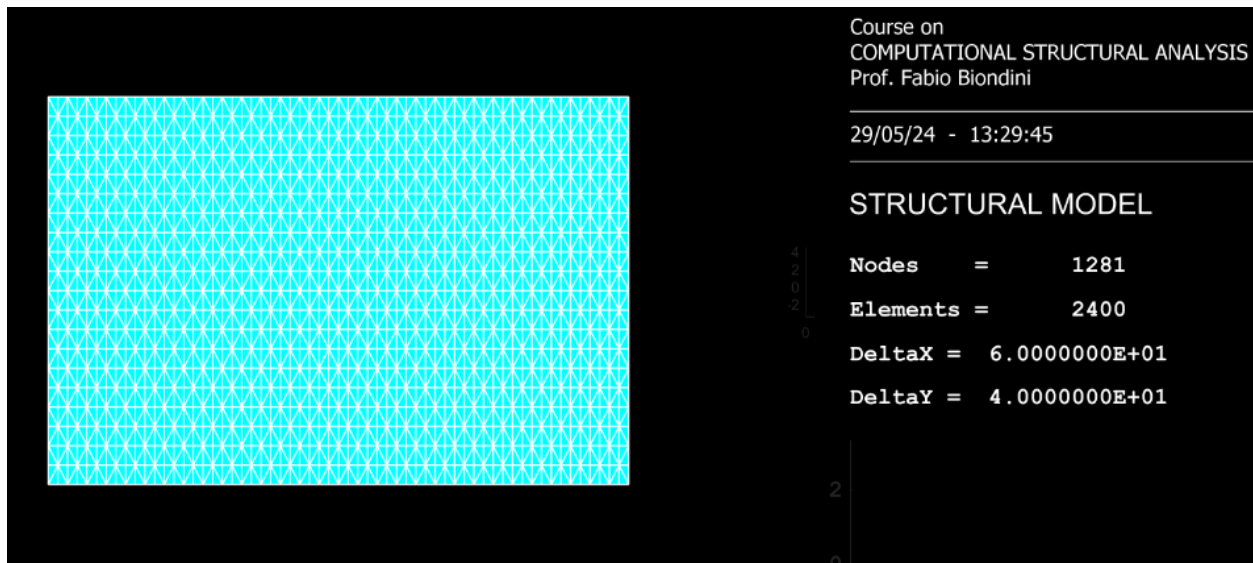




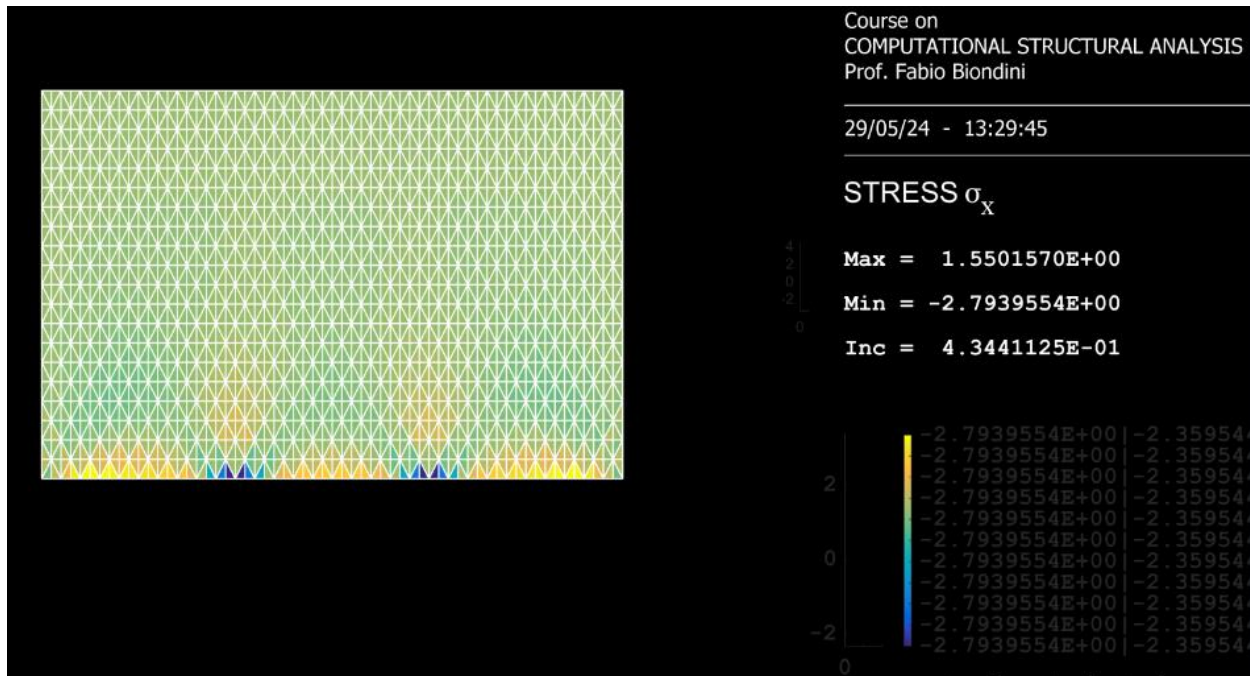
**Figure 13.28:** Horizontal stress distribution - Support - Case B

### 13.4.3 Case C with $h = \infty$

In the third case the depth of the beam is equal to four times the mid-span between the supports. The values of the horizontal stresses  $\sigma_x$  is normalized by the value of the surface force applied on the top.

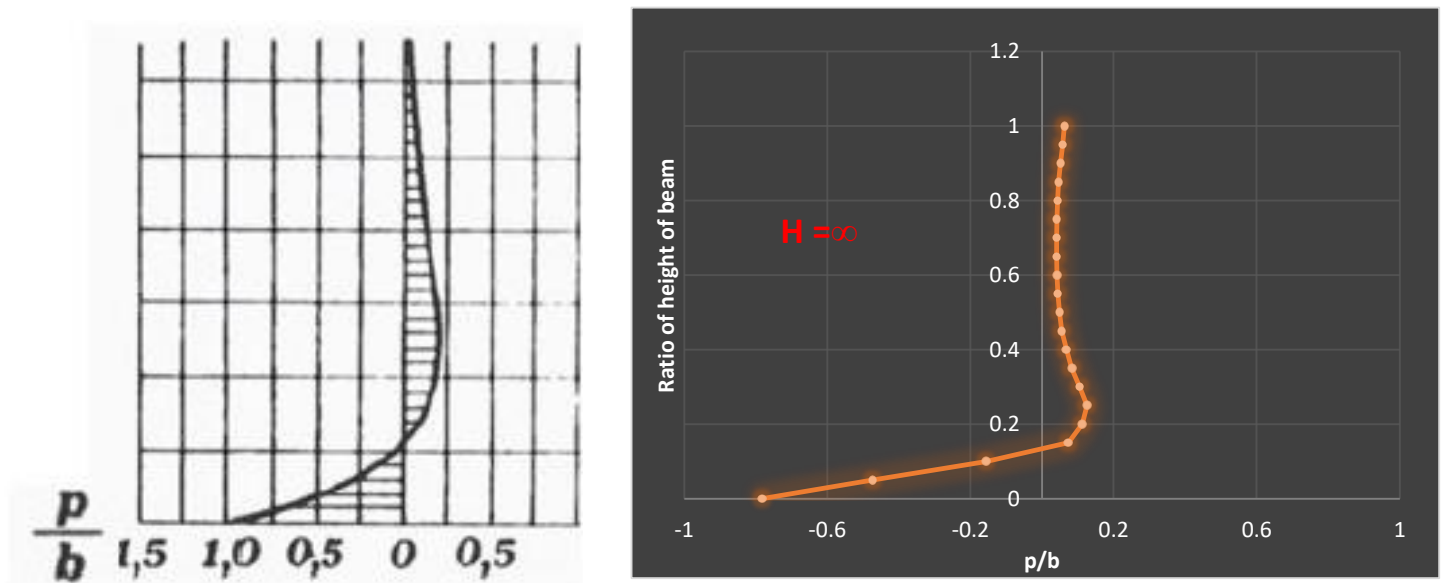


(a) Mesh model for  $h = \infty$

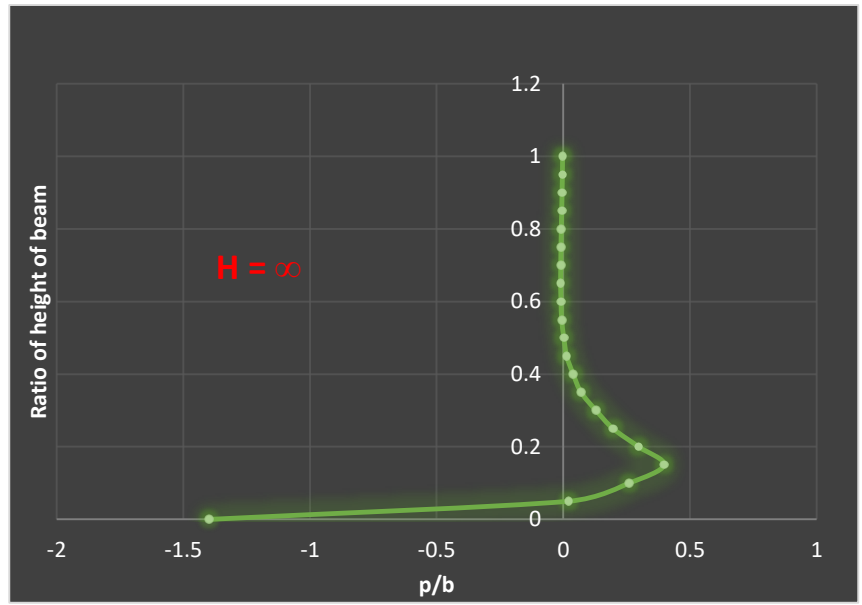
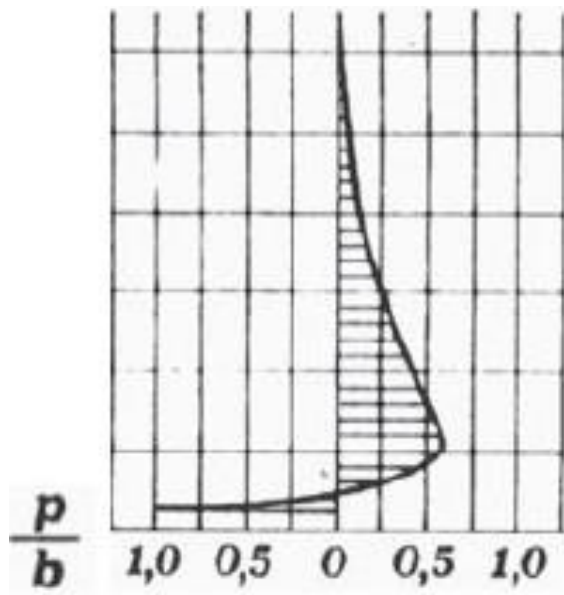


(b)  $\sigma_x$  for  $h = \infty$

**Figure 13.29:** Deformed configuration and stress distribution scaled Case C



**Figure 13.30:** Horizontal stress distribution - Midspan - Case C

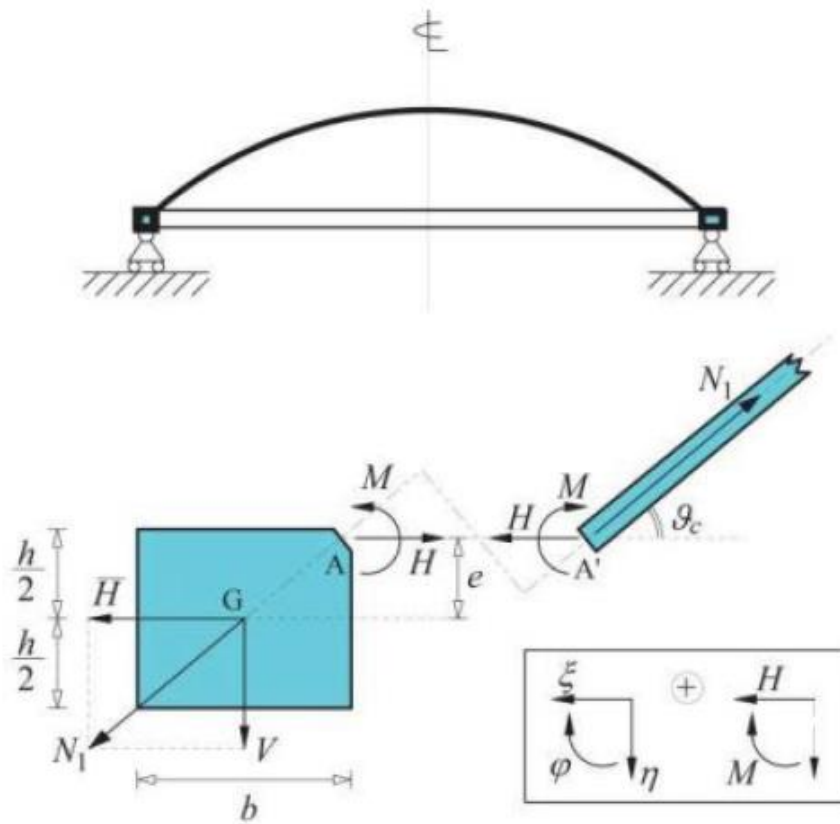


**Figure 13.31:** Horizontal stress distribution - Support - Case C

## Chapter 14

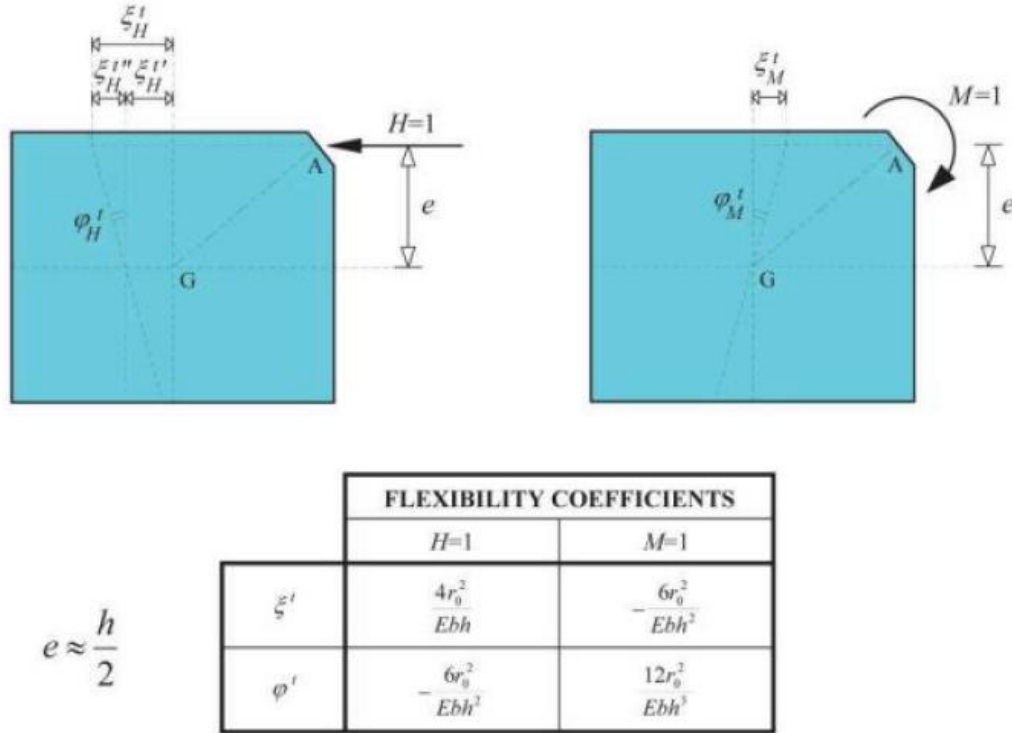
### 3D AXISYMMETRIC BENCHMARK

The CST program, which included the options for 2D plane stress and strain, has been extended to 3D for the case of axisymmetric problems. In this section we use the program to analyse the results for a ring beam under a uniformly distributed radial load, and compare against the analytical formulation. Below is a diagram illustrating the problem as well as a cross section through the beam showing the sign and naming conventions for the problem.



**Figure 14.1:** The structural problem (top) and cross section through the ring beam (bottom)

The following figure below, shows the benchmark to be examined. Given a moment cannot be applied using the CST program, we shall perform a comparison for the flexibility coefficients given by  $H=1$ ; a horizontal radial load applied at the top inner corner of the section.



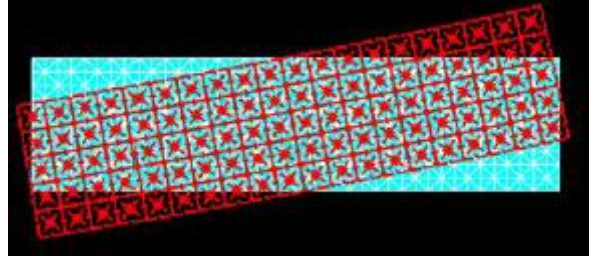
**Figure 14.2:** Flexibility coefficients for the benchmark to be examined

The numerical model consists of a rectangular cross-section of height = 10cm, width = 40cm, and a distance from the centroid to the center-line of the ring beam  $r_0 = 100$ cm. To test the program, we have input a fictional Young's Modulus  $E=100$ , such that our horizontal flexibility would be unity. The calculated flexibility coefficients for  $H=1$  are thus:

$$\xi^t = \frac{4 \cdot r^2}{Ebh} = \frac{4 \cdot 100^2}{100 \cdot 40 \cdot 10} = 1$$

$$\phi^t = -\frac{6 \cdot r^2}{Ebh^2} = -\frac{6 \cdot 100^2}{100 \cdot 40 \cdot 10^2} = -0.15$$

The plotted displacements are shown below:



**Figure 14.3:** Results - Deformed Shape

The results from CST program are:

$$\xi_H^t = 1.03216 \text{ cm}$$

$$\xi_H^{t'} = 0.25794 \text{ cm}$$

Determining rotation:

$$\xi_H^{t''} = \xi_H^t = \xi_H^{t'} = 0.77423 \text{ cm}$$

$$\phi_H^t = -\frac{\xi_H^{t''}}{H/2} = 0.15485 \text{ cm}$$

The difference between CST program result and reference are:

| Output Parameter | Numerical<br>(CST) | Reference<br>(Benchmark) | Error (%) |
|------------------|--------------------|--------------------------|-----------|
| $\xi_H^t$        | 1.03216            | 1                        | 3%        |
| $\phi_H^t$       | -0.15485           | -0.15                    | 3%        |

**Table 14.1:** Axisymmetric Ring Beam Problem - Results Comparison

Both the displacement and rotation flexibility are in very good agreement with the benchmark provided.

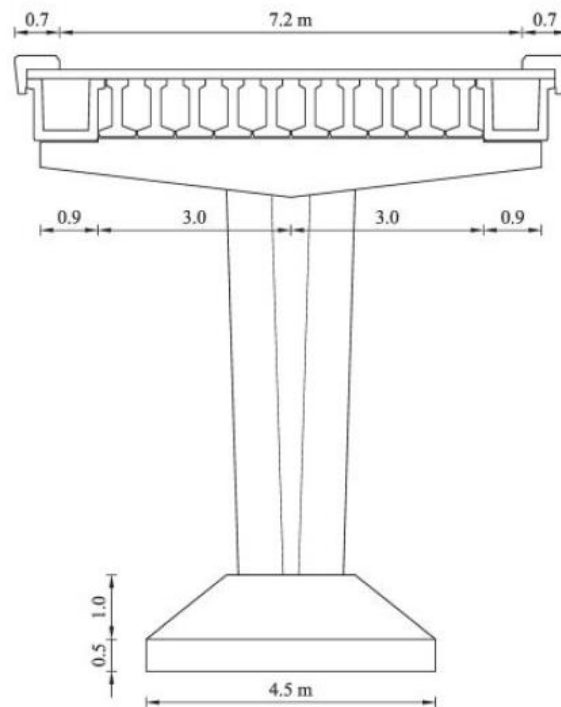
## Chapter 15

### APPLICATION TO A REAL STRUCTURE: PC BEAM

#### 15.1 Introduction

The project under the problem is a part of the extensive experimental and numerical work carried out under the BRIDGE—50 Research Project: Residual Structural Performance of a 50-Year-Old viaduct. The details of the Corso Grosseto Viaduct are as below along with schematic of the cross-section:

- 80-span double deck bridge. Total span of 16-24 m
- Built-in 1970, Demolished in 2019
- Total viaduct length: 1.4 km
- Bridge deck: Simply supported girder system using ten inner precast PC I-beams, two lateral PC box beams, and cast-in-situ RC slab.

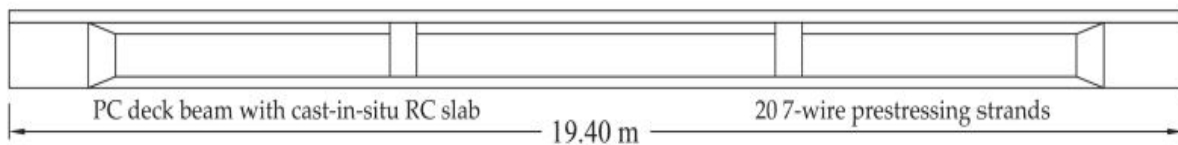


**Figure 15.1:** Schematic of Corso Grosseto Viaduct

The structural analysis to be carried out of PC Beam of the bridge. The analysis of a prestressed reinforced concrete beam under a 3 and 4-point bending tests have been carried out as part of experimentation. Extensive laboratory and NDTs have been carried out to characterize the material of the bridge. The PC beam is required to be analyzed using both FRAME CODE and CST CODE to compare the efficacy of the code models to depict behavior of actual structure.



**Figure 15.2:** PC Beam



**Figure 15.3:** Schematized PC Beam

Specifically we want to study the flexural behaviour under 2 load conditions differentiated by location of the applied load:

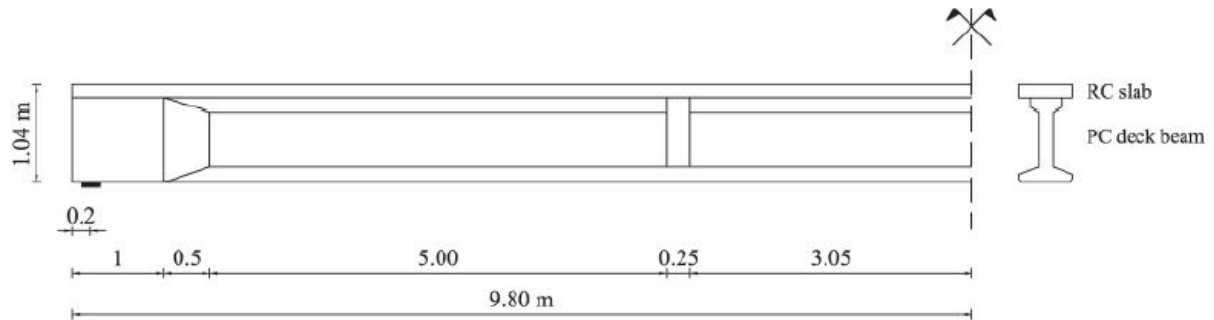
- LOAD CASE 1:  $\alpha = 1$ , the load is applied at the midspan
- LOAD CASE 2:  $\alpha = 0.68$ , the load is near the supports and we can appreciate the shear behaviour

Another aspect to investigate is the behaviour of the beam with and without the top slab. At the end we need 4 models for each finite element analysis for a total of 8 models to compare.



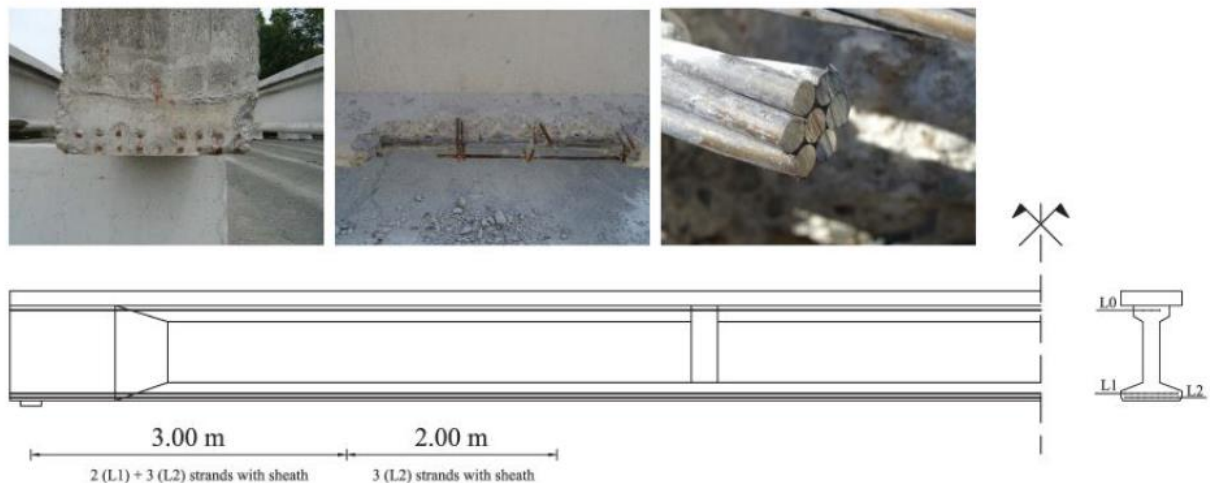
## 15.2 Beam Model - Frame and CST Element

The subject PC beam is schematized as follow considering the symmetry of the problem.



**Figure 15.4:** Selected PC Scheme

Another important aspect to notice is that the prestressing is not constant along the length. In fact, since the prestressing cables are straight, in order to avoid unnecessary negative bending moment on the support where the bending moment given by the external load is null, some cables have sheath towards the support section. By application of sheath, there is no bond between concrete and strands and for those cables the prestressing effect is null within the sheathed length.

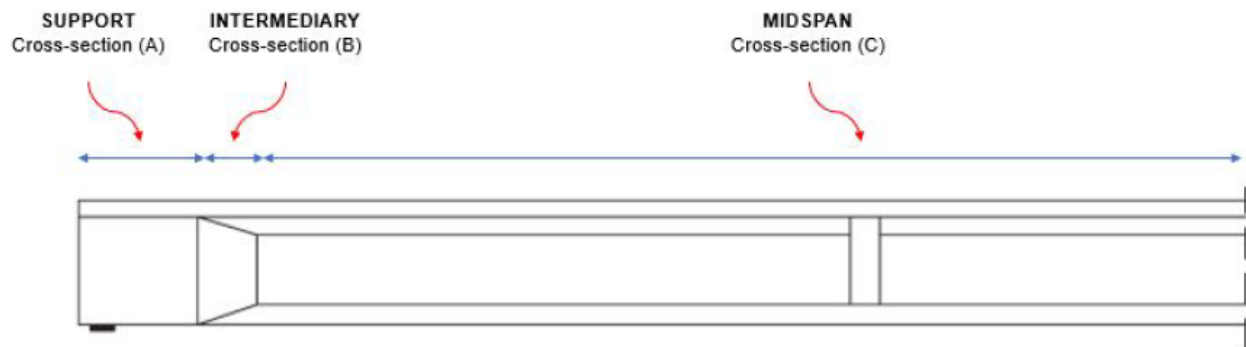


**Figure 15.5:** Prestressing Scheme

### 15.2.1 Cross-sections

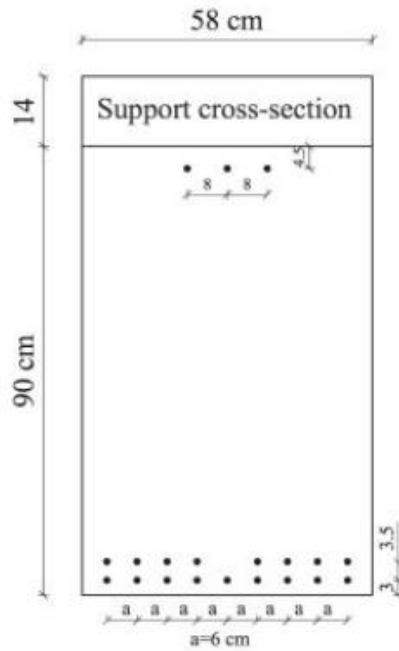
The beam comprise of three different spans with respect to the cross-section. Specifically we have 3 different cross-sections. At the left most side (Support Section) of the model beam shown above, the cross-section is rectangular, the right most side (Midspan Section) is an I-section and an intermediate span has linearly varying cross-section representing the transition from rectangular section to I-section. The first two spans (Support and Midspan sections) are modelled exactly as per their cross-sectional geometrical properties while the intermediary section is modelled as I-section with cross-sectional properties for section at the mid of intermediary section.

The stiffened part at the distance of 6.5 meter from the support has cross-section equal to that of support section (solid) but considering the size of the stiffened part, its contribution is neglected and mid-span cross-section is considered. In addition to this, while calculating the geometrical properties for the different cross-section, two cases are considered; the beam with and without top slab. The details for each of the cross-sections are given in the following tables.



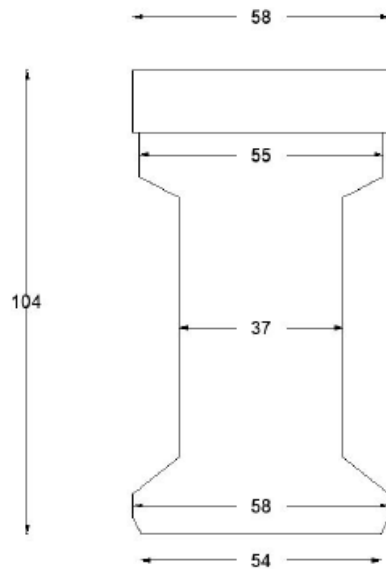
**Figure 15.6:** Selected cross-sections along the beam

Support cross-section (A)



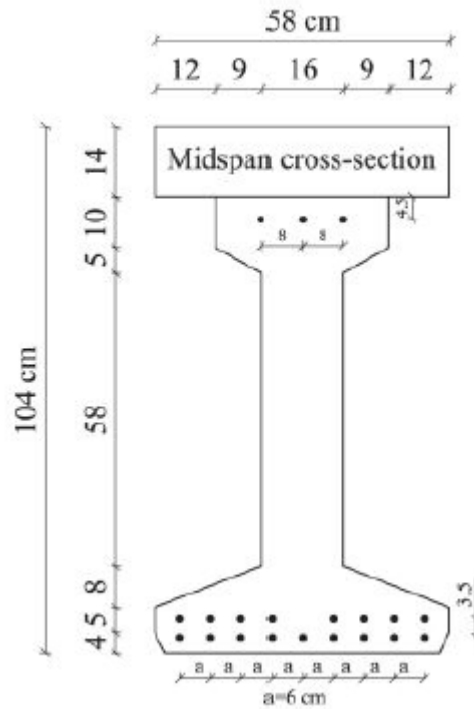
**Figure 15.7:** Support cross-section (A)

Intermediary cross-section (B)



**Figure 15.8:** Support cross-section (B)

### Midspan cross-section (C)



**Figure 15.9:** Support cross-section (C)

| Without Top Slab Case                    | Cross-Sections |         |         |
|------------------------------------------|----------------|---------|---------|
|                                          | A              | B       | C       |
| Moment of inertia ( $I_{xx}$ ) [ $m^4$ ] | 0.03523        | 0.02981 | 0.02042 |
| Moment of inertia ( $I_{yy}$ ) [ $m^4$ ] | 0.0146         | 0.0     | 0.00244 |
| Area (A) [ $m^2$ ]                       | 0.5214         | 0.3821  | 0.2202  |
| Centroid (from bottom) [ $m$ ]           | 0.450          | 0.446   | 0.396   |
| Depth (d) [ $m$ ]                        | 0.90           | 0.90    | 0.90    |

**Table 15.1:** Cross-Section Properties - Without Top Slab Case

| With Top Slab Case                       | Cross-Sections |         |         |
|------------------------------------------|----------------|---------|---------|
|                                          | A              | B       | C       |
| Moment of inertia ( $I_{xx}$ ) [ $m^4$ ] | 0.05427        | 0.04821 | 0.04004 |
| Moment of inertia ( $I_{yy}$ ) [ $m^4$ ] | 0.0168         | 0.0     | 0.0     |
| Area (A) [ $m^2$ ]                       | 0.6022         | 0.4629  | 0.3010  |
| Centroid (from bottom) [m]               | 0.520          | 0.538   | 0.550   |
| Depth (d) [m]                            | 1.04           | 1.04    | 1.04    |

**Table 15.2:** Cross-Section Properties - With Top Slab Case

### 15.2.2 Mechanical Properties

The mechanical properties were taken as given in the assignment. The below Table reproduces the properties:

| Concrete | Beam       | Slab      |
|----------|------------|-----------|
| $f_c$    | -29.94 MPa | -26.0 MPa |
| $E_c$    | 27.22 GPa  | 27.0 GPa  |
| $f_{ct}$ | 3.30 MPa   | 2.35 MPa  |
| $E_{cu}$ | -0.35%     | -0.35%    |

**Table 15.3:** Mechanical Properties of Concrete

| Steel              | Bars    | Strands  |
|--------------------|---------|----------|
| $f_{sy}$           | 450 MPa | 1486 MPa |
| $E_s$              | 200 GPa | 195 GPa  |
| $f_{su}$           | 540 MPa | 1722 MPa |
| $\varepsilon_{su}$ | 8%      | 5%       |

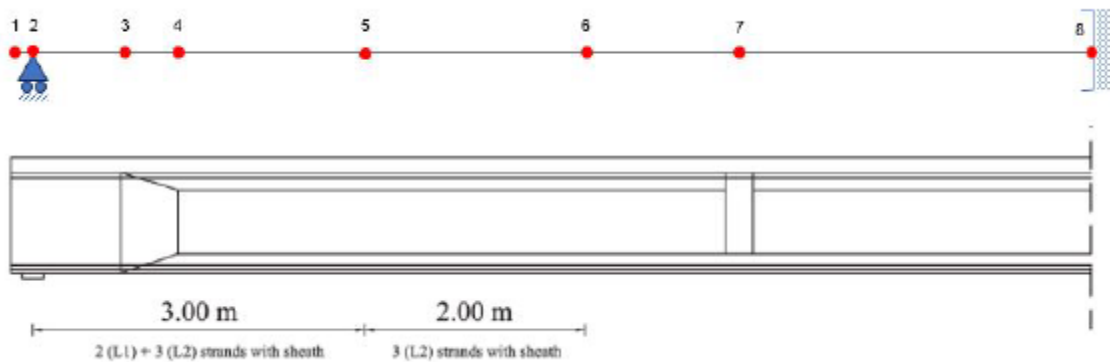
**Table 15.4:** Mechanical Properties of Steel

Apart from this, the value of the Poisson ratio and Self-weight for the concrete was taken as 0.2 and  $25 \text{ kN/m}^3$ , respectively.

### 15.2.3 Frame Model

Initially, as per the requirement, the beam is studied as FRAME ELEMENT. The discretization of the frame is carried out considering the following change points.

- Points where sections are changing
- Points where prestressing tendons are curtailed
- Point of application of the second load case ( $\alpha = 0.68$ )



**Figure 15.10:** Frame model

The model is depicted below. The selected nodes are shown along with following description:

- Node 1: Start of support section
- Node 2: Roller support
- Node 3: End of support section, Start of intermediary section
- Node 4: End of intermediary section, Start of I-section (midspan section)
- Node 5: Curtailment of 2(L1) and 3(L2) prestressing strands
- Node 6: Curtailment of 3(L2) prestressing strands
- Node 7: Load point for load case 2
- Node 8: Load point for load case 1

The modelled support conditions are as following:

- Support (Left): Roller vertical support at a single node
- Support (Right): Slider support at a single node (midspan)

#### 15.2.4 CST Model

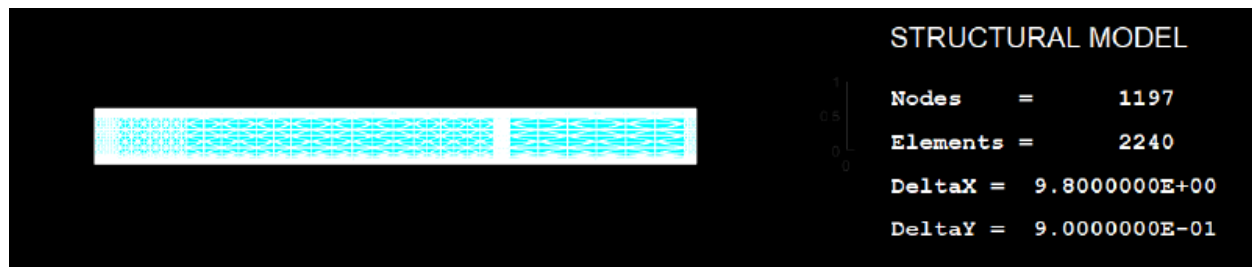
The beam is studied as a set of plane elements under plane stress conditions. The plane elements are Constant Strain Triangles. Considering the beam as a plane, the discretization is carried out in both horizontal and vertical directions. The mesh size is more refined at the support and load application points. In addition to this, sections at the points about the application of prestressing forces are meshed with finer elements. About the horizontal direction, the beam is divided into 9 domains to account for the above-mentioned reasons, cross-sectional changes and points of loading (pre-stressing + applied load). For the vertical direction, the cross-section is divided into 7 and 8 domains for the case without and with slab, respectively. The reason for defining such domains is to reflect the varying thickness at a given cross-section and define the points for the application of prestressing forces. The mesh is shown below along with a tabulation of element thickness along the cross-section.

|                          | Support Section | Intermediate Section | Midspan Section |
|--------------------------|-----------------|----------------------|-----------------|
| Domain 1 ( $y = 0$ )     | 0.58            | 0.56                 | 0.56            |
| Domain 2 ( $y = 0.04$ )  | 0.58            | 0.58                 | 0.58            |
| Domain 3 ( $y = 0.09$ )  | 0.58            | 0.48                 | 0.37            |
| Domain 4 ( $y = 0.17$ )  | 0.58            | 0.37                 | 0.16            |
| Domain 5 ( $y = 0.75$ )  | 0.58            | 0.46                 | 0.25            |
| Domain 6 ( $y = 0.8$ )   | 0.58            | 0.55                 | 0.34            |
| Domain 7 ( $y = 0.855$ ) | 0.58            | 0.55                 | 0.34            |

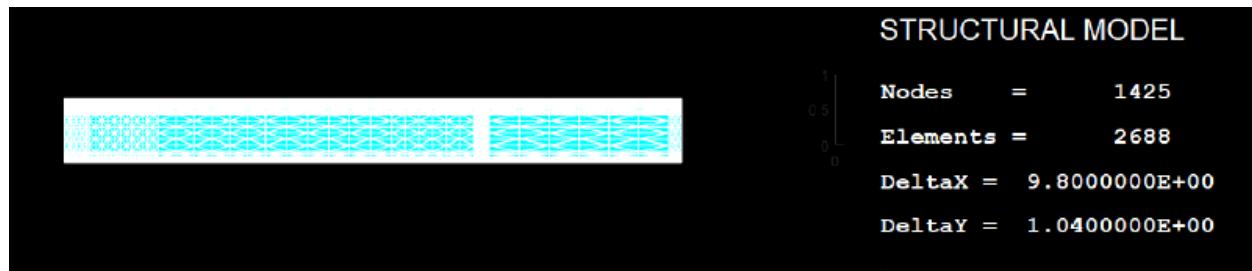
**Table 15.5:** Element Thickness - Without Slab Case

|                          | Support Section | Intermediate Section | Midspan Section |
|--------------------------|-----------------|----------------------|-----------------|
| Domain 1 ( $y = 0$ )     | 0.58            | 0.56                 | 0.56            |
| Domain 2 ( $y = 0.04$ )  | 0.58            | 0.58                 | 0.58            |
| Domain 3 ( $y = 0.09$ )  | 0.58            | 0.48                 | 0.37            |
| Domain 4 ( $y = 0.17$ )  | 0.58            | 0.37                 | 0.16            |
| Domain 5 ( $y = 0.75$ )  | 0.58            | 0.46                 | 0.25            |
| Domain 6 ( $y = 0.8$ )   | 0.58            | 0.55                 | 0.34            |
| Domain 7 ( $y = 0.855$ ) | 0.58            | 0.55                 | 0.34            |
| Domain 8 ( $y = 0.855$ ) | 0.58            | 0.58                 | 0.58            |

**Table 15.6:** Element Thickness - With Slab Case



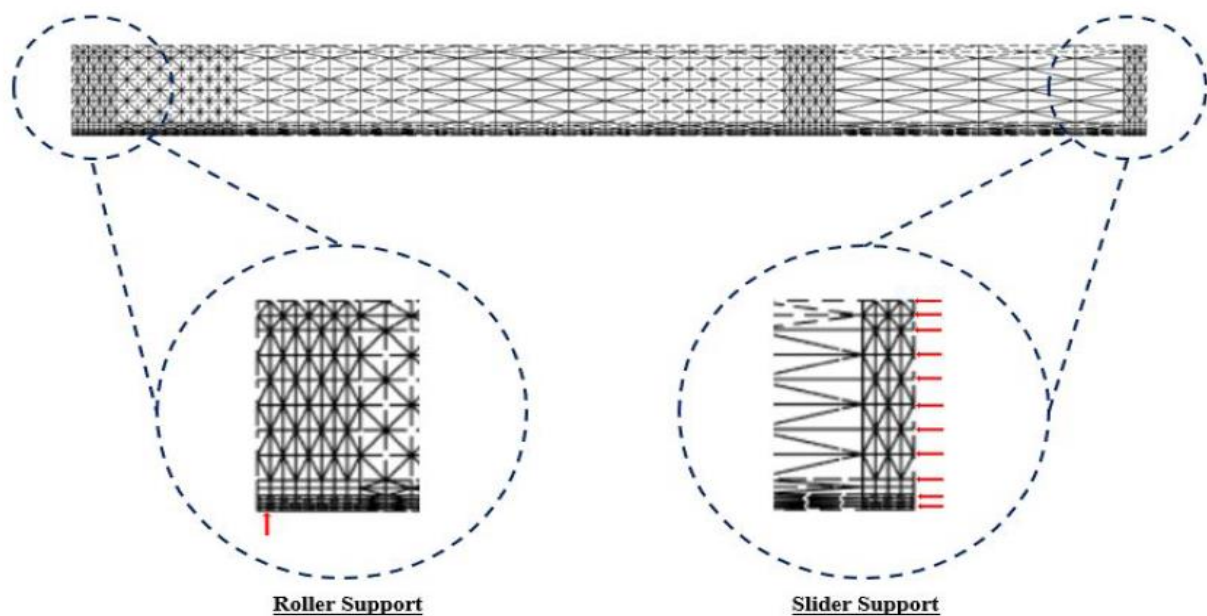
**Figure 15.11:** CST Model - Without Slab



**Figure 15.12:** CST Model - With Slab

The support conditions were modelled as following with description in the following figure.

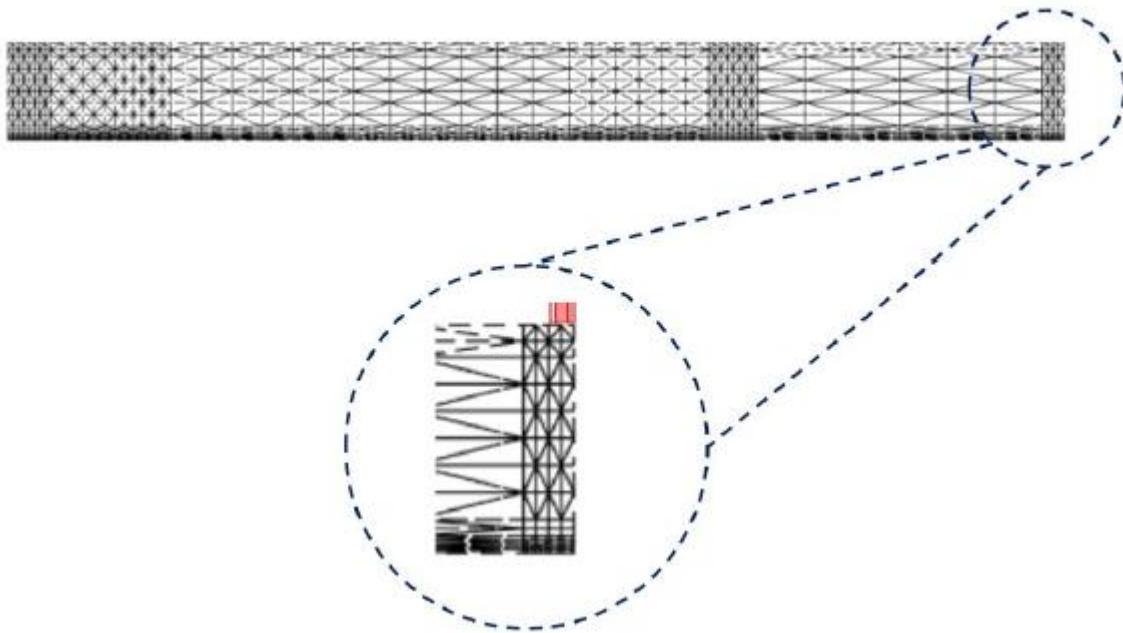
- Support (Left): Roller vertical support at a single node of CST element
- Support (Right): Distributed horizontal support to model slider support



**Figure 15.13:** Support Model



The loading was applied as distributed load along edges of two CST elements (cumulative length of 10 cm). The dimensions of the CST elements and intensity of load was kept as such that the resultant loading was equal to the applied concentrated load. The modelled is adopted considering real conditions where applied load is distributed instead of concentrated. The loading is depicted in the following figure for the case of Load Case 1.



**Figure 15.14:** Load Model

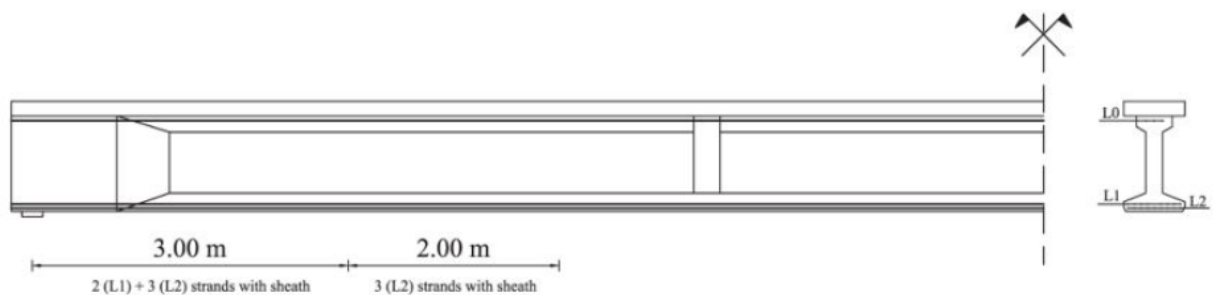
### 15.3 Prestressing

The prestressing system was discussed in the preceding sections along with the layout. It is characterized by twenty-seven strands with a nominal diameter equal to 12.70 mm (effective area  $99 \text{ mm}^2$ ). The estimated pre-stressing stress, net of instantaneous and long-term losses is  $\sigma_p = 614 \text{ MPa}$ . As already mentioned, the prestressing is not uniform along the beam. A table with the prestressing forces at each point (concerning abscissa) and the schematic of prestressing are given below:

| Coordinate        | Position | Eccentricity [m] | Prestressing Force[kN] | $\Delta F$ [kN] |
|-------------------|----------|------------------|------------------------|-----------------|
| $x = 0 \text{ m}$ | $L_0$    | 0.355            | 182.258                | 182.258         |
|                   | $L_1$    | 0.455            | 364.716                | 364.716         |
|                   | $L_2$    | 0.49             | 364.716                | 364.716         |
| $x = 3 \text{ m}$ | $L_0$    | 0.355            | 182.258                | 0               |
|                   | $L_1$    | 0.455            | 486.288                | 121.527         |
|                   | $L_2$    | 0.49             | 364.716                | 0               |
| $x = 5 \text{ m}$ | $L_0$    | 0.355            | 182.258                | 0               |
|                   | $L_1$    | 0.455            | 486.288                | 0               |
|                   | $L_2$    | 0.49             | 547.074                | 182.358         |

**Table 15.7:** Prestressing Force Distribution

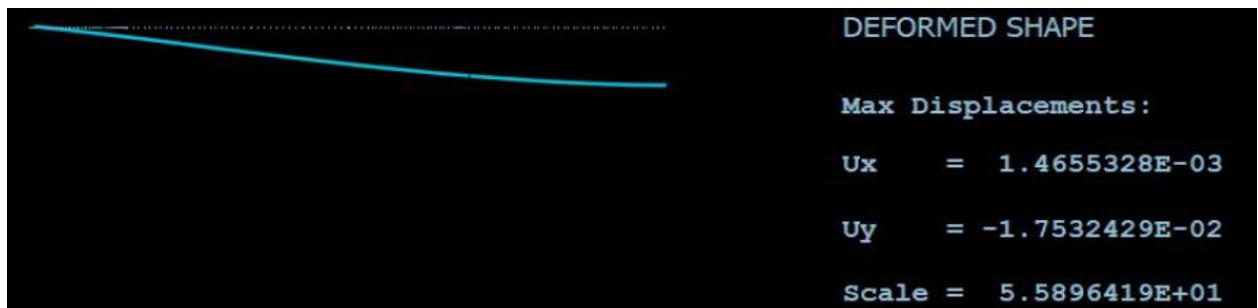
$\Delta F$  is the value that has to be applied in the node for the CST model progressively as the prestressing force changes as per below scheme. Whereas, the prestressing force is the one that has to be applied at the beam element in the beam model.



**Figure 15.15:** Prestressing Scheme

## 15.4 Results

### 15.4.1 Load Case 1 - Without Slab



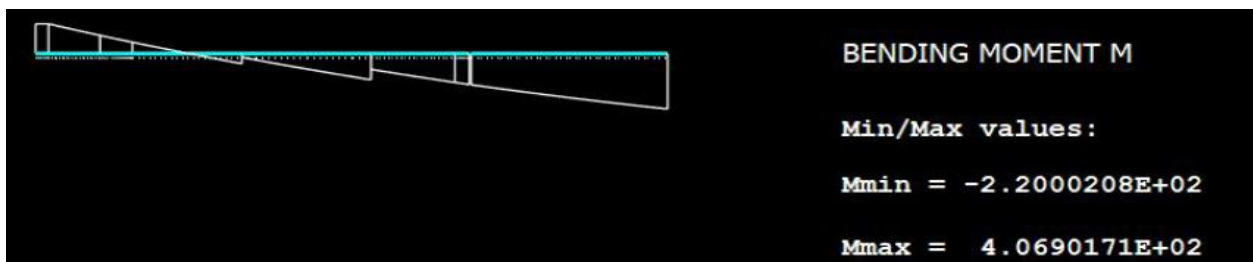
(a) Deformed Shape



(b) Axial Force Diagram

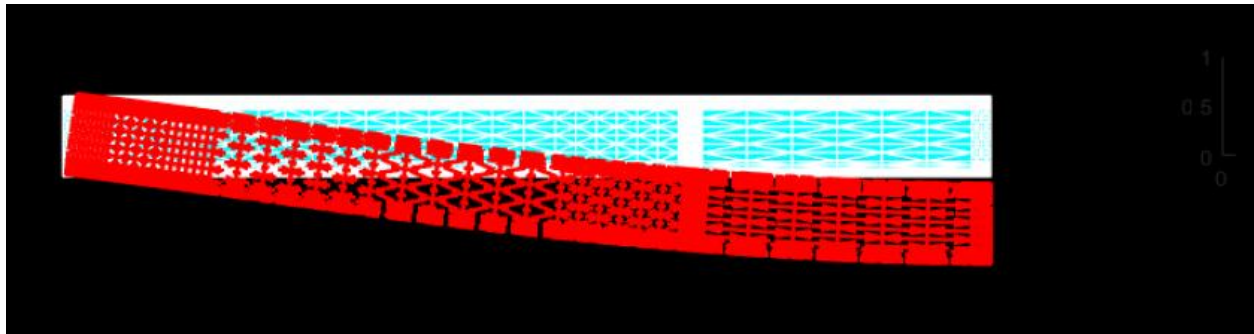


(c) Shear Force Diagram

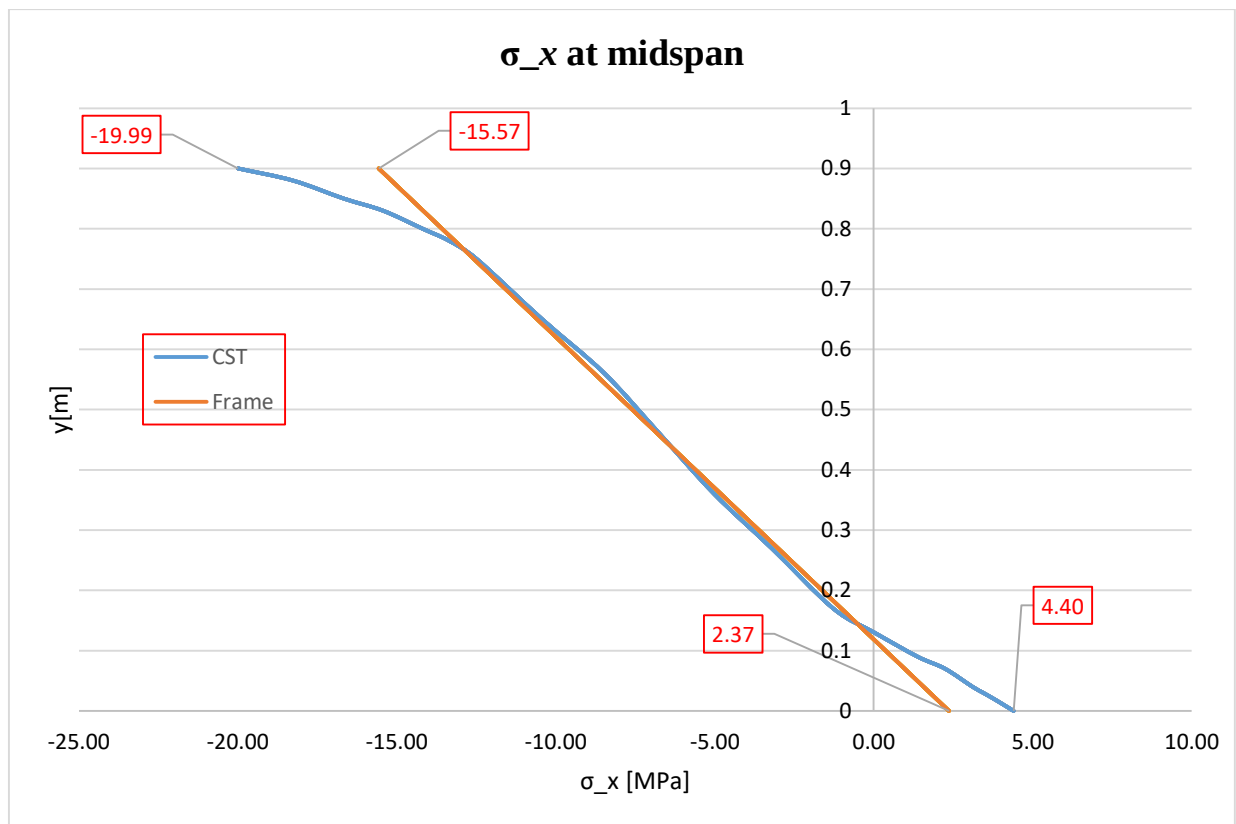


(d) Bending Moment Diagram

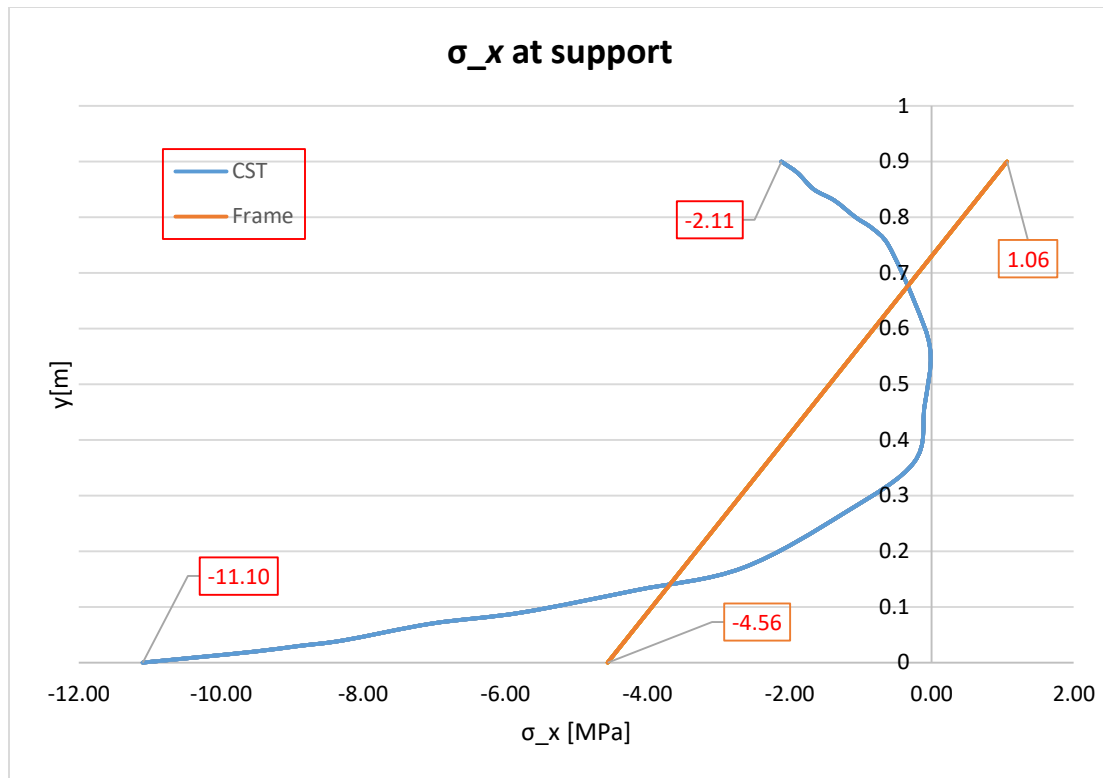
**Figure 15.16:** Frame Code Results - LC1 without slab



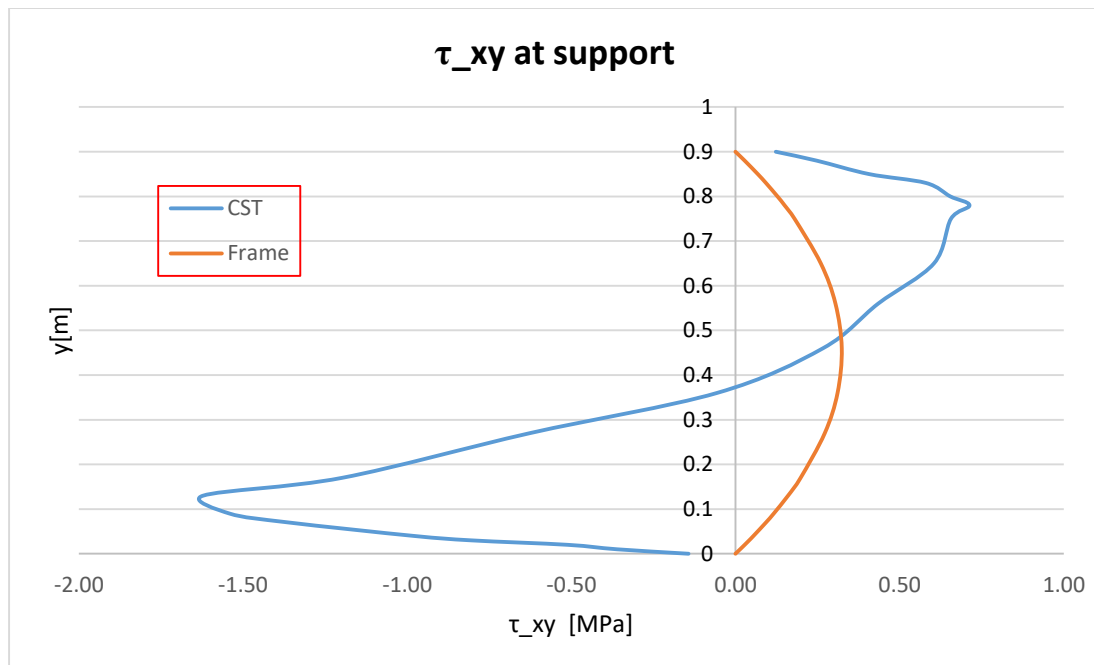
**Figure 15.17: CST Code –Deformed Shape - LC1 without slab**



(a)  $\sigma_x$  at midspan [MPa]



(b)  $\sigma_x$  at Support [MPa]



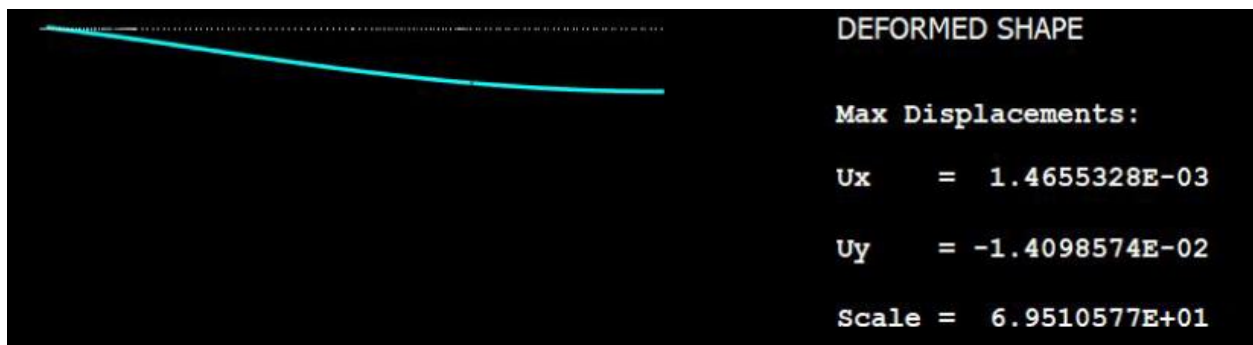
(c)  $\tau_{xy}$  at midspan [MPa]

**Figure 15.18:** Stress comparison –CST vs. Frame Code – LC1 without slab

| Displacement [mm]   |               |             |          |
|---------------------|---------------|-------------|----------|
|                     | Frame Element | CST Element | Error[%] |
| Midspan Deflection  | 18.99         | 19.9        | 4.6 %    |
| Roller Displacement | 1.58          | 1.66        | 5.1 %    |

**Table 15.8:** Results – Load case 1 – Without Slab

#### 15.4.2. Load Case 2 - Without Slab



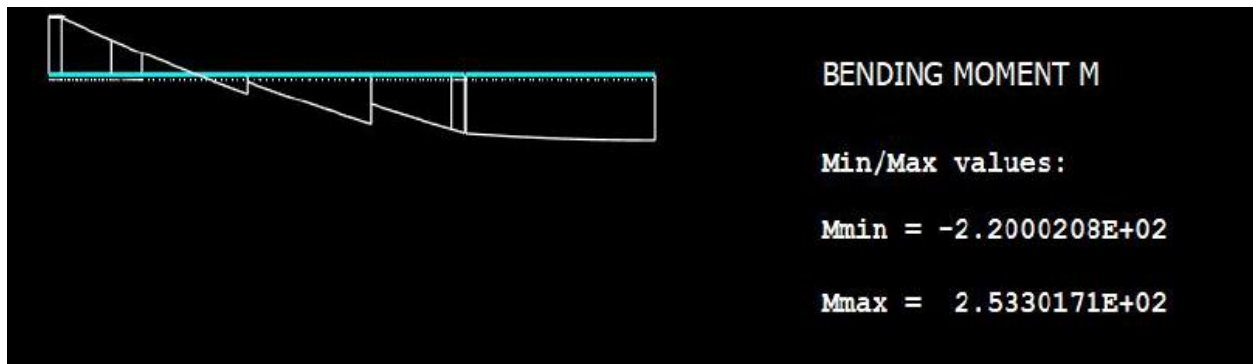
(a) Deformed Shape



(b) Axial Force Diagram

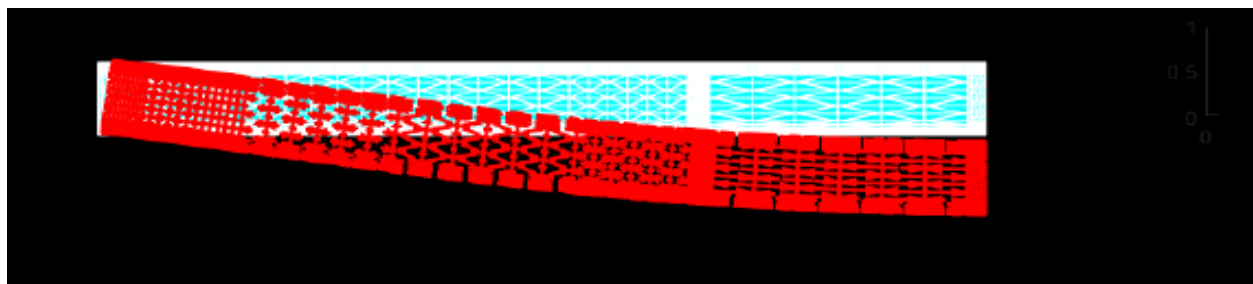


(c) Shear Force Diagram

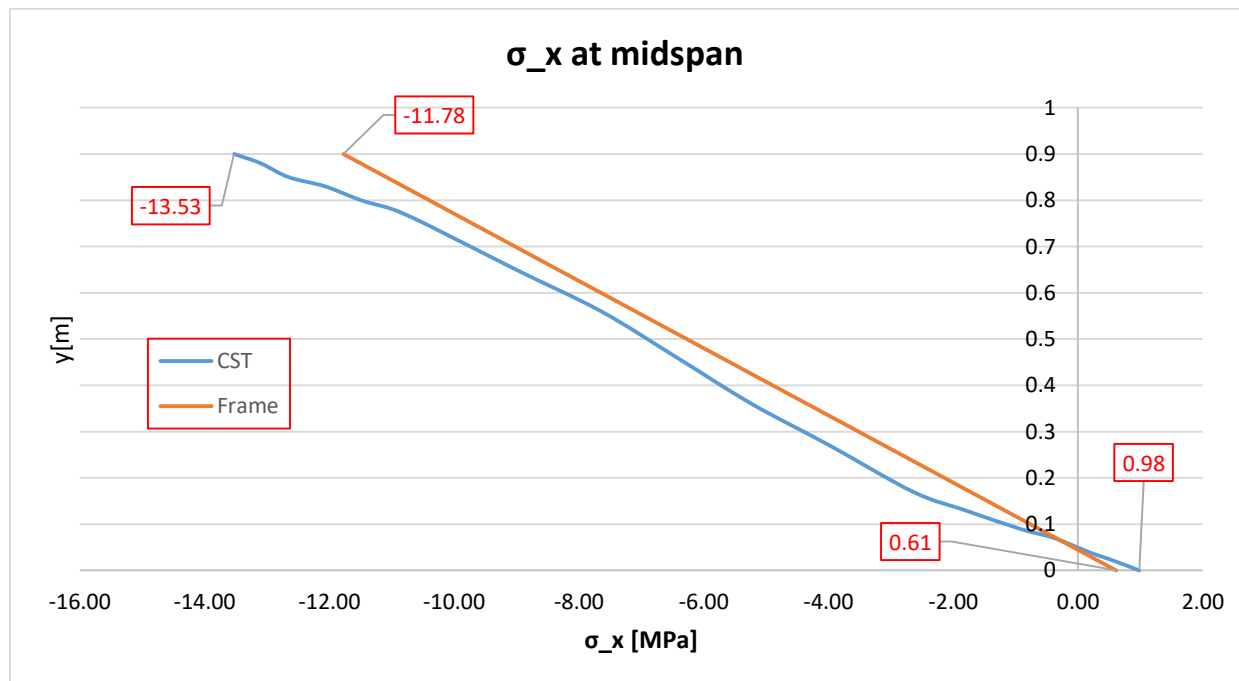


(d) Bending Moment Diagram

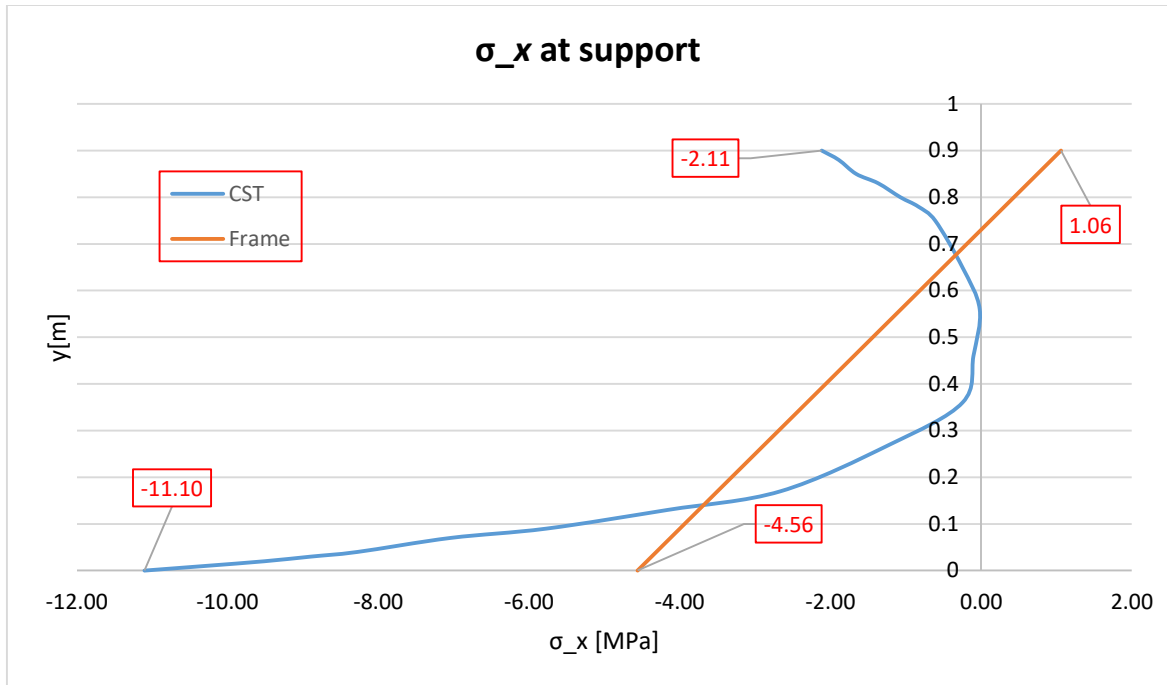
**Figure 15.19:** Frame Code Results – LC2 without slab



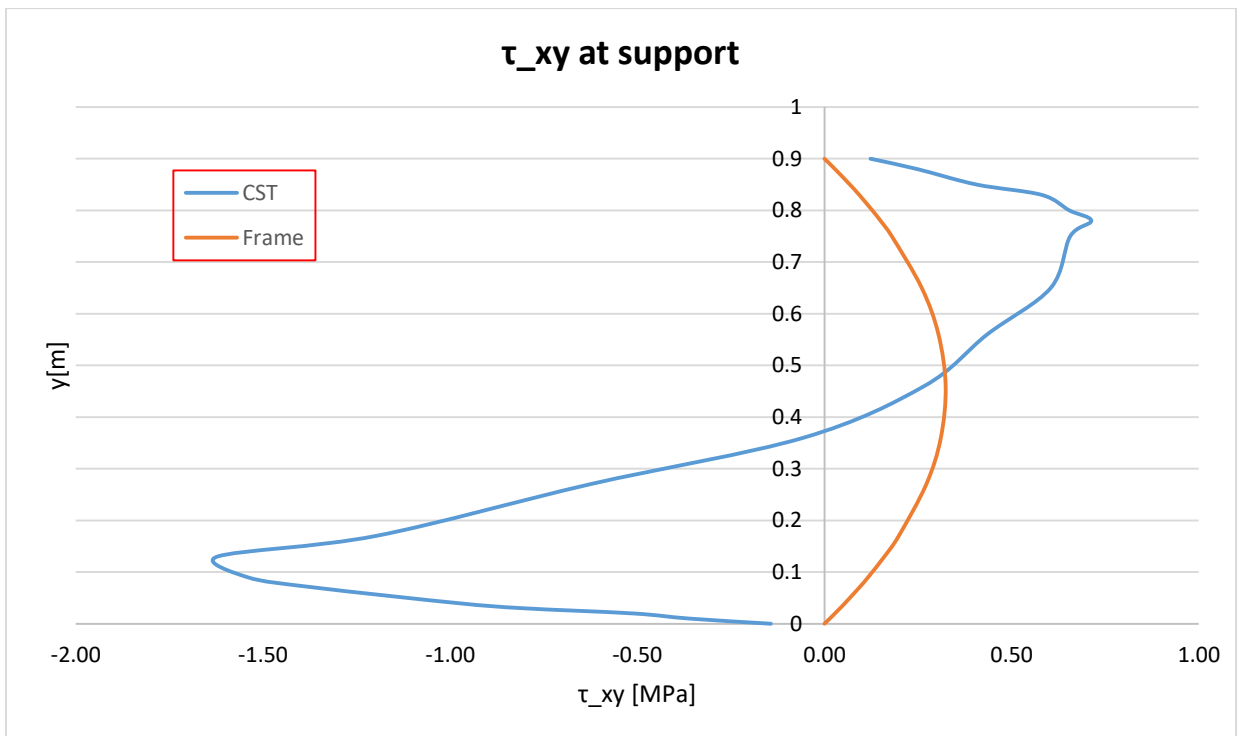
**Figure 15.20:** CST Code –Deformed Shape – LC2 without slab



(a)  $\sigma_x$  at midspan [MPa]



(b)  $\sigma_x$  at support [MPa]



(c)  $\tau_{xy}$  at midspan [MPa]

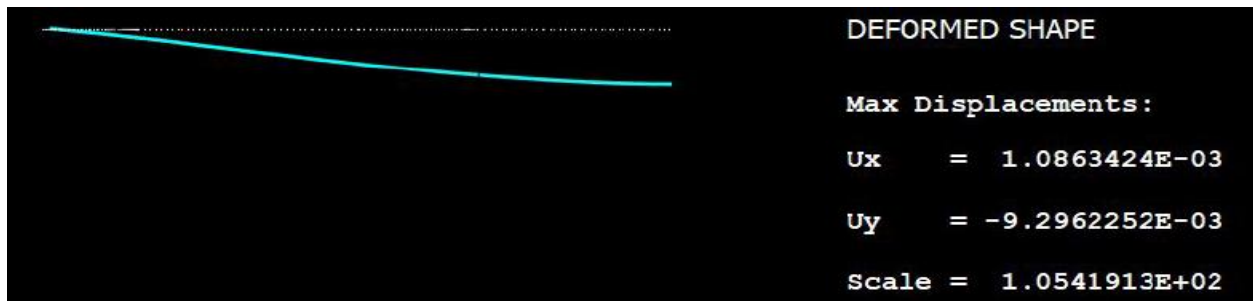
**Figure 15.21:** Stress comparison –CST vs. Frame Code – LC2 without slab



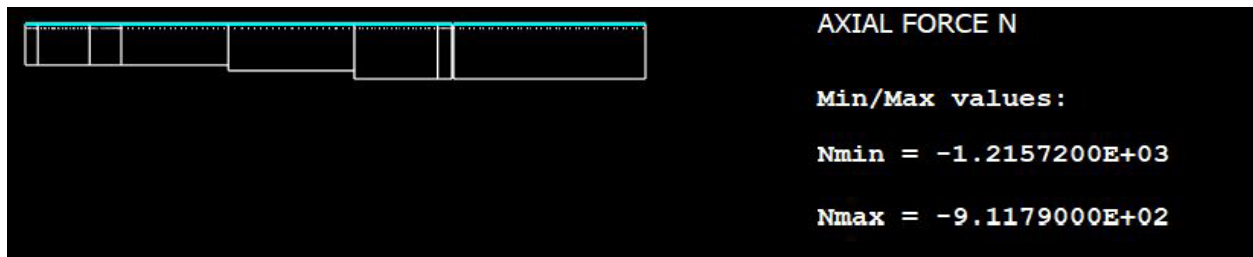
| Displacement [mm]   |               |             |          |
|---------------------|---------------|-------------|----------|
|                     | Frame Element | CST Element | Error[%] |
| Midspan Deflection  | 15.27         | 16.49       | 7.4 %    |
| Roller Displacement | 1.58          | 1.66        | 5.1 %    |

**Table 15.9:** Results – Load case 1 – Without Slab

### 15.4.3. Load Case 1 - With Slab



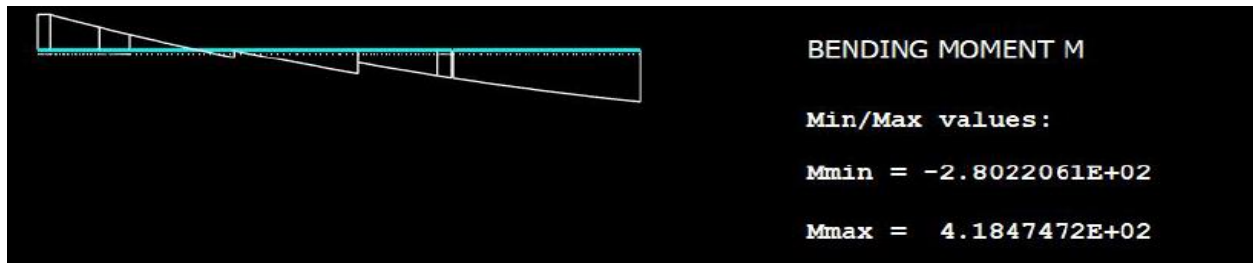
(a) Deformed Shape



(b) Axial Force Diagram

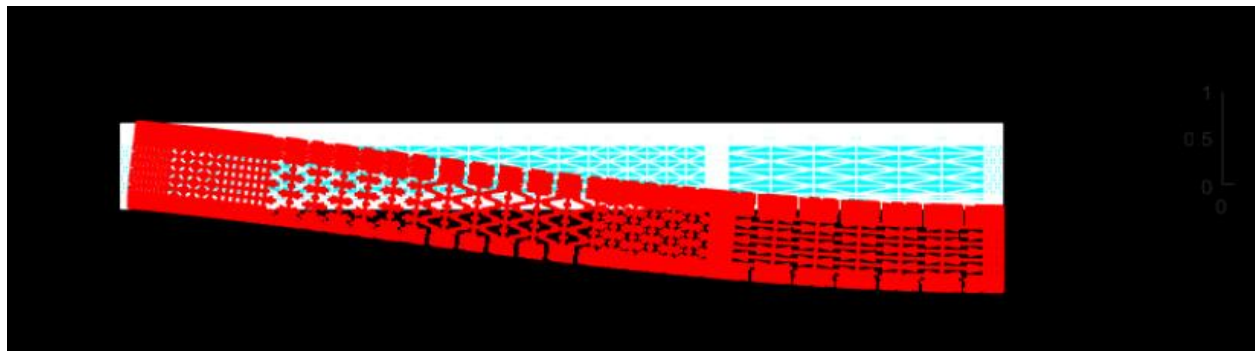


(c) Shear Force Diagram

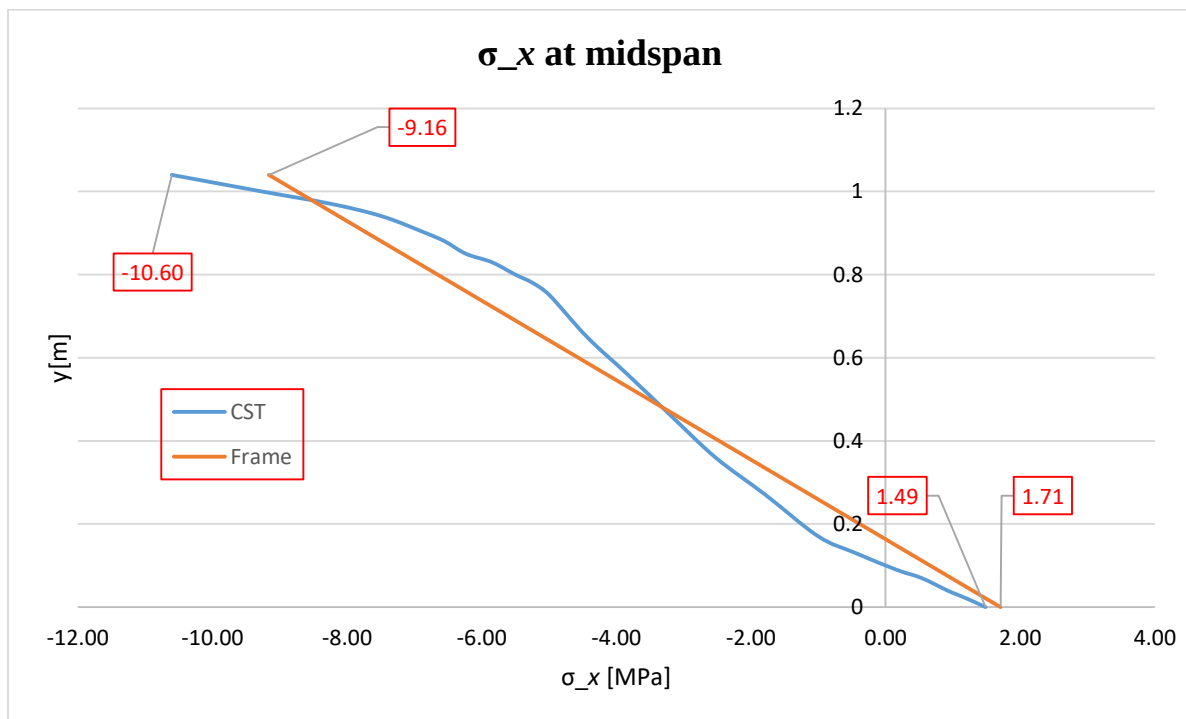


(d) Bending Moment Diagram

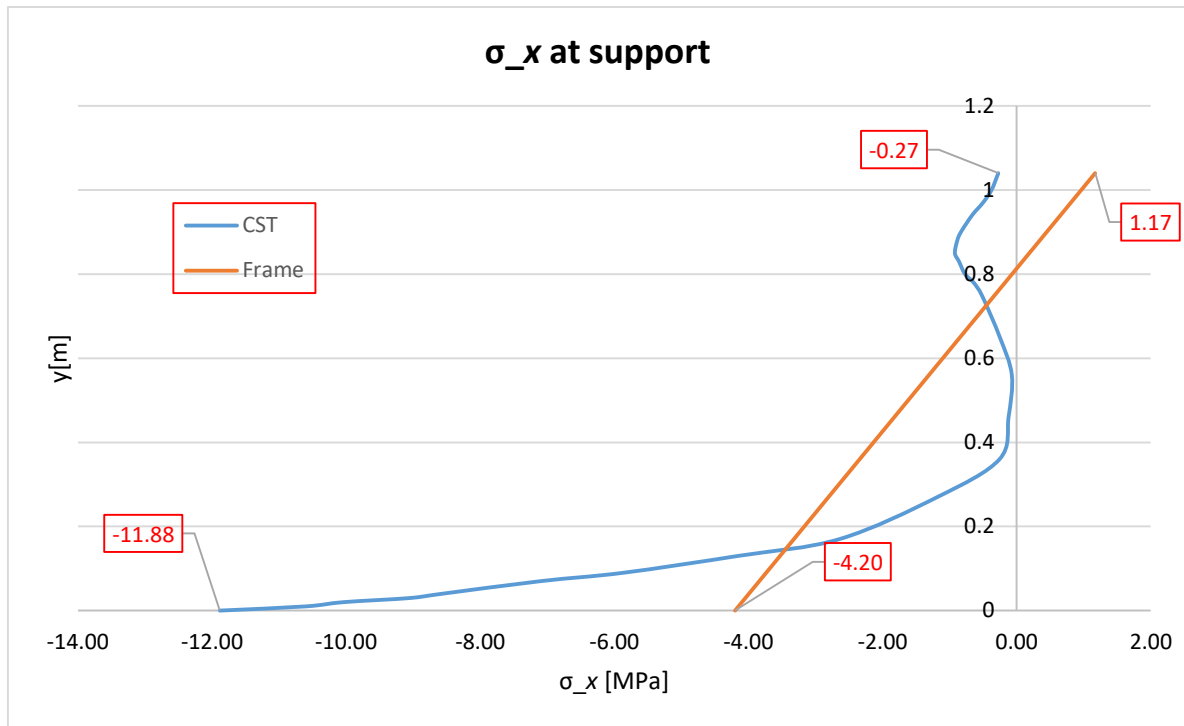
**Figure 15.22:** Frame Code Results – LC1 with slab



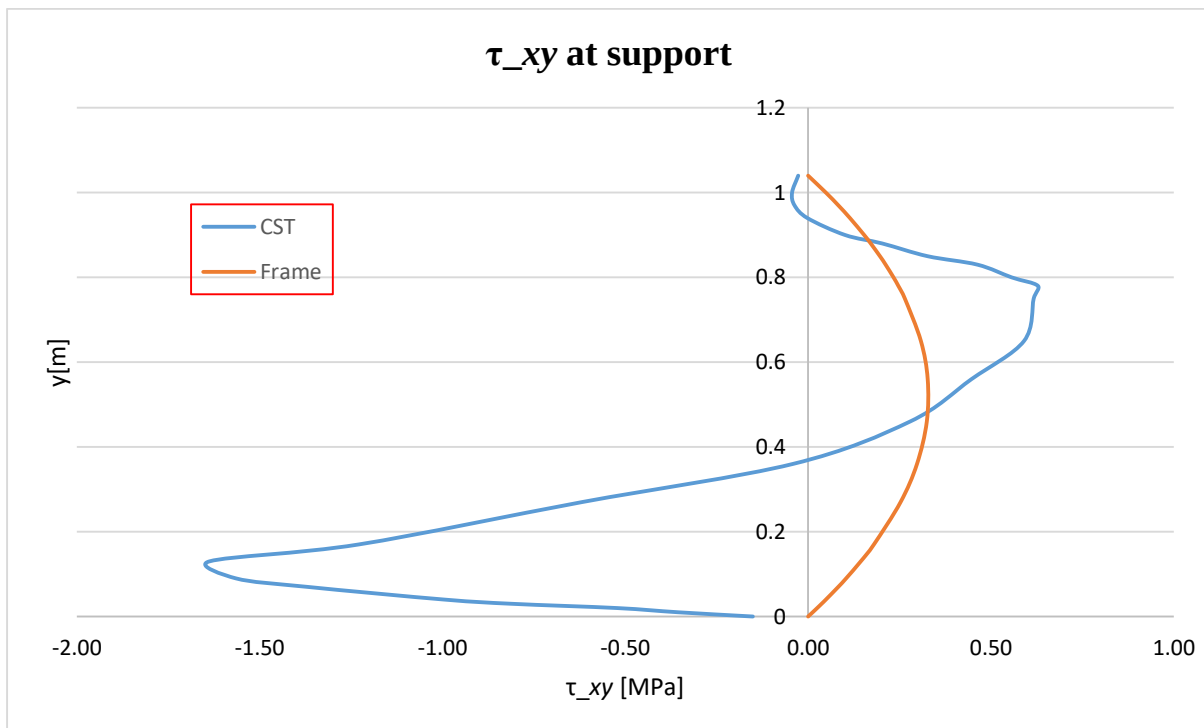
**Figure 15.23:** CST Code –Deformed Shape – LC1 with slab



(a)  $\sigma_x$  at midspan [MPa]



(b)  $\sigma_x$  at support [MPa]



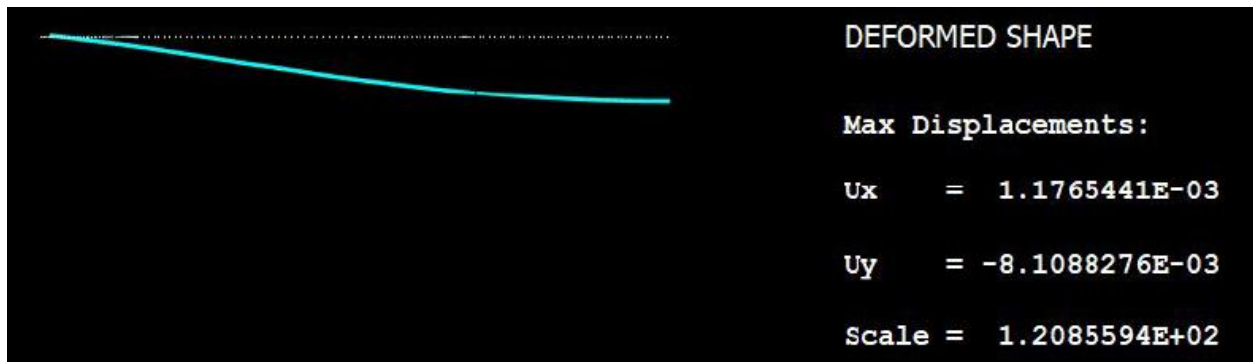
(c)  $\tau_{xy}$  at midspan [MPa]

**Figure 15.24:** Stress comparison –CST vs. Frame Code – LC1 with slab

| Displacement [mm]   |               |             |          |
|---------------------|---------------|-------------|----------|
|                     | Frame Element | CST Element | Error[%] |
| Midspan Deflection  | 10.07         | 9.52.       | -5.8 %   |
| Roller Displacement | 1.17          | 1.28        | 8.6 %    |

**Table 15.10:** Results – Load case 1 – With Slab

#### 15.4.4. Load Case 2 - With Slab



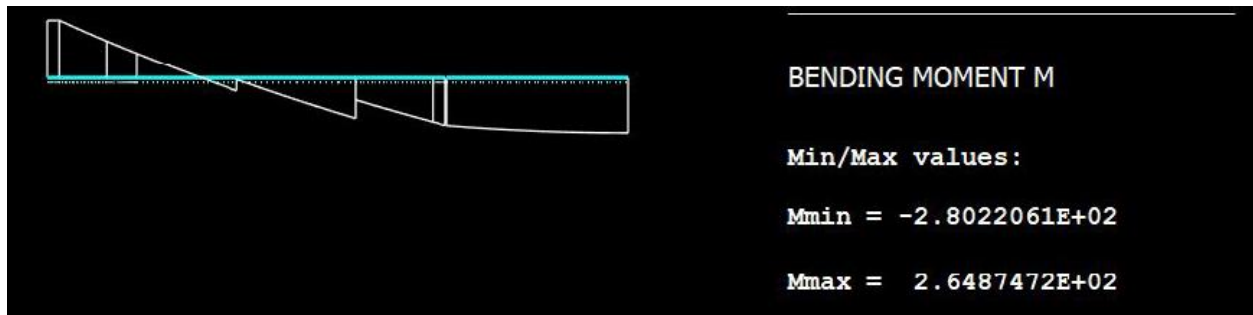
(a) Deformed Shape



(b) Axial Force Diagram



(c) Shear Force Diagram



(d) Bending Moment Diagram

Figure 15.25: Frame Code Results – LC2 with slab

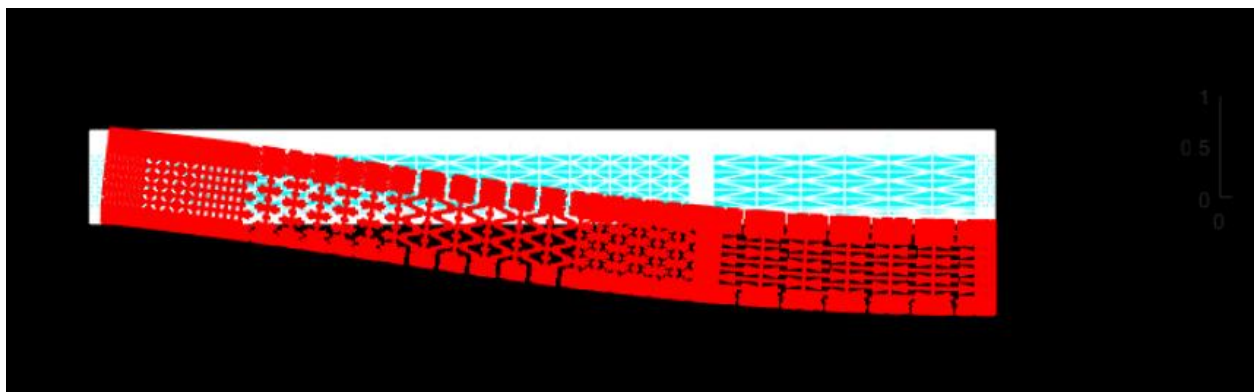
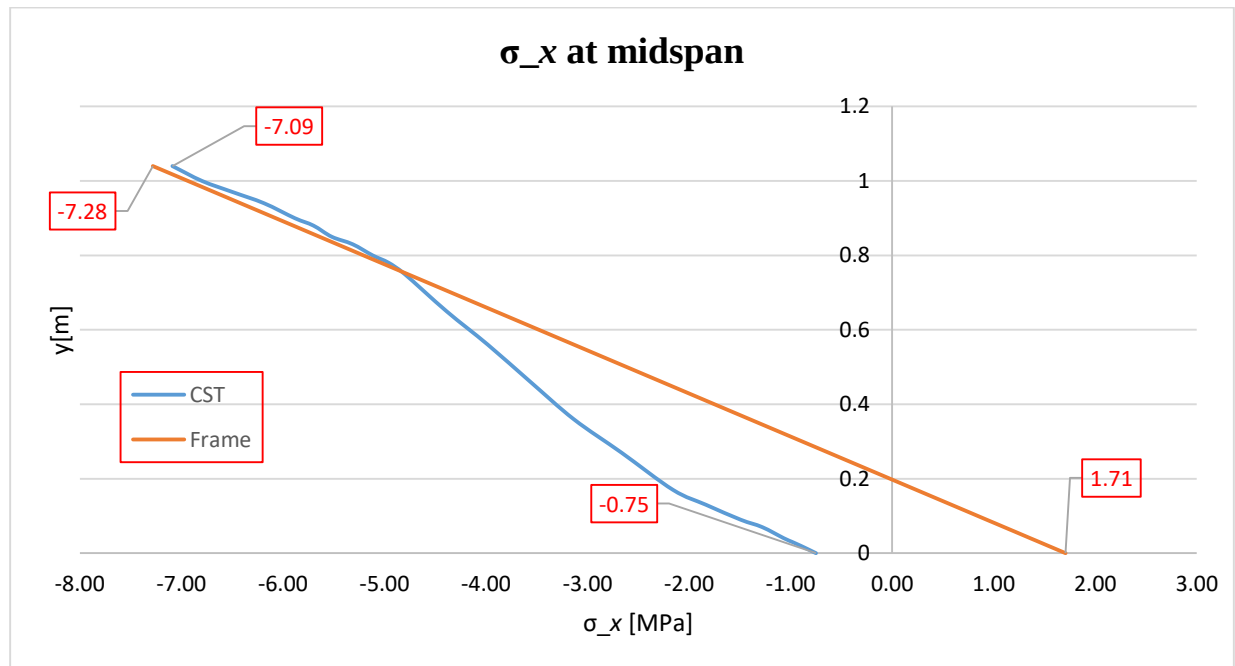
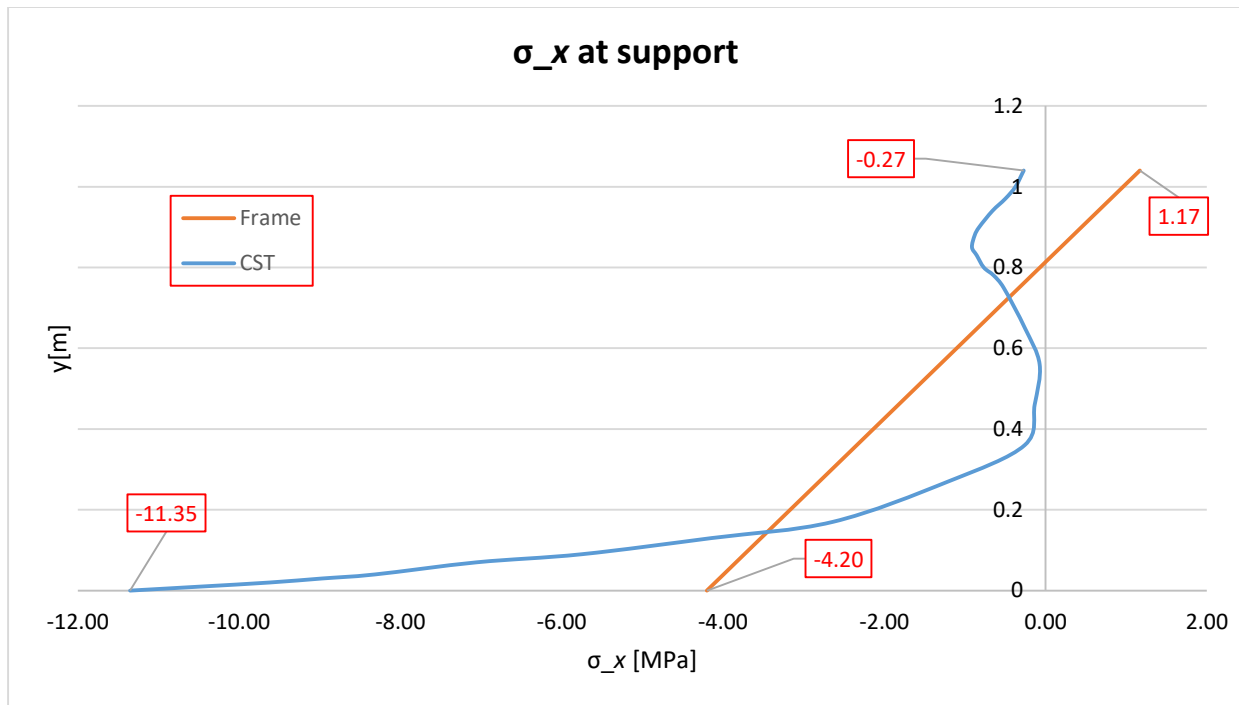


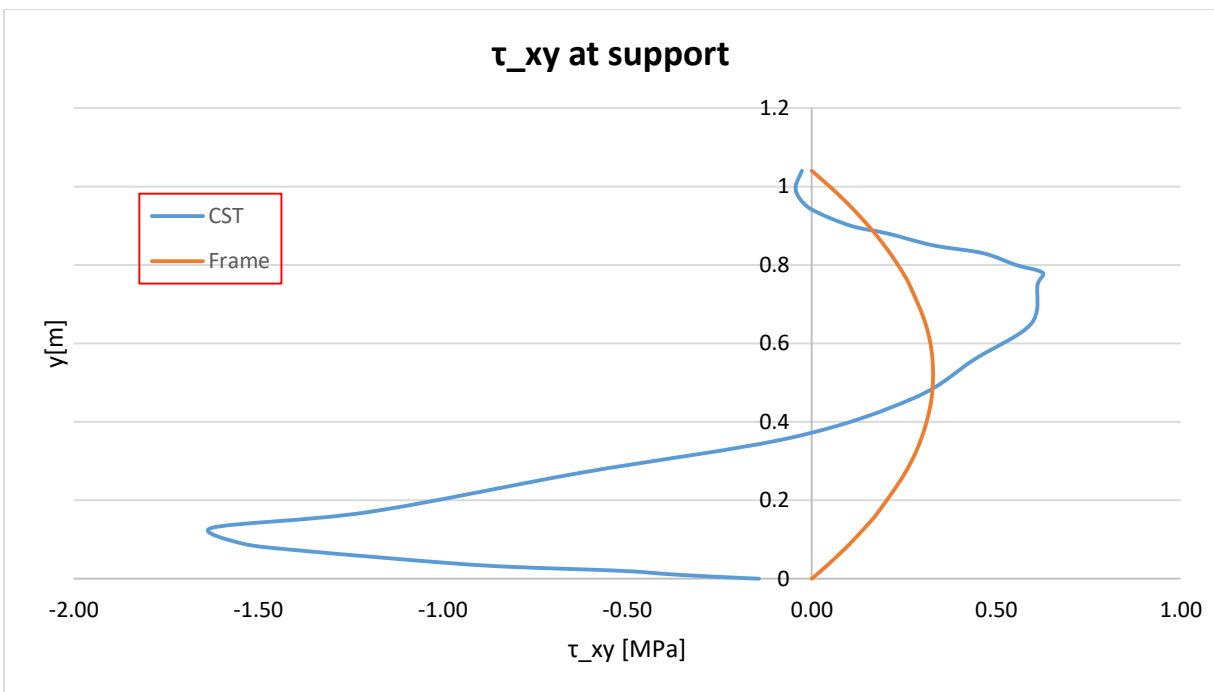
Figure 15.26: CST Code –Deformed Shape – LC2 with slab



(a)  $\sigma_x$  at midspan [MPa]



(b)  $\sigma_x$  at support [MPa]



(c)  $\tau_{xy}$  at midspan [MPa]

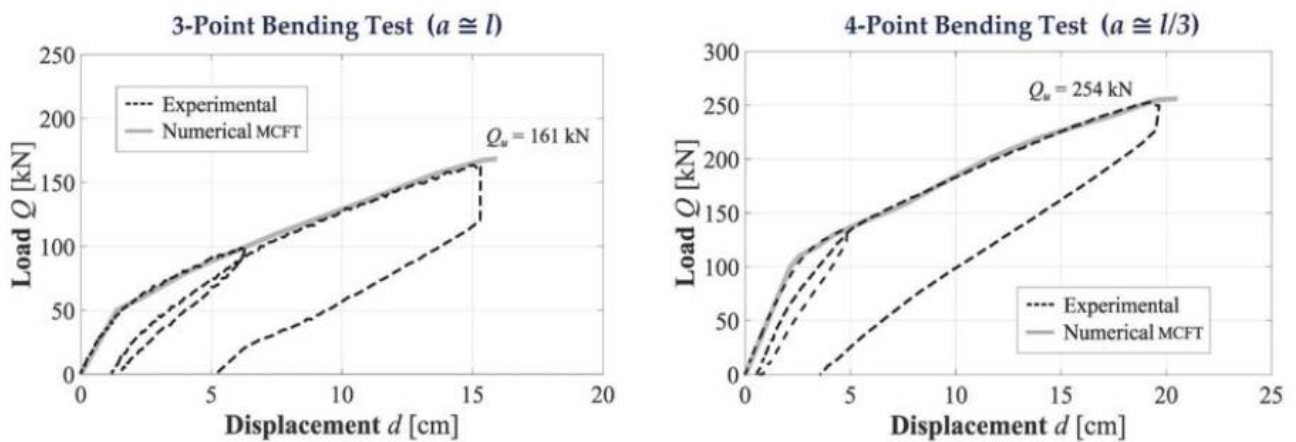
**Figure 15.27:** Stress comparison –CST vs. Frame Code – LC2 with slab

| Displacement [mm]   |               |             |          |
|---------------------|---------------|-------------|----------|
|                     | Frame Element | CST Element | Error[%] |
| Midspan Deflection  | 8.11          | 7.37        | -10.1 %  |
| Roller Displacement | 1.17          | 1.28        | 8.6 %    |

**Table 15.11:** Results – Load case 2 – With Slab

### 15.5 Comparison with experimental results

The experimental results were available for the cases of 'Load Case 1 - With Slab' and 'Load Case 2 - With Slab' from the shared research article<sup>1</sup> and lectures notes in graphical form. The graphs are depicted below.



**Figure 15.28:** Experimental Results - LC1 and LC2 With Slab Section

The results obtained from the Frame and CST models are compared with experimental results in the below table.

|                         | Experimental | Frame Model | CST Model |
|-------------------------|--------------|-------------|-----------|
| Midspan Deflection [mm] | 15.51        | 10.07       | 9.52      |
| Error [%]               | -            | 34.1%       | 38.6 %    |

**Table 15.12:** Comparison - Load Case 1 (With top slab) vs. Experimental Results

|                         | Experimental | Frame Model | CST Model |
|-------------------------|--------------|-------------|-----------|
| Midspan Deflection [mm] | 10.36        | 8.11        | 7.37      |
| Error [%]               | -            | 21.68%      | 28.8 %    |

**Table 15.13:** Comparison - Load Case 2 (With top slab) vs. Experimental Results

In this project, the results are from the analysis of using the FRAME code and CST code to the given experimental test data of the BRIDGE—50 research project. It can be observed that the experimental results of midspan deflection are much higher than that of FRAME and CST Model. The deviation of model results can be attributed to modelling strategy adopted for I-beam and Top Slab connection.

In the beam and CST model, the I-beam and Top Slab are assumed to be rigidly connected as monolithic element. However, it is evident from the given data that the slab was cast-insitu and therefore it is not rigidly connected to the I-beam. The consequence of this fact is that the actual deflection shall be much higher than the model deflection as the cross-section is not rigid as modelled in the Frame and CST code.

In order to tackle this behavior and to define a true model, the authors suggest to develop a model incorporating vertical elastic links between top of I-beam and bottom of Top slab. In this way, a representative model can be developed which correctly reflects the actual behavior/interaction of I-beam and Top Slab.



## 15.6 Discussion and Conclusion

Following conclusions can be drawn from the results of FRAME and CST codes.

- The displacement results of the both codes are comparable
- The stress results from the both codes differentiate significantly
- Considering the above two conclusions, it can be inferred that FRAME CODE has been able to capture the global behavior of the structure but fails to reflect the local phenomena
- The reason for above point can be due to the fact that, in the CST CODE, the section analysed are the ones where we have some sort of disturbance (load applied for midspan, reaction force of the support and prestressing for the support cross-section). The beam element formulation gives us the global behaviour of the element since, to compute stresses, we look at the effect of the generalized forces (axial, shear and bending moment) at that point. With the CST model instead we are able to capture better some localized disturbances as prestressing as we can see, for example in the graphs of  $\sigma_x$  on the support for the top part of the section where there's a localized compressive area given by the top prestressing cables
- The top slab provides the extra stiffness to the cross-section allowing lower deflection as compared to the PC beam only case
- The experimental results vary significantly from the model results (both FRAME and CST). The possible reason for the deviation is the treatment of the connection between the top slab and the beam. The both model consider the connection to be rigid whereas the connection is characterized by vertical elastic connection with a finite stiffness. Due to this, the experimental results show higher deflection as compared to the models due to the lower stiffness.

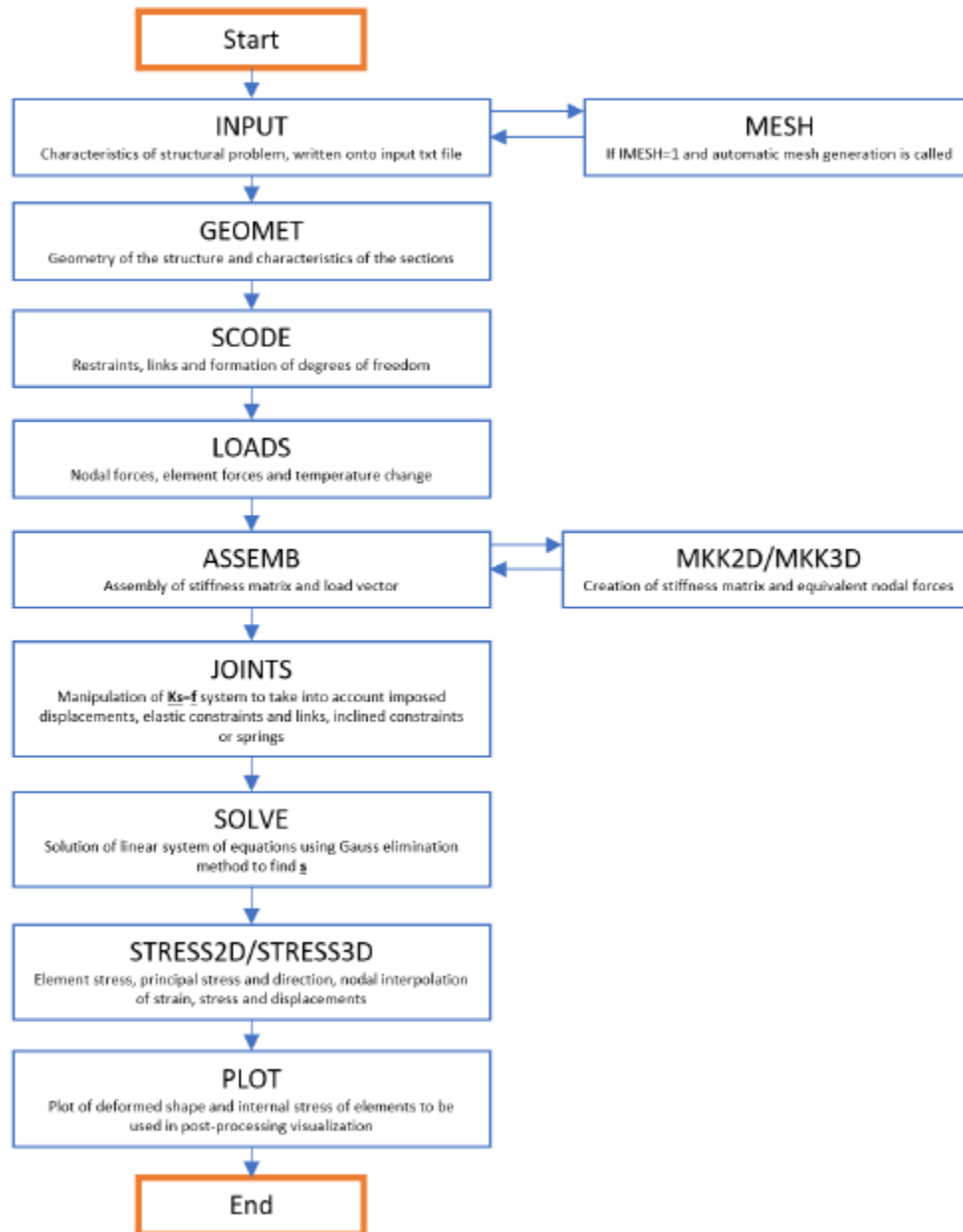
## **16. References**

Biondini, F. (2022). Computational Structural Analysis: Lecture Notes.

Marshal, A. C. (2010). Introduction to Computer Programming using Fortran95.

# Annex

## A1: Flowchart of the program



## A2: User Manual

The user manual presented here serves as a step-by-step guide for generating the results pertaining to the analysis of 2D frame structures. The files required and generated by the program are listed as follows:

- **input.txt:** Contains the description of the structural system
- **output.txt:** Stores the results produced from the analysis
- **plot.dat:** Enables visualisation of the results

The process for developing the input file is comprehensively detailed in the following section.

### B.1 Input

The input file provides the program with all the specifications associated to the structural model, and hence plays a critical role in the analysis. To begin with, the user has to specify a title for the problem along with the units of measurement adopted. Consistency in the units of measurements is pivotal for obtaining the correct results. Following this, the focus is laid upon the definition of the physical model and the parameters required are as noted below. It is worth mentioning that for each input line, <·> denotes a new line of input to be provided.

**OPTIMIZATION PARAMETERS <CHECK> <EFACTOR, RR0, ER, ALPHAN, RRMAX, RMVMAX, MAXITER>**

In case evolutionary structural optimization is desired, a value of 1 is input. Following this, the values of the optimization parameters, such as the EFactor for the material removal, the initial rejection ratio, the evolutionary rate, the steady state factor, the maximum rejection ratio, the maximum removed volume of the material and the maximum number of iterations has to be defined.

**TYPE OF MESH (0 = MANUAL, 1 = AUTOMATIC) <IMESH>**

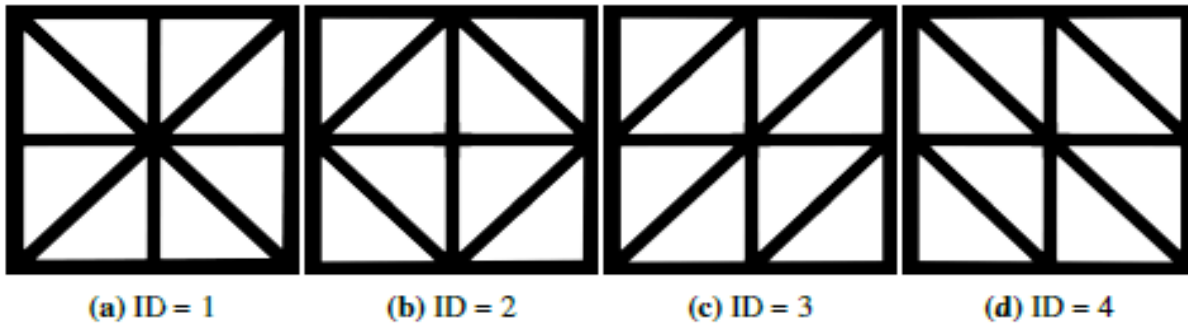
The choice of automatic or manual meshing is done by stating the relative codes as mentioned in the input line.

**SHAPE OF THE MESH (1,2,3,4) <IDMESH>**

The code allows the implementation of four different shape of the mesh and the user has to input the code pertaining to them as illustrated in the figure below.

#### ORIGIN OF THE MESH $\langle x_0, y_0 \rangle$

The user has to provide the coordinates of the origin of the mesh.



#### DIMENSION OF THE MESH $\langle \text{WIDTH, HEIGHT, NSUBX, NSUBY} \rangle$

The width, height, and the number of subdivisions of the domain in the x and y direction, respectively.

#### TYPE OF PROBLEM $\langle \text{ICODE} \rangle$ (0=PLANE STRESS,1=PLANE STRAIN)

The code corresponding to the type of formulation the user desires should be indicated here.

#### WEIGHT (0=ARITHMETIC,1=VOLUME,2=ENERGY)

The weight type to be adopted for the computation of the average nodal strains and stresses are noted here. The value to be given as input in accordance to the various weight types are as mentioned in the input line.

#### NODES $\langle \text{NNODE} \rangle \langle \text{IDNODE, X, Y} \rangle$

If the meshing is manual, the user provides the information on the nodes that constitute the structural model. It should be noted that besides the nodes defining the geometry of the structure, additional nodes that are needed to capture specific features from the analysis should be accounted for. Firstly, the total number of nodes are stated and then in subsequent lines, the ID and coordinate of each node is to be specified.

### ELEMENT CHARACTERISTICS <NTYPE> <IDTYPE, YOUNG, POISSON, THERMAL, WEIGHT, THICKNESS>

The total number of distinct materials to be assigned to the structure are defined first. Then on the next line, the ID representative of each material is allocated, along with the characteristics such as the Young's modulus, the Poisson's ratio, the coefficient of thermal expansion, the specific weight, and the thickness. The coefficient of thermal expansion and the self weight can be neglected, if necessary, by providing null values.

### ELEMENTS <NELE> <IDELE, NODE1, NODE2, NODE 3, IDTYPE>

For the case of manual meshing, the number of elements constituting the structural model are prescribed first. Subsequently, an ID is designated for each element with information on the ID of the three nodes connecting it. Furthermore, the user now assigns the ID of the characteristics corresponding to each element.

### RESTRAINTS <NRES> <NNODE, DIRECTION>

The total number of restraints imposed in the structure, bearing in mind that each direction to be restricted counts as an individual restraint, is provided here. Next, the restricted node ID and the ID relative to the restricted direction is given.

### LINKS <NLINK> <MASTER, SLAVE, DIRECTION>

The number of master slave links connecting certain elements have to be indicated here. Additionally, the ID of the master and the slave node, and the corresponding direction of this link has to be mentioned.

### NODAL LOADS <NLOADS> <NODE, DIR, FORCE>

The number of nodal loads that the structure is subjected to is provided first. Then, the associated node ID, direction of load application and its value is stated in the subsequent line. It is worth noting that the value of the load must be provided in accordance with the sign convention adopted, i.e., positive in the upward direction and the right side.

### SURFACE LOADS <NLOADS> <NODE1, NODE2, PX1, PX2, PY1, PY2>

The total number of surface loads acting on the model should be given followed by the node ID in the next line. The code allows for a linear variation of the surface loads in the x and y direction. As a result, the variable (if required) intensity at either ends of the element in the x and y direction should be specified.

#### **THERMAL LOAD <NTLOADS> <IDELE, DT>**

The number of thermal loads present in the structure are indicated here along with the ID of the corresponding element and the value of the temperature differential experienced by it.

#### **ELASTIC RESTRAINTS <NRES> <NODE, DIR, STIFFNESS>**

The total number of elastic restraints, such as springs, are given first. Following this, the associated node ID, the direction of the restraint, and the stiffness of the elastic restraint has to be specified.

#### **ELASTIC LINKS <NLINKS> <NODE1, NODE2, DIR, STIFFNESS>**

The number of elastic restraints present in the system, along with the ID of the nodes connected to the links, the direction of the connection, and the stiffness value should be noted here.

#### **IMPOSED ABSOLUTE DISPLACEMENT <NADIS> <NODE, DIR, DISPL>**

The total number of imposed absolute displacements that the structural system is subjected to is quantified first. Then, the ID of the node that the absolute displacement is imposed on is mentioned, along with the direction, and the value of this displacement.

#### **IMPOSED RELATIVE DISPLACEMENT <NRDIS> <NODE1, NODE2, DIR, DISPL>**

The number of imposed relative displacements between two nodes is specified, followed by the associated node IDs, the direction, and the value of the relative displacement.

#### **INCLINED JOINTS <NINCL> <IDNODE, BETA, ITYPE, ADIS>**

The total number of inclined joints present in the structural model is noted here. The user has to provide the ID of the associated node, accompanied by information pertaining to the angle of inclination of the constraint, the type of the constraint (0 if elastic and 1 for imposed displacement), and the corresponding value of stiffness or displacement for an elastic constraint and an imposed displacement, respectively.

## A3: Fortran Code

```
MODULE GLOBALVAR
  IMPLICIT double precision (A -H ,O -Z )

  ! GEOMET
  INTEGER :: NNODE , NELE , NMAT , ATYPE , MTYPE, SL
  REAL (2) , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: COORD , CMAT
  INTEGER , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: INC

  ! SCODE
  INTEGER :: NDOF
  INTEGER , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: IDOF

  ! LOADS
  REAL (2) , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: ELOADS
  REAL (2) , DIMENSION (: ) , ALLOCATABLE , SAVE :: VLOADS , THERM

  ! ASSEMBLY
  REAL (2) , DIMENSION (6 ,6) :: ST
  REAL (2) , DIMENSION (: ) , ALLOCATABLE , SAVE :: EQFG
  REAL (2) , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: VK

  ! SOLVE
  REAL (2) , DIMENSION (: ) , ALLOCATABLE , SAVE :: VDISP

  ! STRESS
  REAL (2) , DIMENSION (: ,:) , ALLOCATABLE , SAVE :: EPSNOD , SIGNOD
  REAL (2) , DIMENSION (: ) , ALLOCATABLE , SAVE :: SUMWEIGHT
END MODULE GLOBALVAR

PROGRAM CST
  USE GLOBALVAR
  CHARACTER ( LEN =50) :: TITLE, UNIT1, UNIT2 , REC
  OPEN (11, FILE ="PROJECT2_LC2_S.txt", STATUS = 'UNKNOWN') !11 = input file
  OPEN (12, FILE ="OUTPUT_PROJECT2_LC2_S.txt", STATUS = 'UNKNOWN') !12 =
output file
  OPEN (13, FILE ="MESH2_FINAL_S.txt", STATUS = 'UNKNOWN') !13 = mesh file
  OPEN (9 , FILE ="utility.txt", STATUS = 'UNKNOWN') !9 = temporary file
  READ (11 ,*) TITLE , UNIT1 , UNIT2 ! read 1st line of input file
  WRITE (12 , '( / ,10 X ,45(" -")) ')
  WRITE (12 , '(15 X , A25 ,A6 , A6 ) ') TITLE , UNIT1 , UNIT2
  WRITE (12 , '(10 X ,45(" -")) ')
  READ (11 ,*) REC
  READ (11 ,*) REC
  READ (11 ,*) REC
  READ (11 ,*) ATYPE
  IF( ATYPE ==0) THEN
  WRITE (12 , '( / ,2X , " PLANE ? STRESS ? ANALYSIS ") ')
  ELSEIF ( ATYPE ==1) THEN
  WRITE (12 , '( / ,2X , " PLANE ? STRAIN ? ANALYSIS ") ')
```



```

END IF
CALL GEOMET
CALL SCODE
CALL LOADS
CALL ASSEMB
CALL SOLVE
IF( ATYPE ==0 .OR. ATYPE ==1) THEN
CALL STRESS2D
CALL PLOT2D
END IF
CLOSE (11)
CLOSE (12)
CLOSE (13)
CLOSE (9)
END PROGRAM CST

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!
! !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!! SUBROUTINES
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE GEOMET
SUBROUTINE GEOMET
USE GLOBALVAR
INTEGER :: IDNODE , IDELE , i , j , ICODE
CHARACTER ( LEN =15) , DIMENSION (:) , ALLOCATABLE , SAVE :: IDMAT

CHARACTER ( LEN =50) :: REC
DO I =1 , 9
READ (11 ,*) REC
END DO
READ (11 ,*) N
READ (11 ,*) REC
DO I =1 , N
READ (11 ,*) REC
END DO
READ (11 ,*) REC
READ (11 ,*) N
READ (11 ,*) REC
DO I =1 , N
READ (11 ,*) REC
END DO
READ (11 ,*) REC
READ (11 ,*) REC

WRITE (12 , ' (/ , 3X , 12 (" -") ) ')
WRITE (12 , ' (1 X , A12 ) ') " GEOMETRY "
WRITE (12 , ' (3 X , 12 (" -") , / ) ')

```

```

! READING / WRITING MATERIALS
READ (11 ,*) NMAT
READ (11 ,*) REC ! SKIP LABELS
WRITE (12 , ' (2 X , " Number ? of ? materials ? =" , 1 X , I3 , /) ') NMAT
WRITE (12 , ' (3 X , 6 A15 , /) ') 'MATERIAL', 'E', 'v', 'alpha', 'gamma',
'Thickness'
ALLOCATE ( CMAT ( NMAT , 7) ) !.....
ALLOCATE ( IDMAT ( NMAT ) )
DO i =1 , NMAT
READ (11 ,*) IDMAT ( i ) , CMAT ( i , :)
!WRITE (12 , ' (5 X , I5 , F5.5 , F5.5 , E15 .2 , F15 .2 , F15 .6 , 2 F15 .4) ')
IDMAT ( i) , CMAT ( i , :) !....
WRITE (12 , *) IDMAT ( i) , CMAT ( i , :) !....
ICODE =0
IF( CMAT ( i , 3) <=0) ICODE =1
IF ( ( CMAT ( i , 4) <0) .OR. ( CMAT ( i , 4) >0.5) ) ICODE =2
IF( CMAT ( i , 5) <0) ICODE =3
IF( CMAT ( i , 6) <0) ICODE =4
IF( ICODE >0) THEN
WRITE (12 , ' (/ , 2X , " ERROR ? IN ? MATERIAL " , 1X , I1 , 1X , " , PROP ." , 1
X , I1 ) ') i , ICODE
END IF
END DO
WRITE (12 , ' (/ , 3X , 90 ( " _ " ) , /) ')
READ (11 ,*) REC

! READING / WRITING AND CONNECT . MATRIX

READ (13 ,*) REC
READ (13 ,*) REC
READ (13 ,*) REC
READ (13 ,*) REC
READ (13 ,*) NNODE
READ (13 ,*) REC
WRITE (12 , ' (2 X , " Number ? of ? nodes ? =" , 1X , I3 , /) ') NNODE
WRITE (12 , ' (3 X , 3 A15 , /) ') 'Node' , 'Coord .? X' , 'Coord .?Y'
ALLOCATE ( COORD ( NNODE , 2) )
DO i =1 , NNODE
READ (13 ,*) IDNODE , COORD ( IDNODE , :)
WRITE (12 , ' (3 X , I15 , 2 F15 .2) ') IDNODE , COORD ( IDNODE , :)
END DO
WRITE (12 , ' (/ , 3X , 90 ( " _ " ) , /) ')
READ (13 ,*) REC

! CONNECTIVITY MATRIX
READ (13 ,*) REC
READ (13 ,*) NELE
READ (13 ,*) REC
IF( NELE <1) THEN
WRITE (12 , ' (/ , 1X , " DATA ? ERROR ? -? INCORRECT ? NUMBER ? OF ? ELEMENTS
" ) ')
WRITE (* , ' (/ , 1X , " DATA ? ERROR ? -? INCORRECT ? NUMBER ? OF ? ELEMENTS
" ) ')

```

```

STOP
RETURN
END IF
WRITE (12 , '(2 X , " Number ? of ? elements ? =" , 1X , I3 , /)') NELE
WRITE (12 , '(3 X , 5 A15 , /)') 'Element', 'Node1', 'Node2', 'Node3' , 'ID
? material'
ALLOCATE ( INC ( NELE , 4))
DO i =1 , NELE
READ (13 , *) IDELE , ( INC (i , j), j =1 , 4)
WRITE (12 , '(3 X , 5 I15 ) ') IDELE , INC (i , :)
END DO
WRITE (12 , '(/ , 3X , 90 (" _ " ) , /)')
RETURN

END SUBROUTINE GEOMET

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE CODE
SUBROUTINE SCORE
USE GLOBALVAR
INTEGER :: i , j , NRES , IDNODE , JDIR
CHARACTER ( LEN =80) :: REC
ALLOCATE ( IDOF ( NNODE , 2))
IDOF (: , :)=0 ! initialization

! RESTRAINS
READ (11 , *) REC
READ (11 , *) REC
READ (11 , *) REC
READ (11 , *) REC

READ (11 , *) REC
READ (11 , *) NRES
READ (11 , *) REC
WRITE (12 , '(2 X , " Number ? of ? restrains ? =" , 1 X , I3 , /) ') NRES
WRITE (12 , '(3 X , 2 A15 , /) ') ' NODE ' , ' DIRECTION '
DO i =1 , NRES
READ (11 , *) IDNODE , JDIR
WRITE (12 , '(3 X , 2 I15 ) ') IDNODE , JDIR
IDOF ( IDNODE , JDIR )= -1
END DO
WRITE (12 , '(/ , 3X , 90 (" _ " ) , /) ')
READ (11 , *) REC

! DOF NUMBERING
NDOF =0
WRITE (12 , '(/ , 2X , 12 (" -") ) ')
WRITE (12 , '(1 X , A12 ) ') " DOF ? MATRIX "
WRITE (12 , '(2 X , 12 (" -") , /) ')
WRITE (12 , '(3 A15 ) ') ' NODE ' , 'Ux ' , 'Uy '

```

```

EXT : DO i =1 , NNODE !ext
INT: DO j =1 ,2 ! int
IF( IDOF (i ,j ) <0) CYCLE INT
IF( IDOF (i ,j )==0) THEN
NDOF = NDOF +1
IDOF (i ,j) = NDOF
ELSE
IDNODE = IDOF (i ,j )
IDOF (i ,j) = IDOF ( IDNODE , j)
END IF
END DO INT
WRITE (12 , '(4 I15 )') I , IDOF (i ,:)
END DO EXT
WRITE (12 , '( / ,3X ,90(" _ " ) ,/ )')
RETURN
END SUBROUTINE SCODE

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE LOADS
SUBROUTINE LOADS
USE GLOBALVAR
INTEGER :: i , NNL , IDNODE , JDIR , JDOF , NEL , N1 , N2 , NTERM
REAL (2) :: NFORCE , ELOAD (4) , AL , AREA , B (3) , C (3)
CHARACTER ( LEN =80) :: REC

WRITE (12 , '( / ,2X ,27(" -" ) )')
WRITE (12 , '(1 X , A27 )') " NODAL ? AND ? ELEMENTAL ? NODES "
WRITE (12 , '(2 X ,27(" -" ) ,/ )')

READ (11 ,*) REC
READ (11 ,*) REC
READ (11 ,*) REC

ALLOCATE ( VLOADS ( NDOF ))
ALLOCATE ( THERM ( NELE ))
VLOADS (:)=0.
THERM (:)=0.

! READING / WRITING NODAL LOADS
READ (11 ,*) REC
READ (11 ,*) NNL
READ (11 ,*) REC
IF( NNL <0) THEN
WRITE (12 , '(2 X , " Number ? of ? nodal ? loads ? incorrect " )')
WRITE (* , '(2X , " Number ? of ? nodal ? loads ? incorrect " )')
STOP
RETURN
END IF
IF( NNL >0) THEN
WRITE (12 , '(2 X , " Number ? of ? nodal ? loads ? =" ,1X , I3 ,/ )') NNL

```

```

WRITE (12 , '(3 X ,3 A15 ,/) ') ' NODE ' , 'DIRECTION ' , 'FORCE '
DO i =1 , NNL
READ (11 ,*) IDNODE , JDIR , NFORCE
WRITE (12 , '(3X ,2 I15 , F15 .2)') IDNODE , JDIR , NFORCE
JDOF = IDOF ( IDNODE , JDIR )
VLOADS ( JDOF ) = VLOADS ( JDOF ) + NFORCE
END DO
END IF
IF( NNL ==0) THEN
WRITE (12 , '(2 X , " No ? nodal ? forces ")')
END IF
WRITE (12 , '(/ ,3X ,90(" _ " ) ,/ )')

! READING / WRITING ELEMENTS LOADS
READ (11 ,*) REC
READ (11 ,*) NEL
READ (11 ,*) REC
IF( NEL <0) THEN
WRITE (12 , '(1 X , " Number ? of ? element ? loads ? incorrect ")')
WRITE (* , '(1X , " Number ? of ? element ? loads ? incorrect ")')
STOP
RETURN
END IF
IF( NEL >0) THEN
WRITE (12 , '(2 X , " Number ? of ? surface ? loads ? =" ,1X ,I3 ,/ )') NEL
WRITE (12 , '(3 X ,6 A15 ,/) ') ' Node1', ' Node2 ', ' Px1 ', ' Px2 ', '
Py1', 'Py2'
DO i =1 , NEL
READ (11 ,*) N1 , N2 , ELOAD
WRITE (12 , '(3X ,2 I15 ,4 F15 .2)') N1 , N2 , ELOAD
X1 = COORD (N1 ,1)
X2 = COORD (N2 ,1)
Y1 = COORD (N1 ,2)
Y2 = COORD (N2 ,2)
AL = SQRT ( ( X2 - X1 )**2.+( Y2 - Y1 )**2.)
JDOF1 = IDOF (N1 ,1)
JDOF2 = IDOF (N1 ,2)
JDOF3 = IDOF (N2 ,1)
JDOF4 = IDOF (N2 ,2)
PX1 = ELOAD (1)
PX2 = ELOAD (2)
PY1 = ELOAD (3)
PY2 = ELOAD (4)
IF( JDOF1 >0) VLOADS ( JDOF1 ) = VLOADS ( JDOF1 ) + AL /6.*(2.* PX1 + PX2 )
IF( JDOF3 >0) VLOADS ( JDOF3 ) = VLOADS ( JDOF3 ) + AL /6.*(2.* PX2 + PX1 )
IF( JDOF2 >0) VLOADS ( JDOF2 ) = VLOADS ( JDOF2 ) + AL /6.*(2.* PY1 + PY2 )
IF( JDOF4 >0) VLOADS ( JDOF4 ) = VLOADS ( JDOF4 ) + AL /6.*(2.* PY2 + PY1 )
END DO
END IF
IF( NEL ==0) THEN
WRITE (12 , '(2 X , " No ? surface ? forces ") ')
READ (11 ,*) REC
END IF

```

```

WRITE (12 , ' (/ , 3X , 90 (" _ " ) , /) ')
!SELF - WEIGHT
DO I =1 , NELE
IT = INC ( I , 4)
WEIGHT = CMAT ( IT , 6)
THICK = CMAT (IT , 7)
N1 = INC ( I , 1)
N2 = INC ( I , 2)
N3 = INC ( I , 3)
X1 = COORD (N1 , 1)
X2 = COORD (N2 , 1)
X3 = COORD (N3 , 1)
Y1 = COORD (N1 , 2)
Y2 = COORD (N2 , 2)
Y3 = COORD (N3 , 2)
A1 =( X2 *Y3 - X3 * Y2 )
A2 =( X3 *Y1 - X1 * Y3 )
A3 =( X1 *Y2 - X2 * Y1 )
AREA =( A1 + A2 + A3 )/2.
FORCE =- WEIGHT * THICK * AREA /3.
DO J =1 , 3
JDOF = IDOF ( INC ( I , J ) , 2)
IF ( JDOF >0) THEN
VLOADS ( JDOF )= VLOADS ( JDOF )+ FORCE
END IF
END DO
END DO
! READING / WRITING THERMAL LOADS
READ (11 ,*) NTERM
READ (11 ,*) REC
IF( NTERM <0) THEN
WRITE (12 , ' (1 X , " Number ? of ? thermal ? loads ? incorrect " ) ')
WRITE (* , ' (1X , " Number ? of ? thermal ? loads ? incorrect " ) ')
STOP
RETURN
END IF
IF( NTERM >0) THEN
WRITE (12 , ' (2 X , " Number ? of ? thermal ? loads ? =" , 1X , I3 , /) ')
NTERM
WRITE (12 , ' (3 X , 2 A15 , /) ') ' Element ' , 'Delta_T '
DO i =1 , NTERM
READ (11 ,*) IDELE , THERM
WRITE (12 , ' (3X , I15 , 5 F15 .2) ') IDELE , THERM
N1 = INC ( IDELE , 1)
N2 = INC ( IDELE , 2)
X1 = COORD (N1 , 1)
Y1 = COORD (N1 , 2)
X2 = COORD (N2 , 1)
Y2 = COORD (N2 , 2)
X3 = COORD (N3 , 1)
Y3 = COORD (N3 , 2)
B (1)= Y2 - Y3
B (2)= Y3 - Y1

```

```

B (3)= Y1 - Y2
C (1)= -( X2 - X3 )
C (2)= -( X3 - X1 )
C (3)= -( X1 - X2 )
IT = INC ( IDELE ,4)
EE = CMAT ( IT ,3)
POIS = CMAT (IT ,4)
ALPHA = CMAT (IT ,5)
IF ( ICODE ==0) THEN
CK =1. - POIS
ELSE
CK =(1.+ POIS ) *(1. -2.* POIS )
END IF
COEFF = EE * ALPHA * THERM ( IDELE ) * THICK / (2.* CK )
DO J =1 ,3
FX = COEFF *B( J)
FY = COEFF *C( J)
JX = IDOF ( INC ( IDELE , J ) ,1)
JY = IDOF ( INC ( IDELE , J ) ,2)
IF ( JX >0) VLOADS ( JX )= VLOADS ( JX )+ FX
IF ( JY >0) VLOADS ( JY )= VLOADS ( JY )+ FY
END DO
END DO
END IF
IF( NTERM ==0) THEN
WRITE (12 , ' (2 X , " No ? thermal ? forces " ) ' )
END IF
WRITE (12 , ' ( / , 3X , 90 ( " _ " ) , / ) ' )
READ (11 , *) REC
RETURN
END SUBROUTINE LOADS

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE MKK2D
SUBROUTINE MKK2D (NE)
USE GLOBALVAR
INTEGER :: N1 , N2 , N3 , IT
REAL (2) :: AA , EE , POIS , B1 , B2 , B3 , C1 , C2 , C3
REAL (2) :: TMP (3 ,6) , B (3 ,6) , D (3 ,3)

! GEOMET .
N1 = INC (NE ,1)
N2 = INC (NE ,2)
N3 = INC (NE ,3)
X1 = COORD (N1 ,1)
Y1 = COORD (N1 ,2)
X2 = COORD (N2 ,1)
Y2 = COORD (N2 ,2)
X3 = COORD (N3 ,1)
Y3 = COORD (N3 ,2)

```

```

! COMP . MATRIX B
A1 = ( X2 * Y3 - X3 * Y2 )
A2 = ( X3 * Y1 - X1 * Y3 )
A3 = ( X1 * Y2 - X2 * Y1 )
AA = ( A1 + A2 + A3 )/2.
B1 = Y2 - Y3
B2 = Y3 - Y1
B3 = Y1 - Y2
C1 = -( X2 - X3 )
C2 = -( X3 - X1 )
C3 = -( X1 - X2 )

B (: ,:)=0.
B (1 ,1) = B1 /(2.* AA )
B (2 ,2) = C1 /(2.* AA )
B (3 ,1) = C1 /(2.* AA )
B (3 ,2) = B1 /(2.* AA )
B (1 ,3) = B2 /(2.* AA )
B (2 ,4) = C2 /(2.* AA )
B (3 ,3) = C2 /(2.* AA )
B (3 ,4) = B2 /(2.* AA )
B (1 ,5) = B3 /(2.* AA )
B (2 ,6) = C3 /(2.* AA )
B (3 ,5) = C3 /(2.* AA )
B (3 ,6) = B3 /(2.* AA )

WRITE (9 , '(/ ,2X , " ELEMENT " ,1X , I5 )')NE
WRITE (9 , '(2X ,13 (" -" ) ,/)' )
WRITE (9 , '(3X , " COMPATIBILITY ? MATRIX " ,/)' )
DO I =1 ,3
WRITE (9 , '(6 E20 .10)' ) B(I ,:)
END DO

! CHAR . OF THE ELEMENT
IT = INC (NE ,4)
THICK = CMAT (IT ,7)
EE = CMAT ( IT ,3)
POIS = CMAT (IT ,4)
WEIGHT = CMAT ( IT ,6)
D (: ,:)=0.
IF ( ATYPE ==1) THEN
EE = EE /(1. - POIS **2.)
POIS = POIS /(1. - POIS )
END IF
D (1 ,1) = EE /(1. - POIS **2.)
D (1 ,2) = EE * POIS /(1. - POIS **2.)
D (2 ,1) = D (1 ,2)
D (2 ,2) = D (1 ,1)
D (3 ,3) = EE /(2.*(1.+ POIS ))
WRITE (9 , '(/ ,3X , " MATERIAL ? STIFFNESS ? MATRIX " ,/)' )
DO I =1 ,3
WRITE (9 , '(6 E20 .10)' ) D(I ,:)

```



```

END DO

TMP (: ,:)=0.
ST (: ,:)=0.
TMP = MATMUL (D ,B )
ST = MATMUL ( TRANSPOSE (B ) , TMP )
ST = ST * THICK * AA
WRITE (9 , ' (/ ,3X , " ELEMENT ? STIFFNESS ? MATRIX " , /) ')
DO I =1 , 6
WRITE (9 , ' (6 E20 .10) ') ST (I ,:)
END DO
WRITE (9 , ' (/ ,3X , 90 (" _ " ) , /) ')
RETURN
END SUBROUTINE MKK2D

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE ASSEMB
SUBROUTINE ASSEMB
USE GLOBALVAR
INTEGER :: i , j , NCODE (6) , NE , N1 , N2 , IC , JC
ALLOCATE ( VK ( NDOF , NDOF ) )
VK (: ,:) = 0.
DO NE =1 , NELE
CALL MKK2D (NE)
N1 = INC (NE ,1)
N2 = INC (NE ,2)
N3 = INC (NE ,3)
NCODE (1:2) = IDOF (N1 ,1:2)
NCODE (3:4) = IDOF (N2 ,1:2)
NCODE (5:6) = IDOF ( N3 ,1:2)
! assembly
EXT : DO i = 1 , 6
IC = NCODE (i)
IF (IC <0) CYCLE EXT
INT : DO j = 1 , 6
JC = NCODE (j)
IF (JC <0) CYCLE INT
VK (IC , JC ) = VK (IC , JC ) + ST (i ,j)
END DO INT
END DO EXT
END DO

! WRITE (12 , ' (/ ,2X , 23 (" -") ) ')
! WRITE (12 , ' (2X , " GLOBAL STIFFNESS MATRIX " ) ')
! WRITE (12 , ' (2X , 23 (" -") , /) ')
! DO i=1 , NDOF
! DO j=1 , NDOF
! WRITE (12 , ' (3X , E16 .4) ' , advance ='no ') VK(i,j)
! END DO
! WRITE (12 , * ) ''

```

```

! WRITE (12 , ' (1X) ')
! END DO
! WRITE (12 , ' (1X) ')
!
! WRITE (12 , ' (/ ,2X ,26(" -")) ')
! WRITE (12 , ' (2X , " GLOBAL NODAL FORCES VECTOR ") ')
! WRITE (12 , ' (2X ,26(" -") ,/) ')
!
! DO i=1 , NDOF
! WRITE (12 , ' (3X, E20 .4 ,/) ') VLOADS (i)
! END DO
! WRITE (12 , ' (1X) ')
! WRITE (12 , ' (3X ,90(" _") ,/) ')
RETURN
END SUBROUTINE ASSEMB

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! IMPLEMENT SUBROUTINE SOLVE
SUBROUTINE SOLVE
USE GLOBALVAR
INTEGER :: i , j , k , JDOF
REAL (2) :: DNOD (3) , DMX (3) , DMN (3) , PVAL
ALLOCATE ( VDISP ( NDOF ))

WRITE (12 , ' (/ ,2X ,7(" -")) ')
WRITE (12 , ' (2 X , " RESULTS ") ')
WRITE (12 , ' (2 X ,7(" -") ,/) ')

! Initialization of the vector of the constant terms
VDISP = VLOADS

! Gaussian elimination method
DO I =1 , NDOF
DO K =I +1 , NDOF
VK (I ,K) = VK (I ,K )/ VK (I , I)
END DO
VDISP (I ) = VDISP (I )/ VK (I , I)
VK (I ,I) = 1.
DO J =I +1 , NDOF
DO K = I +1 , NDOF
VK (J ,K) = VK (J ,K )- VK (J , I )* VK (I ,K)
END DO
VDISP (J ) = VDISP (J) - VK (J , I )* VDISP (I )
VK (J ,I) = 0.
END DO
END DO
! Backward substitution
DO I = NDOF ,1 , -1
DO K =I +1 , NDOF
VDISP (I ) = VDISP (I) - VK (I , K )* VDISP (K )

```

```

END DO
END DO
WRITE (12 , '(3 X , " Nodal ? displacement " , /) ')
WRITE (12 , '(3 X , 4 A15 , /) ') ' Node ' , 'Ux' , 'Uy ' , ' theta '

DNOD = 0.
DMX = -1.E6
DMN = 1.E6

! Compute the displacement of each node
DO i = 1, NNODE
DO j = 1 , 2
IF( IDOF (i , j ) <=0) THEN
DNOD ( j) = 0.
ELSE
JDOF = IDOF (i , j)
DNOD ( j) = VDISP ( JDOF )
END IF
! Evaluate min and max displacement
PVAL = DMX (j)
DMX ( j) = MAX( PVAL , DNOD ( j ))
PVAL = DMN (j)
DMN ( j) = MIN( PVAL , DNOD ( j ))
END DO
WRITE (12 , '(3 X , I15 , 3 E15 .4 , /) ') i , ( DNOD ( j) , j =1 , 3)
END DO

WRITE (12 , '( / , 3X , A15 , 3 E15 .4 , /) ') 'Max ' , ( DMX (j) , j =1 , 3)
WRITE (12 , '(3 X , A15 , 3 E15 .4 , /) ') ' Min' , ( DMN ( j) , j =1 , 3)
WRITE (12 , '( / , 3X , " DOF ? displacement " , /) ')
WRITE (12 , '(3 X , 2 A15 , /) ') ' DOF ' , ' Displacement'
DO i = 1, NDOF
WRITE (12 , '(3 X , I15 , E15 .6 , /) ') I , VDISP ( I)
END DO
WRITE (12 , '(1 X ) ')
WRITE (12 , '(3 X , 90 (" _ " ) , /) ')

RETURN
END SUBROUTINE SOLVE

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!!!

! IMPLEMENT SUBROUTINE STRESS2D
SUBROUTINE STRESS2D
USE GLOBALVAR
INTEGER :: I , J , K , NCODE (6)
REAL (2) :: PIGR = 3.14159265359 , VOL
REAL (2) :: QS (6) , B (3 , 6) , D (3 , 3) , ED1EL ( NELE ) , TMP (6) , EPS0
(3) , ALPHA , EPS (4) , SIG (4) , EPSP (2) , SIGP (2)
CHARACTER ( LEN =80) :: REC
REWIND (9)

```

```

READ (11 ,*) REC
READ (11 ,*) REC
READ (11 ,*) REC
READ (11 ,*) MEAN
WRITE (12 , ' (/ ,2X ,21(" -")) ')
WRITE (12 , '(2 X , A21 ) ') " NODAL ? STESSES / STRAINS "
WRITE (12 , '(2 X ,21(" -") ,/) ')
WRITE (12 , '(2 X ," Nodal ? stresses / strains ? based ? on :")')
IF ( MEAN ==0) THEN
WRITE (12 , '(2 X ," Arithmetic ? mean " ,/) ')
ELSEIF ( MEAN ==1) THEN
WRITE (12 , '(2 X ," Mean ? on ? the ? volume " ,/)')
ELSE
WRITE (12 , '(2 X ," Mean ? on ? energy " ,/)')
END IF

! INITIALIZATION
EDTOT = 0.
EDTOTW = 0.
VOLUMETOT = 0.
EDPTOT = 0.
ALLOCATE ( SIGNOD ( NNODE ,3) )
ALLOCATE ( EPSNOD ( NNODE ,3) )
ALLOCATE ( SUMWEIGHT ( NNODE ) )
SUMWEIGHT = 0.
SIGNOD = 0.
EPSNOD = 0.

! EVALUATION DEF . ENERGY
EDTOTW = DOT_PRODUCT ( VDISP , VLOADS )
EDTOTW = EDTOTW /2.

! DEFORMATION AND STRESSES
DO NE =1 , NELE
DO i =1 , 3
READ (9 ,*) REC
END DO
DO I =1 , 3
READ (9 ,*) B(I ,:)
END DO
READ (9 ,*) REC
DO I =1 , 3
READ (9 ,*) D(I ,:)
END DO
READ (9 ,*) REC
DO I =1 , 6
READ (9 ,*) ST (I ,:)
END DO
READ (9 ,*) REC

N1 = INC (NE ,1)
N2 = INC (NE ,2)
N3 = INC (NE ,3)

```

```

NCODE (1:2) = IDOF (N1 , :)
NCODE (3:4) = IDOF (N2 , :)
NCODE (5:6) = IDOF ( N3 , :)

DO I =1 , 6
NDOF = NCODE ( I )
IF( NDOF == -1) THEN
QS ( I ) = 0.
ELSE
QS ( I ) = VDISP ( NDOF )
END IF
END DO
DO I =1 , 3
EPS ( I ) = 0.
EPS0 = 0.
DO J =1 , 6
EPS ( I ) = EPS ( I ) + B(I , J ) * QS ( J )
END DO
END DO

IT = INC (NE , 4)
EE = CMAT ( IT , 3)
POIS = CMAT (IT , 4)
ALPHA = CMAT (IT , 5)
IF( ATYPE ==0) THEN
EPS0 (1) = ALPHA * THERM (NE)
EPS0 (2) = EPS0 (1)
ELSE
EPS0 (1) = (1. - POIS ) * ALPHA * THERM (NE)
EPS0 (2) = EPS0 (1)
END IF

DO I =1 , 3
SIG ( I ) = 0.
DO J =1 , 3
SIG ( I ) = SIG ( I ) + D(I , J ) * ( EPS ( J ) - EPS0 ( J ) )
END DO
END DO

! DEPENDENT STRAINS / STRESSES
IF( ATYPE ==0) THEN
SIG (4) = 0.
EPS (4) = - POIS / (1. - POIS ) * ( EPS (1) + EPS (2) ) + (1. + POIS ) / (1. - POIS ) * ALPHA * THERM (NE)
ELSE
EPS (4) = 0.
SIG (4) = POIS * ( SIG (1) + SIG (2) ) - EE * ALPHA * THERM (NE)
END IF

! PRINCIPAL STRESS / STRAINS
C1 = ( SIG (1) + SIG (2) ) / 2.
C2 = DSQRT ( ( ( SIG (1) - SIG (2) ) / 2. ) ** 2. + SIG (3) ** 2. )
SIGP (1) = C1 + C2

```

```

SIGP (2) = C1 - C2
C1 = ( EPS (1)+ EPS (2))/2.
C2 = DSQRT ((( EPS (1) - EPS (2))/2. )**2.+ EPS (3)**2.)
EPSP (1) = C1 + C2
EPSP (2) = C1 - C2

IF( SIG (1)/= SIG (2)) THEN
THETA = DATAN (2.* SIG (3)/( SIG (1) - SIG (2)))/2.
ELSE
THETA = PIGR /4.
IF( SIG (3) <0) THEN
THETA = - THETA
END IF
END IF
IF( SIG (1) < SIG (2)) THEN
THETA = THETA + PIGR /2.
ELSE
IF( SIG (3) <0) THEN
THETA = THETA + PIGR
END IF
END IF
THETA = THETA *180/ PIGR

WRITE (12 , ' (3 X , " ELEMENT : " , 1 X , I5 ) ' ) NE
WRITE (12 , ' (3 X , 13 ( " - " ) , / ) ' )
WRITE (12 , ' (3 X , 8 A15 ) ' ) ' epsX' , ' epsY ' , ' gammaXY ' , ' epsZ ' , '
eps1 ' , ' eps2 ' , ' theta '
WRITE (12 , ' ( / , 3X , 8 E15 .5 ) ' ) ( EPS ( I ) , I =1 , 4 ) , ( EPSP ( I ) , I =1 , 2 )
, THETA

WRITE (12 , ' ( / , 3X , 8 A15 , / ) ' ) ' sigX ' , ' sigY ' , ' tauXY ' , ' sigZ ' , '
sig1 ' , ' sig2 ' , ' theta '
WRITE (12 , ' ( / , 3X , 8 E15 .5 , / ) ' ) ( SIG ( I ) , I =1 , 4 ) , ( SIGP ( I ) , I =1
, 2 ) , THETA

THICK = CMAT (IT , 7)
N1 = INC (NE , 1)
N2 = INC (NE , 2)
N3 = INC (NE , 3)
X1 = COORD (N1 , 1)
Y1 = COORD (N1 , 2)
X2 = COORD (N2 , 1)
Y2 = COORD (N2 , 2)
X3 = COORD (N3 , 1)
Y3 = COORD (N3 , 2)
A1 = ( X2 * Y3 - X3 * Y2 )
A2 = ( X3 * Y1 - X1 * Y3 )
A3 = ( X1 * Y2 - X2 * Y1 )
AA = ( A1 + A2 + A3 ) /2.
VOL = AA * THICK
VOLUMETOT = VOLUMETOT + VOL

ED1EL (NE) = 0.

```

```

TMP = MATMUL (ST , QS )
ED1EL (NE) = DOT_PRODUCT (QS , TMP )
ED1EL (NE) = ED1EL (NE )/2.
ED1ELP = ED1EL (NE )*100./ EDTOTW
EDTOT = EDTOT + ED1EL (NE)
EDPTOT = EDPTOT + ED1ELP

WRITE (12 , '( / , 3X , 3 A15 ) ') ' Volume ' , ' Energy ' , ' Energy ?% '
WRITE (12 , '( / , 3X , 2 E15 .5 , F15 .4 , /) ') VOL , ED1EL (NE) , ED1ELP
! NODAL STRAINS / STRESSES
IF( MEAN ==0) THEN
WMED = 1.
ELSEIF ( MEAN ==1) THEN
WMED = VOL
ELSE
WMED = ED1EL (NE)
END IF

DO I =1 , 3
NODE = INC (NE , I)
SUMWEIGHT ( NODE ) = SUMWEIGHT ( NODE ) + WMED
DO J =1 , 3
EPSNOD ( NODE , J) = EPSNOD ( NODE , J) + EPS ( J ) * WMED
SIGNOD ( NODE , J) = SIGNOD ( NODE , J) + SIG ( J ) * WMED
END DO
END DO
END DO
DO I = 1, NNODE
DO J =1 , 3
EPSNOD (I , J) = EPSNOD (I , J) / SUMWEIGHT (I )
SIGNOD (I , J) = SIGNOD (I , J) / SUMWEIGHT (I )
END DO
END DO
WRITE (12 , '( / , 2X , " NODAL ? DEFORMATIONS " ) ')
WRITE (12 , '( 2 X , 18 ( " -" ) , /) ')
WRITE (12 , '( 3 X , 4 A15 , /) ') ' Node ' , ' epsX ' , ' epsY ' , ' gammaXY '
DO K =1 , NNODE
WRITE (12 , '( 3 X , I15 , 3 E15 .5 ) ') K , ( EPSNOD (K , I ) , I =1 , 3)
END DO

WRITE (12 , '( / , 2X , " NODAL ? STRESSES " ) ')
WRITE (12 , '( 2 X , 14 ( " -" ) , /) ')
WRITE (12 , '( 3 X , 4 A15 , /) ') ' Node ' , ' sigX ' , ' sigY ' , ' tauXY '
DO K =1 , NNODE
WRITE (12 , '( 3 X , I15 , 3 E15 .5 ) ') K , ( SIGNOD (K , I ) , I =1 , 3)
END DO
WRITE (12 , '( / , 3X , " ENERGY ? STORED ? IN ? THE ? STRUCTURE " ) ')
WRITE (12 , '( 3 X , 30 ( " -" ) , /) ')
WRITE (12 , '( 3 X , 3 A25 ) ') ' Total ? volume ' , ' Total ? strain ? energy '
, ' Total ? strain ? energy ? % '
WRITE (12 , '( 3 X , 2 E25 .5 , F25 .2 ) ') VOLUMETOT , EDTOT , EDPTOT
RETURN
END SUBROUTINE STRESS2D

```

```

!
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!!!!!!!!!

! SUBROUTINE PLOT2D
SUBROUTINE PLOT2D
USE GLOBALVAR
INTEGER :: I ,J , NCODE (6) , JDof
REAL (2) :: PIGR =3.14159265359
REAL (2) :: QS (6) , NDISP (2) , B (3 ,6) , D (3 ,3) , ED1EL ( NELE ) , TMP
(6) , EPS0 (3) , ALPHA , EPS (4) , SIG (4) , EPSP (3) , SIGP (3)
CHARACTER ( LEN =80) :: REC
!
! Rewind of the temporary file
!
REWIND (9)
!
! Opening file " PLOT . dat " ( for post - processing )
!
OPEN (33 , FILE = 'PLOT_FINAL_P2.DAT', STATUS = 'UNKNOWN')
!
! Type of analysis
! 0 = PLAIN STRESS
! 1 = PLAIN STRAIN
! 2 = AXISYMMETRIC
!
WRITE (33 ,*) ATYPE
!
! Number of nodes and coordinates
!
WRITE (33 ,*) NNODE
DO I =1 , NNODE
WRITE (33 ,*) ( COORD (I ,J) ,J =1 ,2)
END DO
!
! Number of elements and incidence
!
WRITE (33 ,*) NELE
DO I =1 , NELE
WRITE (33 ,*) ( INC (I , J),J =1 ,3)
END DO
!
! Strains and stresses in the elements
!
DO NE =1 , NELE
DO i =1 ,3
READ (9 ,*) REC
END DO
DO I =1 ,3
READ (9 ,*) B(I ,:)
END DO
READ (9 ,*) REC

```



```

DO I =1 ,3
READ (9 ,*) D(I ,:)
END DO
READ (9 ,*) REC
DO I =1 ,6
READ (9 ,*) ST (I ,:)
END DO
READ (9 ,*) REC

N1 = INC (NE ,1)
N2 = INC (NE ,2)
N3 = INC (NE ,3)
NCODE (1:2) = IDOF (N1 ,:)
NCODE (3:4) = IDOF (N2 ,:)
NCODE (5:6) = IDOF ( N3 ,:)
DO I =1 ,6
NDOF = NCODE (I )
IF ( NDOF == -1) THEN
QS (I ) = 0.
ELSE
QS (I ) = VDISP ( NDOF )
ENDIF
END DO
DO I =1 ,3
EPS ( I ) = 0.
EPS0 ( I ) = 0.
DO J =1 ,6
EPS ( I ) = EPS ( I )+ B(I ,J )* QS ( J)
END DO
END DO
IT = INC (NE ,4)
EE = CMAT ( IT ,3)
POIS = CMAT (IT ,4)
ALPHA = CMAT (IT ,5)
IF ( ATYPE ==0) THEN
EPS0 (1) = ALPHA * THERM (NE)
EPS0 (2) = EPS0 (1)
ELSE
EPS0 (1) = (1. - POIS )* ALPHA * THERM (NE)
EPS0 (2) = EPS0 (1)
ENDIF
DO I =1 ,3
SIG ( I ) = 0.
DO J =1 ,3
SIG ( I ) = SIG ( I )+ D(I ,J )*( EPS (J )- EPS0 ( J ))
END DO
END DO
!
! Dependent stresses / deformations
!
IF ( ATYPE ==0) THEN
SIG (4) = 0.

```

```

EPS (4) = - POIS / (1. - POIS ) * ( EPS (1) + EPS (2) ) + (1. + POIS ) / (1. - POIS
) * ALPHA * THERM (NE)
ELSE
EPS (4) = 0.
SIG (4) = POIS * ( SIG (1) + SIG (2) ) - EE * ALPHA * THERM (NE)
ENDIF
!
! Principal stresses / deformations
!
C1 = ( SIG (1) + SIG (2) ) / 2.
C2 = SQRT ( ( ( SIG (1) - SIG (2) ) / 2. ) ** 2. + SIG (3) ** 2. )
SIGP (1) = C1 + C2
SIGP (2) = C1 - C2
C1 = ( EPS (1) + EPS (2) ) / 2.
C2 = SQRT ( ( ( EPS (1) - EPS (2) ) / 2. ) ** 2. + EPS (3) ** 2. )
EPSP (1) = C1 + C2
EPSP (2) = C1 - C2
IF ( SIG (1) /= SIG (2) ) THEN
THETA = ATAN ( 2. * SIG (3) / ( SIG (1) - SIG (2) ) ) / 2.
ELSE
THETA = PIGR / 4.
IF ( SIG (3) < 0 ) THEN
THETA = - THETA
ENDIF
ENDIF
IF ( SIG (1) < SIG (2) ) THEN
THETA = THETA + PIGR / 2.
ELSE
IF ( SIG (3) < 0 ) THEN
THETA = THETA + PIGR
ENDIF
ENDIF
THETA = THETA * 180. / PIGR
WRITE (33,*) ( EPS (J), J = 1, 4 ), ( EPSP (J), J = 1, 2)
WRITE (33,*) ( SIG (J), J = 1, 4 ), ( SIGP (J), J = 1, 2)
END DO
!
! Nodal displacements
!
DO I = 1, NNODE
DO J = 1, 2
IF ( IDOF (I, J) <= 0 ) THEN
NDISP (J) = 0.
ELSE
JDOF = IDOF (I, J)
NDISP (J) = VDISP (JDOF)
ENDIF
END DO
WRITE (33,*) ( NDISP (J), J = 1, 2)
END DO
EDTOT = 0.
EDTOTW = 0.
EDPTOT = 0.

```

```

!
! Evaluation of the external work
!
EDTOTW = DOT_PRODUCT ( VDISP , VLOADS )
EDTOTW = EDTOTW /2.
!
! Evaluation of internal energy
!
REWIND (9)
DO NE =1 , NELE
DO i =1 , 3
READ (9 ,*) REC
END DO
DO I =1 , 3
READ (9 ,*) B(I ,:)
END DO
READ (9 ,*) REC
DO I =1 , 3
READ (9 ,*) D(I ,:)
END DO
READ (9 ,*) REC
DO I =1 , 6
READ (9 ,*) ST (I ,:)
END DO
READ (9 ,*) REC

N1 = INC (NE ,1)
N2 = INC (NE ,2)
N3 = INC (NE ,3)
NCODE (1:2) = IDOF (N1 ,:)
NCODE (3:4) = IDOF (N2 ,:)
NCODE (5:6) = IDOF ( N3 ,:)
DO I =1 , 6
NDOF = NCODE (I )
IF ( NDOF == -1) THEN
QS (I ) = 0.
ELSE
QS (I ) = VDISP ( NDOF )
ENDIF
END DO
ED1EL (NE) = 0.
TMP = MATMUL ( ST , QS )
ED1EL (NE) = DOT_PRODUCT (QS , TMP )
ED1EL (NE) = ED1EL (NE )/2.
ED1ELP = ED1EL (NE )*100./ EDTOTW
EDTOT = EDTOT + ED1EL (NE)
EDPTOT = EDPTOT + ED1ELP
WRITE (33 ,*) ED1ELP
END DO
WRITE (33 ,*) EDTOT
CLOSE (33)
RETURN
END SUBROUTINE PLOT2D

```

